

Regression Analysis

7 Simple Regression

Why and how to carry out simple regression analysis, and how to interpret its results

Motivation

You want to identify hotels in a city that are good deals: underpriced for their location and quality. You have scraped the web for data on all hotels in the city, and you have cleaned the data. You have carried out exploratory data analysis that revealed that hotels closer to the city center tend to be more expensive, but there is a lot of variation in prices between hotels at the same distance. How should you identify hotels that are underpriced relative to their distance to the city center? In particular, how should you capture the average price–distance relationship that would provide you a benchmark, to which you can compare actual price to find good deals?

The analysis of hotel prices and distance to the city center reveals that hotels further away from the center are less expensive by a certain amount, on average. Can you use this result to estimate how much more revenue a hotel developer could expect if it were to build a hotel closer to the center rather than farther away?

Part II of our textbook is about regression analysis, the most widely used method in data analysis. Regression is a model for the conditional mean: the mean of y for different values of one or more x variables. Regression is used to uncover patterns of association. That, in turn, is used in causal analysis, to uncover the effect of x on y, and in predictions, to arrive at a good guess of what the value of y is if we don't know it, but we know the value of x.

Simple regression analysis uncovers patterns of association between two variables: y and a single x variable. We spend this chapter, and the next two chapters, on simple regression. We turn to multiple regression, which includes more x variables, in Chapter 10. While simple regression is rarely used as the only method in actual data analysis, we spend a lot of time to understand it in detail. We do so because simple regression is a good starting point to most analyses, it is the building block of more complicated methods, and what we learn about it will carry over to multiple regression with appropriate, and usually small, modifications.

In this chapter, we introduce simple non-parametric regression and simple linear regression, and we show how to visualize their results. We then discuss simple linear regression in detail. We introduce the regression equation, how its coefficients are uncovered (estimated) in actual data, and we emphasize how to interpret the coefficients. We introduce the concepts of predicted value and residual and of goodness of fit, and we discuss the relationship between regression and correlation. We end with a note on the relationship between causation and regression.

The case study in this chapter, **Finding a good deal among hotels with simple regression**, uncovers the pattern of association between hotel prices in a European city and the distance of the hotel to the city center. We use the same data as in Chapters 1 and 2, the `hotels-vienna` dataset.

In this chapter we use this case study to illustrate simple non-parametric and linear regression, and we will develop this study further in subsequent chapters.

Learning outcomes

After working through this chapter, you should be able to:

- identify problems that call for and lend themselves to regression analysis;
- estimate simple non-parametric regression and simple linear regression using standard statistical software;
- interpret the results of simple linear regression analysis in a precise yet useful and accessible way;
- visualize patterns uncovered by regression analysis.

7.1 When and Why Do Simple Regression Analysis?

In Chapter 4 we introduced several tools to uncover patterns of association between two variables: whether and how the values of y differ when the values of x differ. In this chapter we focus on the most important pattern of association, mean-dependence. Recall that mean-dependence tells whether, and to what extent, the mean of y is different when the value of x is different.

The method we introduce in this chapter is **regression analysis**, often abbreviated simply to **regression**. Regression is the most widely used method of comparison in data analysis. **Simple regression analysis** uncovers mean-dependence between two variables. It amounts to comparing average values of one variable, called the dependent variable (y) for observations that are different in the other variable, the explanatory variable (x). This and the following chapters provide a thorough introduction and discuss many details, issues, and generalizations of the method. **Multiple regression analysis**, to be introduced in Chapter 10, involves more variables.

Discovering patterns of association between variables is often a good starting point even if our question is more ambitious. We are often interested in causal analysis: uncovering the effect of one variable on another variable. Examples include how sales would change as a result of more or different kinds of advertising, or how the educated workforce of a city would change in the future if the city were to spend more on education. Or, we may be interested in predictive analysis: what to expect of a variable (long-run polls, hotel prices) for various values of another variable (immediate polls, distance to the city center). In both causal analysis and predictions we are often concerned with other variables that may exert influence. Simple regression is a good starting point to answering the main question even if we want to consider more variables, too. Simple regression is also a major building block for more complicated methods.

7.2 Regression: Definition

Regression analysis is a method that uncovers the average value of a variable y for different values of variable x. (In this textbook we use expected value, average, and mean as synonyms.) Regression is a **model** for the **conditional mean**, or, in other words, the **conditional expectation**. Introduced in Chapter 4, Section 4.4, the conditional mean shows the mean value of y for various values of x. The model for the conditional mean gives a rule that takes x values and tells what the mean value of y is.

The precise way to formulate this idea is

$$E[y|x] = f(x) \tag{7.1}$$

This should be read as follows: the expected value of y conditional on x is given by the function f. That function f is the model, which we call regression. For most of the textbook, we use a simpler shorthand notation:

$$y^E = f(x) \tag{7.2}$$

The shorthand y^E here stands for $E[y|x]$. It is less precise because it does not say what variable the function is conditional on (here x). Instead, we should infer this from the context (here the dependence is on x because it equals $f(x)$).

There is another often used notation for the regression. This different notation acknowledges the fact that the actual value of y is equal to its expected value plus a deviation from it. That deviation is called the **error term** of the regression:

$$y = f(x) + e \tag{7.3}$$

We are not going to use this notation in this textbook. Instead, we'll denote a regression as the conditional mean: $y^E = f(x)$. Later, in Chapter 21, Sections 21.3 and 21.12, we'll discuss why a version of the error term notation may be helpful in causal analysis.

Some more conventions in notation: written as $y^E = f(x)$, the expression is a general statement of the model: the relationship between the expected value of y and different values of x. Alternatively, written as $y^E = f(x = x_0)$ means the expected value of y if the variable x equals a particular value x_0 (substitute a suitable number for x_0). It shows the result of plugging a specific x value into the model.

The y and x variables in the $y^E = f(x)$ regression are labeled using various names. Throughout this text, we shall use the following names: **dependent variable** or **left-hand-side variable**, or simply the y variable, and **explanatory variable**, **right-hand-side variable**, or simply the x variable. The box below lists a few more names that are often used. We sometimes say that we "regress y on x," or "run a regression of y on x." By these we mean that we do simple regression analysis with y as the dependent variable and x as the explanatory variable.

Regression analysis finds patterns in the data by comparing observations with different values of the x variable to see whether and how the mean value of the y variable differs across them. Thus, we need data with variation in x. Usually, the more variation in x, the better because it allows for more comparisons.

Regression analysis may reveal that average y tends to be higher at higher values of x. That would be a pattern of positive mean-dependence, or positive association. Or, it may reveal a negative association, or negative mean-dependence, with average y being lower at higher values of x. However, the pattern of association may be non-monotonic, in which average y tends to be higher for higher values of x in a certain range of the x variable and lower for higher values of x in another range of the x variable. Regression analysis may also reveal no association between average y and x, when average y tends to be the same regardless of the value of x. The goal of regression analysis is to uncover this pattern and characterize it in a useful way, by visualizing it or summarizing it in one or more numbers.

Review Box 7.1 Regression and regression analysis

- Regression analysis finds patterns of association between variables.
- Simple regression shows the average value of the variable y for different values of variable x.
- As a formula, $y^E = f(x)$, where y^E is a shorthand for the conditional expectation $E[y|x]$.
- Naming the variables in the regression:
  - y, dependent variable or left-hand-side variable, also explained variable, regressand, outcome variable, predicted variable;
  - x, explanatory variable or right-hand-side variable, also independent variable, regressor, feature variable, predictor variable.

7.3 Non-parametric Regression

Non-parametric regressions describe the $y^E = f(x)$ pattern without imposing a specific **functional form** on f. They let the data dictate what that function looks like, at least approximately. In contrast, **parametric regressions** impose a functional form on f. Parametric examples include linear functions: $f(x) = a + bx$; exponential functions: $f(x) = ax^b$; quadratic functions: $f(x) = a + bx + cx^2$. These are called parametric because the functions have parameters a, b, c. As we shall see, parametric regressions are restrictive, but they produce readily interpretable numbers. In contrast, non-parametric regressions can spot patterns that restrictive parametric functions may miss, but that comes at a price: they do not produce readily interpretable numbers.

Non-parametric regressions come in various forms. When x has few values and there are many observations in the data, the best and most intuitive non-parametric regression for $y^E = f(x)$ shows average y for each and every value of x. There is no functional form imposed on f here. For example, the hotels in our data have 3 stars, 3.5 stars, or 4 stars. If we calculate the average price of hotels with the same numbers of stars and compare these averages across 3, 3.5, and 4 stars, we are carrying out a non-parametric regression analysis.

With many x values, things become more complicated, especially when the data has few observations for some, or all, x values. In such cases, there are two ways to do non-parametric regression analysis: **bins** and **smoothing**.

The first and more straightforward way is based on grouped values of x: bins. Bins are disjoint categories (no overlap) that span the entire range of x (no gaps). How exactly to create bins and how many such bins should be created are decisions for the analyst, similarly to the case of creating a histogram.

In fact, we have already seen such non-parametric regressions in Chapter 4 where we introduced their visualization: **bin scatters**. Bin scatters show the average value of y that corresponds to each bin created from the values of x, in an x–y coordinate system. An alternative visualization of the same non-parametric regression is the **step function**. A step function shows average y over the entire range of the bins. The case study will show some examples.

The second method produces a "smooth" graph that is both continuous and has no kink at any point. There are many ways of smoothing and thus many ways of producing such graphs. In data science these types of graphs are called smoothed conditional means plots. Indeed, non-parametric regression of this kind shows conditional means, smoothed to get a better image. Of the various methods available we consider one, called lowess.

Lowess is one of the most widely used non-parametric regression methods that produces a smooth graph. The name is an abbreviation of locally weighted scatterplot smoothing (sometimes abbreviated as "loess"). A lowess may be thought of as a smooth curve fit around a bin scatter. It may help to relate this method to density plots (introduced in Chapter 3, Section 3.3). Similarly to density plots, we have to set the bandwidth for smoothing, a choice that affects the results. In general, a wider bandwidth results in a smoother graph but may miss important details of the pattern. In contrast, a narrower bandwidth produces a more rugged-looking graph but may uncover more details of the pattern. Producing a lowess graph with default software settings is usually a meaningful way to visualize the patterns of association between two variables.

Smooth non-parametric regression methods, including lowess, do not produce numbers that would summarize the $y^E = f(x)$ pattern. Instead, they provide a value y^E for each of the particular x values that occur in the data, as well as for all x values in-between. The typical way to show the results of non-parametric regressions is via a graph. We interpret these graphs in qualitative, not quantitative ways. They can show interesting shapes in the pattern, such as non-monotonic parts, steeper and flatter parts.

In any case, if we want to visualize a regression, such as a lowess, it is good practice to plot it **together with the scatterplot** of y versus x. As we introduced in Chapter 4, Section 4.5, scatterplots visualize the joint distribution of two variables in the data in an x–y coordinate system. Together with the scatterplot, visualization of a regression shows not only the extent to which the mean of y tends to differ across observations with different values of x, but it also shows how the actual values of y are scattered around those mean y values. That helps in putting the slope of the regression in context.

7.A1 CASE STUDY – Finding a Good Deal among Hotels with Simple Regression

Question, data, non-parametric regression

In this case study we use simple regression analysis to find a good deal among hotels. In particular, we want to find hotels that are underpriced relative to their distance to the city center. The use of regression analysis is to find what the hotel price "should be" for various levels of distance to the city center, and then compare that price to the actual price of each hotel to identify the especially low values.

We look at Vienna hotels for a November 2017 weekday. Based on our investigations in Chapters 2 and 3, we focus on hotels that are located in Vienna actual, are not too far from the center, are classified as hotels, and have 3, 3.5, or 4 stars. We excluded extremely high priced hotels that we had classified as error. In the original dataset, there are 428 hotels for that weekday in Vienna; our focused sample has N = 207 observations.

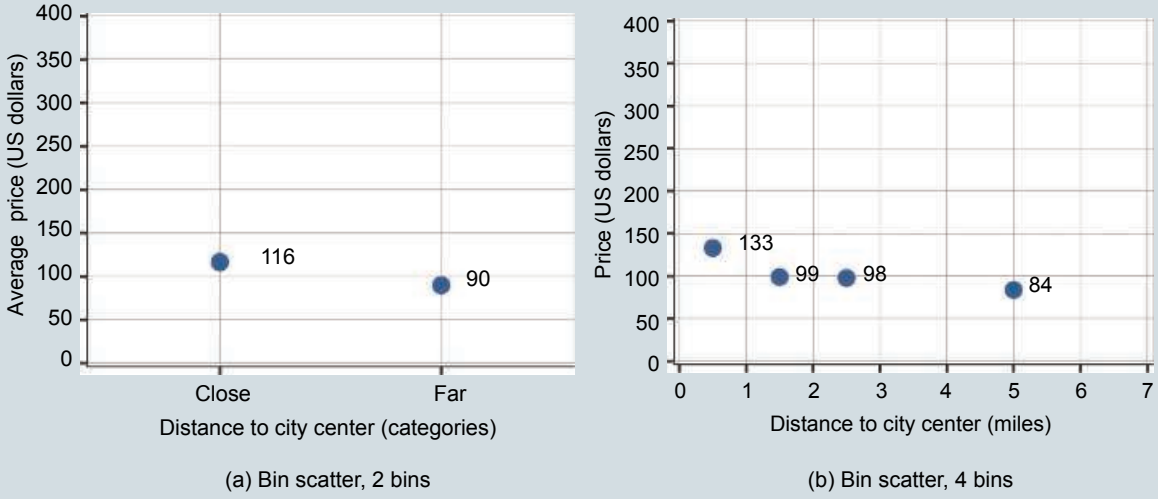
To learn about the average price of hotels with differing distances to the city center, the simplest comparison is to split the data into two parts, or, as we say, two bins: one with hotels that are close to the city center and one with hotels that are further away. At what distance to split the data is, of course, arbitrary. Let's say that the couple is fine walking two miles, so let's put the distance threshold at two miles. The "close" bin contains 157 hotels, with an average distance of 1 mile (minimum 0.1 miles, maximum 1.9 miles); the "far" bin contains 50 hotels, with an average distance of 3.2 miles (minimum 2.0 miles, maximum 6.6 miles).

The average price of the hotels close to the city center is 116 US dollars; the average price of the hotels far from the city center is 90 dollars. The average price difference is therefore −26 dollars. Simple as it may be, this comparison of average prices between "close" and "far" hotels is actually an example of regression analysis. Indeed, as we indicated earlier, regression may be thought of as generalizing this approach of comparing means.

We can create more refined bins, splitting each of the previous close and far bins into two. The four bins we have created have ranges [0,1], [1,2], [2,3], and [3,7] – all expressed in miles. This produces a different non-parametric regression from the previous one. It has the same dependent variable but different bins for the explanatory variable.

The results of the two regressions are visualized in Figures 7.1a and 7.1b. These are bin scatters. The *x* axis of Figure 7.1a has only two points, close and far. We put these two points at arbitrary places in terms of the *x* axis. In contrast, on Figure 7.1b we chose to have the continuous distance on the *x* axis. Here we put the regression points at specific values of distance within the bins. We chose the midpoint of each bin, which is usually good practice.

Both regressions show that hotels that are closer to the city center are, on average, more expensive. In other words, both suggest a negative pattern of association between hotel price and distance to the center. Whether the relationship is linear, convex, concave, or none of these, cannot be established with two values of *x* only (Figure 7.1a). With four distance categories, we can say more (Figure 7.1b). The relationship appears to be monotonic but nonlinear: the differences in average *y* between the adjacent bins are not always the same. We see a larger negative difference between the [0,1] miles and the [1,2] miles bins than between adjacent bins at higher distances.



(a) Bin scatter, 2 bins

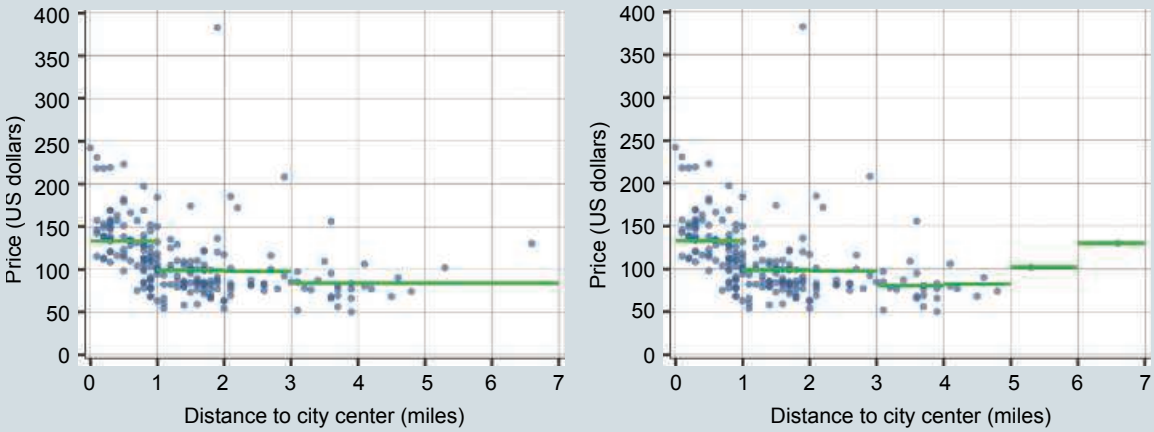
(b) Bin scatter, 4 bins

Figure 7.1 Hotel price and distance to the city center: two examples of bin scatters

Note: Bin scatter non-parametric regression. 2 bins: close if 0–1.9 miles, far if 2–7 miles. 4 bins: [0,1], [1,2], [2,3], [3,7] miles; average price visualized at the midpoint of each distance bin.
Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. N=207.

Figure 7.2a visualizes the same four-bin non-parametric regression in a different way. Instead of a bin scatter, it shows a step function: horizontal line segments spanning the entire range of

each bin. The figure includes the scatterplot as well, with the prices and distances of all 207 hotels. Having the regression results together with the scatterplot puts the regression in perspective. The average pattern is downward sloping, but there is a lot of variation of prices around that average pattern. Also, the scatterplot suggests that the negative pattern may not be valid beyond five miles, although there are only two hotels there.



(a) Non-parametric regression, 4 bins (b) Non-parametric regression, 7 bins

Figure 7.2 Hotel price and distance to the city center: non-parametric regression and scatterplot

Note: Bin scatter non-parametric regression with 4 bins and 7 bins.
Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. N=207.

A non-parametric regression with even more bins is visualized in Figure 7.2b. There are seven bins here, each of width of one mile. This regression looks non-monotonic, with flat parts in the middle and positive slope at the end. If we take a closer look, there is more variation around the step function in the middle than towards the end. At the same time there are very few observations in some of these bins: only one hotel in the [6,7] bin, one in the [5,6] bin, and six in the [4,5] bin. It also turns out that defining the bins in different ways, such as bins with equal numbers of hotels instead of equal-length distances produces a somewhat different shape. You are invited to produce non-parametric regressions with alternative bin definitions as a data exercise.

We can already draw an important conclusion from this analysis: when using bins, the number and size of bins may affect the patterns suggested by the non-parametric regressions.

After bin scatters, let's look at a smooth non-parametric regression. Figure 7.3 shows the lowess non-parametric regression, together with the scatterplot. The default bandwidth selected by statistical software is 0.8 miles. This is very close to the 1 mile bin width of the bin scatter with seven bins shown in Figure 7.2b. The smooth non-parametric regression retains some aspects of that figure but not all. In particular, it, too, is steeper at small distances and flatter at longer distances, and it turns positive at the longest distances. At the same time it has fewer rugged features. It is what it is supposed to be: a smoother version of the corresponding non-parametric regression with disjoint bins of similar width.

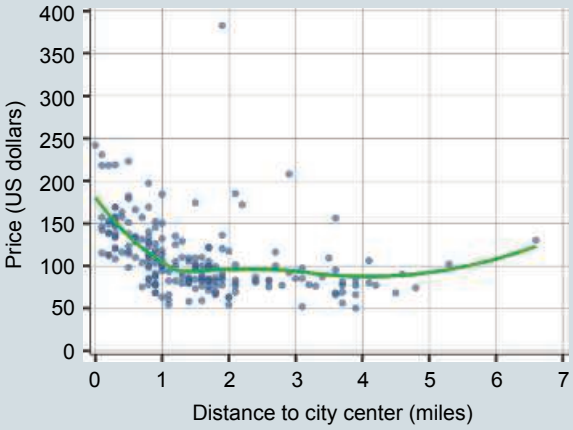


Figure 7.3 Hotel price and distance to the city center: lowess regression and scatterplot

Note: Lowess non-parametric regression, bandwidth=0.8.
Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. N=207.

The most robust pattern we uncovered across these various non-parametric regressions is a negative slope in general. Yes, there are two hotels at large distance that do not fit this pattern. But the other more than 200 hotels do. Thus, hotels further away from the city center are, on average, less expensive. But how much less expensive are they? In particular, if we compare two hotels, one of which is one mile further away from the city center than the other, what price difference should we expect between the two? This is a quantitative question that is not easy to answer with the non-parametric regression results at hand. That would be a lot easier to do with a regression that produces readily interpretable quantitative answers. Linear regression does that; we shall discuss it in the next section.

Review Box 7.2 Non-parametric regression

- Non-parametric regressions show the average of the dependent variable as a function of the explanatory variable without imposing a particular functional form.
- The three most frequent types of non-parametric regressions show the average of the dependent variable computed:
  - for each and every value of the explanatory variable;
  - for bins (disjoint intervals) created from the explanatory variable;
  - as a smooth curve making use of overlapping intervals of the explanatory variable (lowess being the most widely used method).

7.4 Linear Regression: Introduction

Linear regression is the most widely used method in data analysis. It is the simplest regression method, and it is also the major building block of many more complicated regression models. Simple linear regression imposes linearity of the function f in $y^E = f(x)$. Linear functions have two parameters,

also called coefficients: the **intercept coefficient** and the **slope coefficient**. By convention, we use Greek letters for these coefficients so that a linear regression has the form

$$y^E = \alpha + \beta x \tag{7.4}$$

To be more precise, linear regression imposes linearity in terms of its coefficients. We can have any function, including any nonlinear function, of the original variables themselves. It's still a linear regression as long as it consists of an intercept plus a slope coefficient, whether that slope multiplies x or any function of x (think of logarithm, square, and so on). Indeed, in Chapter 8 we will discuss several ways we can use linear regression to capture nonlinear, and potentially non-monotonic, patterns of associations between the original y and x variables. For the remainder of this chapter, we focus on linear regressions with y and x, not their functions. Or, alternatively, whatever we say should be understood for regressions with a left-hand-side variable and a right-hand-side variable, be those the original variables in the data or their transformations.

With y on its left-hand side and x on its right-hand side, linear regression is a line through the x–y scatterplot. With the usual method of calculating the coefficients, to be discussed below in Section 7.7, this line is the best-fitting line one can draw through the scatterplot. It is the best fit in the sense that it is the line that is closest to all points of the scatterplot. We shall discuss the fit of the linear regression in more detail later, in Section 7.9.

One way to look at the linearity in linear regression is as an assumption: by doing linear regression analysis we assume that the regression function is linear in its coefficients. While, as we shall see, this can accommodate various **nonlinear patterns**, it cannot capture all kinds of nonlinear patterns. The linearity assumption may or may not be true, and, unfortunately, that often cannot be established within the framework of linear regression.

Another, more useful way to look at linearity is to treat it as an approximation. Whatever the form of the $y^E = f(x)$ relationship, the $y^E = \alpha + \beta x$ regression fits a line through it. By fitting a line, linear regression approximates the average slope of the $y^E = f(x)$ curve. The average slope has an important interpretation: it is the difference in average y that corresponds to different values of x, averaged across the entire range of x in the data.

7.5 Linear Regression: Coefficient Interpretation

What makes linear regression very powerful is that its coefficients have a clear interpretation based on the idea of comparing conditional means. The linear regression $y^E = \alpha + \beta x$ has two coefficients.

Less important is the **intercept**: α. It is the average value of y when x is zero. Formally: $E[y|x = 0] = \alpha + \beta \times 0 = \alpha$.

The more important coefficient is the **slope**: β. It shows the expected difference in y corresponding to a one unit difference in x. y is higher, on average, by β for observations with a one-unit higher value of x. Or, in a longer version that is sometimes more helpful: comparing two observations that differ in x by one unit, we expect y to be β higher for the observation with one unit higher x. Formally: $E[y|x = x_0 + 1] - E[y|x_0] = (\alpha + \beta \times (x_0 + 1)) - (\alpha + \beta \times x_0) = \beta$.

Sometimes the slope of linear regression is given more ambitious interpretations. One such ambitious interpretation talks about increases or decreases, e.g., saying that "β shows how much y increases, on average, when x is increased." This interpretation may be correct if the data is a time series where a change in y is regressed on a change in x. However, it is not correct in general.

In fact, it can be very misleading for cross-sectional data where we are comparing different observations instead of comparing changes for the same observation. Another, even more ambitious interpretation is calling the slope the "effect" of x on y. Attributing a cause and effect relationship to differences uncovered by regression analysis is a conclusion that may or may not be correct. Typically, it is not correct in observational data – a question that we'll return to at the end of this chapter and in Chapters 19 and 21.

Importantly, the interpretation based on comparing conditional means is always true whether or not the more ambitious interpretations are true. It is good practice to always state this interpretation explicitly. If we want to go further to a more ambitious interpretation, we should state that more ambitious interpretation after this one, adding the additional assumptions behind it and the evidence that substantiates those assumptions.

Review Box 7.3 Simple linear regression

- Simple linear regression shows the average of the dependent variable as a linear function of the explanatory variable:

$$y^E = \alpha + \beta x$$

- This line is best thought of as a linear approximation of an underlying $y^E = f(x)$ pattern of any form.
- The parameters, or coefficients, of the linear regression are:
  - α is the intercept: it is the average value of y when x is zero.
  - β is the slope: y is β units higher, on average, for observations with one unit higher value of x.

7.6 Linear Regression with a Binary Explanatory Variable

An important special case of linear regression is when x is a binary variable, zero or one. The regression formula is the same:

$$y^E = \alpha + \beta x \tag{7.5}$$

The parameters of this regression are interpreted as follows: α is the average value of y when x is zero; β is the difference in average y between observations with $x = 1$ and observations with $x = 0$; and the average value of y when $x = 1$ is $\alpha + \beta$. In formulae:

$$E[y|x = 0] = \alpha \tag{7.6}$$

$$E[y|x = 1] - E[y|x = 0] = \beta \tag{7.7}$$

$$E[y|x = 1] = \alpha + \beta \tag{7.8}$$

In the case of a binary explanatory variable, linear regression is also the only non-parametric regression. The results of non-parametric regression with a binary x are two values: average y when x is zero and average y when x is one. We can get these two numbers with the help of the two parameters of the linear regression. Graphically, the regression line of linear regression goes through two points: average y when x is zero (α) and average y when x is one ($\alpha + \beta$).

Review Box 7.4 Simple linear regression with binary right-hand-side variable

Consider a binary variable x, where $x = 0$ or 1. In this case, the simple linear regression, $y^E = \alpha + \beta x$, has the following interpretations:

- α (intercept): the average value of y is α when x is zero.
- β (slope): average y is β higher among observations with $x = 1$ than among observations with $x = 0$.
- When $x = 1$, the average value of y is $\alpha + \beta$.

7.7 Coefficient Formula

One of the advantages of linear regression is that there are ready formulae for its coefficients. These formulae can help us understand how linear regression works. At a practical level, we simply have to plug in appropriate values in the data in these formulae to get the coefficients. In practice, statistical software does the plugging in. This is an advantage compared to methods that don't have such formulae, which instead require computers to execute long iterative algorithms (we'll discuss such methods in Chapter 11). With the advance of computing, this advantage has lost its appeal for small and moderate-sized data. But it still matters with very big data.

α and β denote the coefficients of linear regression in the abstract. In a particular dataset, the calculated values are denoted by putting a hat on top of the Greek letters of the parameters, such as $\hat{\alpha}$ and $\hat{\beta}$. These calculated values $\hat{\alpha}$ and $\hat{\beta}$ are called **estimates** of the coefficients α and β. This distinction becomes important when we want to generalize the patterns uncovered in a particular dataset to other situations, a question we shall address in Chapter 9. The formulae are called **estimators** as they give the estimates once data is plugged into them.

The slope coefficient is calculated by first calculating the covariance of the dependent variable and the explanatory variable in the data and then dividing that by the variance of the explanatory variable in the data. The slope coefficient formula is

$$\hat{\beta} = \frac{Cov[x, y]}{Var[x]} = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

One way to understand the slope coefficient formula is to view it as a normalized version of the covariance between x and y. Recall from Chapter 4 that the covariance measures the degree of association between the two variables, but it returns values that are hard to interpret. The slope measures the covariance relative to the variation in x. That is why the slope can be interpreted as differences in average y corresponding to differences in x.

The intercept is calculated after the slope is calculated. It is average y minus average x multiplied by the estimated slope $\hat{\beta}$:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} \tag{7.9}$$

The formula of the intercept reveals that the regression line always goes through the point of average x and average y. To see that formally, just rearrange the formula to get $\bar{y} = \hat{\alpha} + \hat{\beta}\bar{x}$. In linear regressions, the expected value of y for average x is indeed average y.

The derivation of the formulae is called **ordinary least squares** and is abbreviated as **OLS**. In Under the Hood section 7.U1, we present the derivation in detail. The idea underlying OLS is to find the values of the intercept and slope parameters that make the regression line fit the scatterplot best.

Data analysts sometimes use the name OLS for the linear regression with coefficients estimated by ordinary least squares. The frequent usage of this name reflects the fact that the simple linear regression is a line fitted on the scatterplot, and OLS finds the best fit.

Figure 7.4 shows the idea: it is a scatterplot with the best-fitting linear regression found by OLS. The figure is based on artificial data with two variables, *x* and *y*, that we created for illustration purposes. Besides the best-fitting OLS regression line it shows a vertical dashed line at the average value of *x* and a horizontal dashed line at the average value of *y*. The regression line goes through the point of average *x* and average *y*.

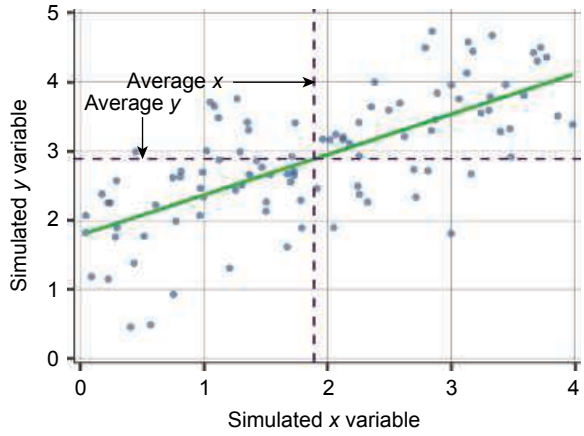


Figure 7.4 Scatterplot and best-fitting linear regression found by OLS

Note: Artificial data with 100 observations. Dashed lines at average *x* and average *y*.

Mathematically, the OLS method finds the values of the coefficients of the linear regression that minimize the sum of squares of the difference between actual *y* values and their values implied by the regression, $\hat{\alpha} + \hat{\beta}x$. In the next section we'll discuss these differences: they are called regression residuals.

$$min_{\alpha,\beta} \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)^2 \tag{7.10}$$

For this minimization problem, we can use calculus to give $\hat{\alpha}$ and $\hat{\beta}$, the values for α and β that give the minimum. Under the Hood section 7.U1 gives the details.

Review Box 7.5 The OLS coefficients of simple linear regression

$$y^E = \alpha + \beta x$$

- The formula for β, the slope: $\hat{\beta} = \frac{Cov[x,y]}{Var[x]} = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$
- The formula for α, the intercept: $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$

- The calculated values are called estimates.
- The formulae used for calculating the values are called estimators.
- The method to calculate the estimates is called OLS (for ordinary least squares).
- The estimates, and estimators, are also called OLS estimates or OLS estimators. Sometimes linear regression with OLS estimation is simply called OLS, too.
- The linear regression is the best-fitting line through the scatterplot when its coefficients are estimated with OLS.

7.A2 CASE STUDY – Finding a Good Deal among Hotels with Simple Regression

Linear regression

The linear regression of hotel prices (in dollars) on distance (in miles) produces an intercept of 133 and a slope of −14. Figure 7.5 shows the regression line together with the scatterplot.

The intercept is 133, suggesting that the average price of hotels right in the city center is 133 US dollars. Note that there is no such hotel right in the city center because that is the middle of a square containing a church: the hotel closest to the city center is a few steps away from it.

The slope of the linear regression is −14. Hotels that are 1 mile further away from the city center are, on average, 14 US dollars cheaper in our data.

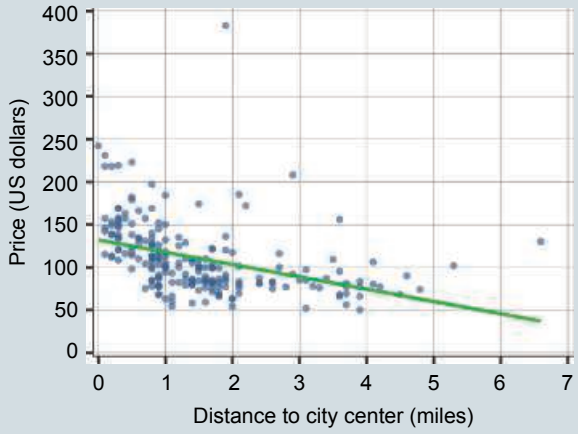


Figure 7.5 Hotel price by distance to the city center: linear regression and scatterplot

Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. N=207.

Simple regressions fit a straight line. Therefore, by design, our simple regression cannot reproduce the convex nonlinear pattern suggested by the non-parametric regressions, i.e., that the same difference in distance is associated with larger price differences closer to the city center.

We can assess how good the linear regression is in approximating the nonlinear relationship between average price and distance we uncovered earlier. The slope of the linear regression is −14; if it gave a perfect approximation to the average slope, it would be close to the average

of the many slopes calculated in a non-parametric regression with many distance bins. Take our non-parametric regression with four non-overlapping bins in distance, shown earlier in Figure 7.2a.

The distance of the midpoints of the first and second bins is 1 mile, the distance between the midpoints of the second and the third bin is 1 mile again, and for the third and fourth bins it's 2.5 miles. The corresponding slopes are between the consecutive values: 99 – 133 = – 34, 98 – 99 = –1, and 84 – 98 = –14. Let's divide these by the distance between midpoints: 1, 1, and 2.5, and get –34, –1, and –6 (rounded). Their unweighted average is –14. But that's not really the right way of calculating the average, as the different bins have different numbers of hotels. A weighted average of the differences is –20.

This average slope of –20 that we have calculated from the four-point bin scatter is one approximation of the average slope of the $f(x)$ function. The slope of the linear regression, –14 (coincidentally the same as the unweighted average), is another approximation. These two numbers are different but not very different.

7.8 Predicted Dependent Variable and Regression Residual

The **predicted value** of the dependent variable is our best guess for its average value if we know the value of the explanatory variable. The predicted value can be calculated from the regression: it is the calculated value of y^E using $f(x)$ for a particular value of x. As the name suggests, the primary use of the predicted value is in predictive analysis.

The predicted value of dependent variable y for observation i is denoted as $\hat{y}_i$. The predicted value is straightforward to compute in a linear regression: $\hat{y}_i$ for observation i, making use of a simple linear regression, is calculated by plugging in the value of the explanatory variable (x_i) and the estimated values of the coefficients to get

$$\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i \tag{7.11}$$

Predicted dependent variables exist in non-parametric regressions, too. They cannot be computed by plugging in values into a formula as there is no formula for non-parametric regressions. Instead, the result of a non-parametric regression is the complete list of predicted values of the dependent variable for each value of the explanatory variable in the data.

The **residual** is another important measure that we can compute from a regression. It is the difference between the actual value of the dependent variable for an observation and its predicted value:

$$e_i = y_i - \hat{y}_i. \tag{7.12}$$

The concept of the residual is general for all regressions, not only linear regression. The residual is always the difference between the actual dependent variable and the value implied by the regression: $y_i - \hat{y}_i$. The residual may be important in its own right. Indeed, we may be interested in identifying observations that are special in that they have a dependent variable that is much higher or much lower than "it should be" as predicted by the regression. That is in fact the case in our case study, as we'll explain below.

The scatterplot with the regression line helps visualizing both concepts. The predicted values of the dependent variable are the points of the regression line itself. The residual for a particular observation

is a difference of two y values: the value of y for the observation (the y value of the scatterplot point) minus its predicted value $\hat{y}$ (the y value of the regression line for the corresponding x value). Thus, the residual is the vertical distance between the scatterplot point and the regression line. For points above the regression line the residual is positive. For points below the regression line the residual is negative. For points right on the regression line the residual is zero.

It turns out that the residuals sum to zero if a linear regression is fitted by OLS. As their sum is zero, the average of the residuals is zero, too. A related fact is that the predicted average is equal to the actual average of the left-hand-side variable: average $\hat{y}$ equals average y. These are all consequences of the way OLS calculates coefficients $\hat{\alpha}$ and $\hat{\beta}$. The derivations are included in Under the Hood section 7.U1.

With estimated $\hat{\alpha}$ and $\hat{\beta}$, the predicted value of y can be computed for any value of x. This is possible for any x value: not only those that correspond to an observation in our data but other values for x, too.

Computing the predicted value of the dependent variable for non-existing values of the explanatory variable in-between existing values in the data is **interpolation**. Computing it for values of the explanatory variable that are outside its range (below the smallest existing value in the data or above the largest existing value) is called **extrapolation**. Usually, interpolation is viewed as a relatively benign exercise. Extrapolation, on the other hand, is often viewed as adventurous. It works fine if the patterns are similar outside the range but not otherwise. However, this is an assumption that we cannot check using the existing data. Hence, we suggest caution with extrapolation.

In contrast with predicted y values, residuals can be computed for existing observations only. That's because a residual is the difference between the actual value of y for an observation and its predicted value. While we can have predicted values for any x, actual y values are only available for the observations in our data.

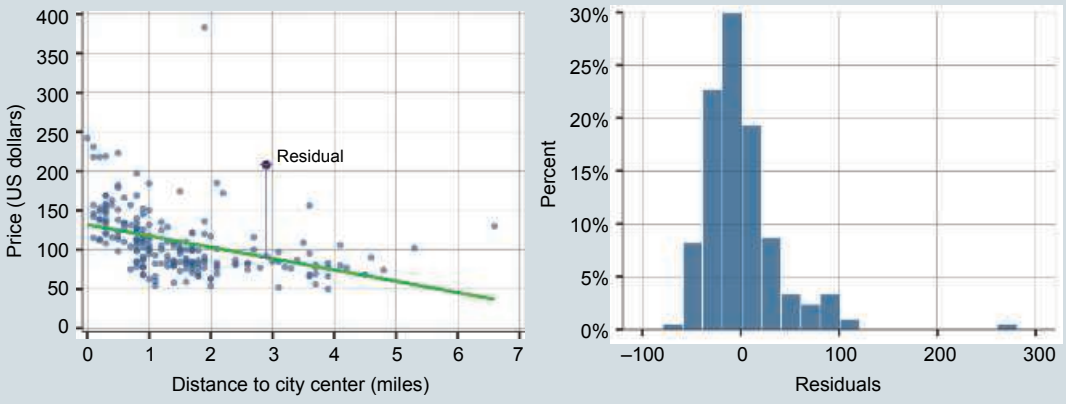
Review Box 7.6 Predicted dependent variable and residual in simple linear regression

- The predicted value of the dependent variable for observation i is $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$, where $\hat{\alpha}$ and $\hat{\beta}$ are the OLS estimates for the coefficients of the linear regression $y^E = \alpha + \beta x$.
- The residual for observation i is $e_i = y_i - \hat{y}_i$.
- Predicted values ($\hat{y}$) can be computed for any x value even if it's not in the data:
  - interpolation means predicting $\hat{y}$ for x not in the data but in-between x values in the data;
  - extrapolation means predicting $\hat{y}$ for x not in the data and outside the range of x in the data.
- Residuals can only be computed for observations in the data.

7.A3 CASE STUDY – Finding a Good Deal among Hotels with Simple Regression

Predicted price and regression residual

Figure 7.6a illustrates the predicted values and residuals of the linear regression of hotel prices as the dependent variable and distance to the city center as the explanatory variable. It is an exact replica of Figure 7.5 with added notes. The regression line shows the predicted values. The vertical distance of each point in the scatterplot from the regression line is the residual corresponding to that observation.



(a) Regression line and the residual (b) Distribution of residuals

Figure 7.6 Hotel price by distance to the city center: linear regression and residuals

Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. N=207.

Figure 7.6b shows the distribution of the residuals. The distribution has mean zero, but it is skewed with a longer right-hand tail. There are more hotels with a negative residual, but those residuals tend to be smaller than the positive residuals.

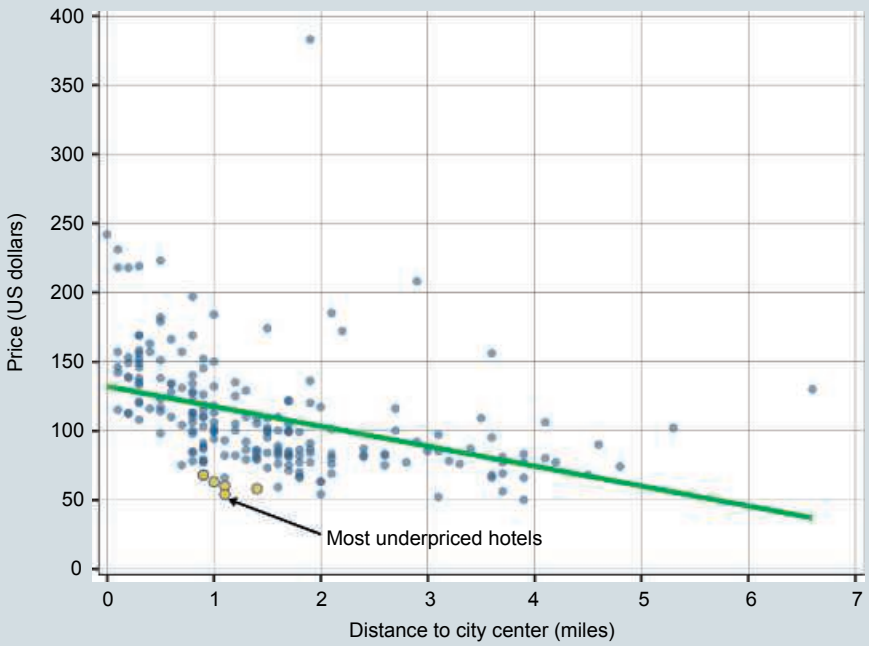


Figure 7.7 Underpriced hotels

Note: Scatterplot, regression line, and hotels with the five lowest values of the residual. Underpriced hotels are below the regression line; overpriced hotels are above the regression line.
Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. N=207.

The residuals help find an answer to the question of the case study: how to find a good deal. Candidates for a good deal are hotels that are underpriced relative to their location. These are hotels that come with a lower price than is expected for their location. Regression residuals show the difference of actual price from what's expected based on the regression. The most underpriced hotels are the ones with the lowest, most negative residuals, as shown in Figure 7.7.

Table 7.1 shows a list of the hotels with the five lowest values of the residual.

Table 7.1 Most underpriced hotels

No.	Hotel_id	Distance	Price	Predicted price	Residual
1	22080	1.1	54	116.17	−62.17
2	21912	1.1	60	116.17	−56.17
3	22152	1	63	117.61	−54.61
4	22408	1.4	58	111.85	−53.85
5	22090	0.9	68	119.05	−51.05

Note: List of the five hotels with the largest negative residual from the linear regression of hotel price on distance.
Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. N=207.

If we thought that the simple linear regression we estimated gave good prediction and good residuals, we would be done. Then this list would offer five good candidates for the couple to consider and investigate more carefully to choose a hotel that is underpriced for its location.

Unfortunately, this simple linear regression may not produce the best possible predictions and residuals. Indeed, it's a little suspicious that this particular list includes hotels that are all around 1 mile from the center. The reason can be seen in Figure 7.6: there is a bunching of large negative-residual observations around 1 mile. But that appears to be a consequence of fitting a line on a nonlinear pattern.

The regression line does not capture the nonlinear pattern suggested by the non-parametric regressions and, indeed, by the scatterplot (steeper slope at shorter distances). In Chapters 8 and 10 we shall further develop this study to get a better list, using residuals from regressions that accommodate the nonlinear pattern and that add other features of hotels besides their distance to the city center.

Finally, let us discuss the data visualization aspect of the graph in Figure 7.7. In particular, let's discuss the visual layers we introduced in Chapter 3, Section 3.5: graph type, encoding, scaffolding, and annotation.

In the graph, we used a plain scatterplot as the number of observations is not very large, and we had several information encoded. First, it is a scatterplot, so each observation is denoted by a dot. Second, the five most underpriced hotels are denoted with a different color. Third, a regression line is shown, drawn with a different color.

For scaffolding: as x and y are quantitative, we used grid lines, as they help read off approximate values. We have prices and distance values along both axes.

Finally, regarding annotations, we used a different color and added an arrow and a textbox to draw attention to the five dots depicting best deals.

7.9 Goodness of Fit, R-Squared

An important property of regression is how well it fits the data, called **goodness of fit**. One result of an estimated regression are the predicted values of y for all values of x in the data. The **fit of a regression** captures how these predicted values compare to the actual values.

The most commonly used measure of the fit of a regression is the **R-squared** (R^2). R-squared captures how close the predicted y values are to the actual values. It does so by measuring how much of the variation in y is captured by the regression, and how much is left for residual variation. R-squared may be defined as how much of the overall variation in y is captured by variation predicted by the regression, that is, the variation in $\hat{y}$. Another way of looking at the same thing considers how much is there that is not residual variation. In formulae:

$$R^2 = \frac{Var[\hat{y}]}{Var[y]} = 1 - \frac{Var[e]}{Var[y]} \tag{7.13}$$

where $Var[y] = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2$, $Var[\hat{y}] = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$, and $Var[e] = \frac{1}{n}\sum_{i=1}^{n}(e_i)^2$. Note that $\bar{\hat{y}} = \bar{y}$, and $\bar{e} = 0$.

This is based on the decomposition of the overall variation in the dependent variable into variation in predicted values (variation "explained by the regression") and the remaining, residual variation (variation "not explained by the regression"):

$$Var[y] = Var[\hat{y}] + Var[e] \tag{7.14}$$

R-squared (or R^2) can be defined for both parametric and non-parametric regressions. Any kind of regression produces predicted $\hat{y}$ values, and all we need to compute R^2 is its variance compared to the variance of y.

The value of R-squared is always between zero and one. R-squared is one if the regression fits the data perfectly. In this case each and every data point is equal to its predicted value from regression, and all residuals are zero. In a linear regression it means that all data points lie exactly on the regression line. The other polar case is an R-squared of zero. In this case all of the predicted $\hat{y}$ values are equal to the overall average value $\bar{y}$ in the data regardless of the value of the explanatory variable x. This corresponds to a slope of zero: the regression line is completely flat.

The fit of a regression depends on two factors: (1) how well the particular version of the regression captures the actual function f in $y^E = f(x)$, and (2) how far actual values of y are spread around what would be predicted using the actual function f. The decisions of the analyst can affect the first: for example, a non-parametric regression almost always gives a better fit than a linear regression. The second factor is beyond the control of the analyst and is determined by the nature of the data. In some data actual y values are close to some $f(x)$; in other data they are widely spread around it. In the second case, even the best-fitting regressions may achieve low R-squared.

When our goal is prediction, R-squared may help in choosing between different versions of regression for the same data. Besides giving information on the fit of a particular regression R-squared is used to compare alternative regressions that use different x variables to predict the same y variable. Comparing two regressions, the one with a higher R-squared does a better job at predicting y in the data.

Review Box 7.7 The fit of a regression

- The fit of a regression captures how close the predicted y values are to the actual values.
- R-squared is a measure of fit of a regression.
- R-squared measures the fit by calculating how much of the variation in y is captured by the regression, and how much is left for residual variation: $R^2 = \frac{Var[\hat{y}]}{Var[y]} = 1 - \frac{Var[e]}{Var[y]}$
- R-squared is based on the fact that $Var[y] = Var[\hat{y}] + Var[e]$.

7.10 Correlation and Linear Regression

Linear regression is closely related to correlation. As we introduced in Chapter 4, the correlation coefficient measures the degree of association between two variables. It is a normalized version of the covariance, dividing it by the standard deviations of the two variables: $Corr[x, y] = \frac{Cov[y,x]}{Std[y]Std[x]}$. The correlation coefficient is between -1 and $+1$, and the measure is symmetric in that x and y are interchangeable.

The OLS formula for the slope estimate of the linear regression $y^E = \alpha + \beta x$ is also a normalized version of the covariance, only here it is divided by the variance of the x variable: $\hat{\beta} = \frac{Cov[y,x]}{Var[x]}$. In contrast with the correlation coefficient, its values can be anything, and y and x are not interchangeable.

Despite their differences, the covariance, the correlation coefficient, and the slope of a linear regression capture similar information: the degree of association between the two variables. Formally, we can express the correlation coefficient in terms of the slope coefficient and vice versa (recall that $Var[x] = (Std[x])^2$):

$$\hat{\beta} = Corr[x, y]\frac{Std[y]}{Std[x]} \quad \text{or} \quad Corr[x, y] = \hat{\beta}\frac{Std[x]}{Std[y]} \tag{7.15}$$

As a side note, let's consider yet one more way to normalize the covariance: dividing it by the variance of y not x. Actually, that would be the OLS estimator for the slope coefficient of the **reverse regression**: switching the role of y and x in the linear regression. Using different Greek letters to avoid confusion, the reverse regression formula is

$$x^E = \gamma + \delta y \tag{7.16}$$

The OLS estimator for the slope coefficient here is $\hat{\delta} = \frac{Cov[y,x]}{Var[y]}$. Again, this captures very similar information to the covariance, the correlation coefficient, and the OLS estimator of the slope of the original linear regression, $\hat{\beta}$. The OLS slopes of the original regression and the reverse regression are related as $\hat{\beta} = \hat{\delta}\frac{Var[y]}{Var[x]}$. The two are different unless $Var[x] = Var[y]$, but they always have the same sign, and both are larger in magnitude the larger the covariance.

It also turns out that R-squared of the simple linear regression is the square of the correlation coefficient. So the R-squared is yet another measure of the association between the two variables. This is harder to see right away, but it comes from the fact that the numerator of R-squared, $Var[\hat{y}]$, can be written out as $Var[\hat{\alpha} + \hat{\beta}x] = \hat{\beta}^2 Var[x]$, and thus $R^2 = \hat{\beta}^2 Var[x]/Var[y] = (\hat{\beta} Std[x]/Std[y])^2$.

Review Box 7.8 Correlation and linear regression

Consider the OLS estimators for the coefficients of the linear regression $y^E = \alpha + \beta x$.

- The slope of a simple linear regression and the correlation coefficient are closely related as

$$\hat{\beta} = Corr[x, y]\frac{Std[y]}{Std[x]} \quad \text{or} \quad Corr[x, y] = \hat{\beta}\frac{Std[x]}{Std[y]}$$

- The R-squared is the square of the correlation coefficient:

$$R^2 = (Corr[x, y])^2$$

7.11 Regression Analysis, Regression toward the Mean, Mean Reversion

The relationship between the slope coefficient of the simple linear regression and the correlation coefficient helps explain the somewhat strange name for the method itself: regression. The concept of the linear regression, and its name, originated with Francis Galton in the second half of the nineteenth century. Galton analyzed the association between the height of parents (x) and the height of their children (y) and estimated the regression line using this data. He found a slope coefficient that was less than one. We now understand why: the standard deviation of y and x were the same in his data, and thus the regression coefficient was also the correlation coefficient (that name was invented somewhat later), which is naturally between -1 and 1.

From this Galton concluded that height exhibits a **regression toward the mean**: parents with height away from the mean tend to have children who also have height away from the mean but less so. This substantive conclusion about a particular phenomenon then stuck to the name of the method of linear regression, and, more generally, regression analysis. To separate the method from the substantive phenomenon, we use the term **mean reversion** instead of the term regression toward the mean to describe the substantive phenomenon: events further away from the mean tend to be followed by events closer to the mean. We reserve the name regression to the method of regression analysis.

7.12 Regression and Causation

Correlation does not imply causation – a warning we can find in every introduction to statistics. The slope of a simple regression describes the same association as correlation, so this message is equivalent to saying that the slope of a simple regression does not imply causation. By that we mean that, in general, the slope coefficient of a linear regression does not show the effect of the explanatory variable on the dependent variable.

Causality is a concept that is not that easy to define. One intuitive approach postulates that x causes y if we could expect y to change if we were to change x. We shall define the cause and effect relationship in a more systematic way in Chapter 19.

Regression does not imply causation because, in general, we cannot infer a cause and effect relationship from differences we see in data. Regression is a method of comparison: it compares observations that are different in x and shows corresponding average differences in y. It is a way to find patterns of association by comparisons. The fact that we can't, in general, infer causation from regression analysis is not the fault of the method. It is a consequence of the nature of the data. In particular, whether we can or cannot infer causation depends on the source of variation in the right-hand-side variable, a concept we introduced in Chapter 4, Section 4.2 and we'll discuss in more detail in Chapter 19.

Regression analysis cannot in general uncover the effect of x on y in observational data because there variation in x is not controlled. In observational data we compare observations that may be different in terms of x for many reasons. As a result, the data does not necessarily inform us of what would happen to y if we were to change x. For example, using observational data to uncover the effect of advertising on sales, we may regress sales on the amount of advertising and see a strong positive coefficient. But that may not show an effect at all. For example, both advertising and sales may be above average in the holiday season. That in itself would lead to a positive slope coefficient without any effect of advertising.

In contrast, the same regression analysis can uncover the effect of x on y in experimental data where variation in x is controlled. In well-designed experiments (to be discussed in Chapter 20), an experimenter induces controlled variation in x: they manipulate the value of x in a way that rules out the influence of other effects and observe differences in the expected value of y as a result. For example, a firm may consciously experiment by allocating varying resources to advertising, in a random fashion, and keep track of sales. A regression of sales on the amount of advertising can uncover the effect of advertising here.

This is why proper interpretation of the slope of a regression is important. Comparing observations that are different in x and seeing the extent to which y differs among them on average is what regression analysis does. The proper interpretation of the slope is necessary whether the data is observational or comes from a controlled experiment. A positive slope in a regression of sales on advertising means that sales tend to be higher when advertising time is higher. This interpretation is true both in observational and experimental data. It may or may not imply that we have learned how we could expect y to change if we were to change x.

Instead of saying that the correlation, or regression, does not imply causation, we should rather say that we should not infer cause and effect from comparisons, especially when the data is observational. That's not as catchy and short. But it has the advantage of being both true and useful.

In any case, when the slope of the $y^E = \alpha + \beta x$ regression is not zero in our data ($\beta \neq 0$) and the linear regression captures the y–x association reasonably well, one of three things – which are not mutually exclusive – may be true:

1. x causes y. If this is the single one thing behind the slope, it means that we can expect y to increase by β units if we were to increase x by one unit.
2. y causes x. If this is the single one thing behind the slope, it means that we can expect x to increase if we were to increase y.
3. A third variable causes both x and y (or many such variables do). If this is the single one thing behind the slope it means that we cannot expect y to increase if we were to increase x.

7.A4 CASE STUDY – Finding a Good Deal among Hotels with Simple Regression

Fit and causation

In our example of linear regression of hotel prices on their distance to the city center, the R-squared of the regression is 0.16 = 16%. This means that of the overall variation in hotel prices, 16% is explained by the linear regression with distance to the city center; the remaining 84% is left unexplained. 16% may not seem much, but it is actually a rather good fit for a cross-sectional regression with a single explanatory variable. In any case it is the fit of the best-fitting line.

In our case study, the slope of the linear regression in question tells us that hotels that are one mile further away from the city center are, on average, 14 US dollars cheaper. The model fits the data decently. Does that mean that a longer distance causes hotels to be cheaper by that amount? Maybe. Our calculations are certainly consistent with such an interpretation. But this is observational data so our result does not imply causality. It is consistent with other interpretations, too.

If we think about it, a causal relationship is not that simple here. One way to think about causality is that it would imply that we would see the price of a hotel decrease by 14 US dollars if we could somehow move that hotel one mile further away from the city center. That's clearly impossible. Perhaps a better way to think about causality is what would happen if we built a very similar hotel one mile further away. How does this effect compare to what we actually do when we calculate the slope coefficient in our regression in our data?

The regression slope coefficient estimate is based on comparing many different hotels that are in different locations. However, these hotels are likely to be different in other features as well. That's clearly not the same as building a very similar hotel one mile further away. The hotels in the data are not identical to each other, and those differences likely affect their price. As we shall see in subsequent chapters, what matters is whether location and differences in those other features are related to each other. In other words, whether hotels that are further away tend to be different from hotels closer to the city center. They may very well be. It may make sense to build higher quality hotels closer to the city center. And that would mean that hotels further away tend to have lower quality, which may explain all or part of the negative association our regression uncovered.

Regression does not imply cause and effect in this data. There may be a third variable, quality, that may explain the negative slope on distance. As we discussed above, this is not the fault of the method. It is a consequence of the nature of our data. It is observational data – a cross-section of different hotels. In Chapter 10 we'll learn how to incorporate variables that measure quality in the regression so we can compare hotels that are similar in quality – at least according to our measures. However, whatever statistical method we use, we are bound to compare the prices of different hotels in the end.

7.13 Main Takeaways

Regression is a method to compare average y across observations with different values of x; it models the conditional mean $E[y|x]$.

- Non-parametric regressions (bin scatter, lowess) can visualize any form of $E[y|x]$, but they don't produce a single interpretable number.
- Linear regression is an approximation of $E[y|x]$.

- In the linear regression $y^E = \alpha + \beta x$, β shows how much larger y is, on average, for observations with a one-unit larger x.
- When β is not zero, one of three things (or any combination of them) may be true:
  - x causes y
  - y causes x
  - a third variable causes both x and y.

PRACTICE QUESTIONS

1. How can you calculate $y^E = f(x)$ in the data if x is a binary variable?
2. What's lowess and what is it good for?
3. How can you get the intercept and slope parameter of a linear regression by OLS? Write down the formulae and explain how in practice you get these numbers using the statistical software of your choice.
4. How do you interpret the parameters of the linear regression $y^E = \alpha + \beta x$ if x is a binary variable?
5. Does it make sense to estimate a simple linear regression between y and x when the pattern of association between them is nonlinear? Why or why not?
6. How is an R-squared computed and what does it show?
7. What are the features of the residuals of a linear regression if $y^E = f(x)$ follows a nonlinear pattern? You may draw a graph, or simply use Figure 7.5.
8. What's the relationship between the covariance, the correlation coefficient, and the slope coefficient of a simple linear regression? Do they measure the same thing?
9. The R-squared of a simple linear regression $y^E = \alpha + \beta x$ is 0.5. Interpret this number. What is the correlation coefficient between x and y (approximately)? What are your answers if the R-squared is 0.8? If the R-squared is 0.01?
10. The estimated slope coefficient on x is positive in a simple linear regression. List the three kinds of causal relations that may be behind this coefficient estimate.
11. The phrase "Correlation doesn't mean causation" is often stated. Argue why this statement should not be about the method to measure association but the kind of data it is applied to.
12. The slope parameter of a simple linear regression $y^E = \alpha + \beta x$ is +2. Interpret this number.
13. The intercept parameter of a simple linear regression $y^E = \alpha + \beta x$ is –2. Interpret this number.
14. An analysis relates the age of used cars (in years) to their price (in USD), using data on a specific type of car. In a linear regression of the price on age, the slope parameter is –700. Interpret this number.
15. An analysis relates the size of apartments (in square meters) to their price (in USD), using data from one city. In a linear regression of price on size, the slope parameter is +600. Interpret this number.

DATA EXERCISES

Easier and/or shorter exercises are denoted by [*]; harder and/or longer exercises are denoted by [**].

1. Use the same `hotels-vienna` dataset that we used for the case study. Analyze the pattern of association between hotel price and average customer rating. First estimate a bin scatter with

two bins, then one with four bins, and then estimate a lowess regression. Visualize the results and summarize the most important findings from these non-parametric regressions. Then move on to estimate a simple linear regression and interpret its coefficients. [*]

2. Use the same `hotels-vienna` dataset that we used for the case study. Analyze the pattern of association between hotel price and stars. First estimate the non-parametric regression. Visualize its results and summarize the most important findings from this non-parametric regression. Then move on to estimate a simple linear regression as if stars were a quantitative variable and interpret its coefficients. [*]
3. Analyze the hotel price–distance to the center pattern for another city. Select a city from the large `hotels-europe` dataset. Keep hotels only, those with 3 to 4 stars, for a November 2017 weekday. Examine the distribution of the distance variable and drop observations if you think you should. First estimate a bin scatter with four bins and then a lowess regression. Visualize the results and summarize the most important findings from these non-parametric regressions. Then move on to estimate a simple linear regression and interpret its coefficients. Compare your results to what we found for Vienna. [**]
4. Collect data on used cars of a specific brand and type, and analyze price and age of the car. First estimate a bin scatter with two bins, then one with four bins, and then estimate a lowess regression. Visualize the results and summarize the most important findings from these non-parametric regressions. Then move on to estimate a simple linear regression and interpret its coefficients. Finally, use the results from the simple linear regression to list candidate cars for the best deal using the residuals of the linear regression. [**]
5. Analyze the pattern of association between how easy it is to do business in a country and how much income the country generates. Download data from the World Bank on GDP per capita in PPP (y) and the time required to start a business (x) for a recent year. Estimate a non-parametric regression, visualize its results, and summarize its main findings. Then estimate a simple linear regression and interpret its results. Do you think the results show the effect of the business-friendliness of a country on how much income it generates? [**]

REFERENCES AND FURTHER READING

There are many econometric textbooks that discuss regression analysis, typically with more technical detail than us. Of them, we especially recommend Cameron & Trivedi (2005), Hansen (2019), Stock & Watson (2015), and Wooldridge (2001, 2012).

On non-parametric regression, see Takezawa (2006).

On the early history of regression analysis, two interesting books are Stigler (1986) and Stigler (2016).

7.U1 UNDER THE HOOD: DERIVATION OF THE OLS FORMULAE FOR THE INTERCEPT AND SLOPE COEFFICIENTS

In this subsection, we'll derive the formulae of the intercept and slope coefficients using ordinary least squares (OLS).

The goal is to find values for α and β that minimize the sum of squared deviations from the regression line:

$$min_{\alpha,\beta} \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)^2 \tag{7.17}$$

The values for α and β that minimize this formula are $\hat{\alpha}$ and $\hat{\beta}$. For them to minimize the formula, they need to satisfy the first-order conditions: the first derivatives should be zero. (That's a necessary condition; we'll check second-order conditions later.)

Let's start with the first-order condition with respect to α:

$$\frac{\partial}{\partial \alpha} \sum_{i=1}^{n} (y_i - \hat{\alpha} - \hat{\beta} x_i)^2 = \sum_{i=1}^{n} \left[2(y_i - \hat{\alpha} - \hat{\beta} x_i)(-1) \right] = 0$$

Divide by two and then by -1 and expand to get

$$\sum_{i=1}^{n} (y_i - \hat{\alpha} - \hat{\beta} x_i) = 0$$

Collect terms, realizing that both $\hat{\alpha}$ and $\hat{\beta}$ are the same for all i so we can move them outside the sum:

$$\sum_{i=1}^{n} y_i - \hat{\alpha} \sum_{i=1}^{n} 1 - \hat{\beta} \sum_{i=1}^{n} x_i = 0$$

Now multiply both sides of the equation by $\frac{1}{n}$ (and realize that $\sum_{i=1}^{n} 1 = n$):

$$\frac{1}{n} \sum_{i=1}^{n} y_i - \hat{\alpha} \frac{1}{n} n - \hat{\beta} \frac{1}{n} \sum_{i=1}^{n} x_i = 0$$

Realize that these are averages for y and x:

$$\bar{y} - \hat{\alpha} - \hat{\beta} \bar{x} = 0$$

Finally, rearrange this to get the OLS formula for $\hat{\alpha}$:

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

Now consider the first-order condition for β:

$$\frac{\partial}{\partial \beta} \sum_{i=1}^{n} (y_i - \hat{\alpha} - \hat{\beta} x_i)^2 = \sum_{i=1}^{n} \left[2(y_i - \hat{\alpha} - \hat{\beta} x_i)(-x_i) \right] = 0$$

Divide by two and expand to get

$$\sum_{i=1}^{n} (-y_i x_i + \hat{\alpha} x_i + \hat{\beta} x_i^2) = 0$$

Now substitute in $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$:

$$\sum_{i=1}^{n}(-y_i x_i + (\bar{y} - \hat{\beta}\bar{x})x_i + \hat{\beta}x_i^2) = 0$$

Collect terms:

$$\sum_{i=1}^{n}(-y_i x_i) + \sum_{i=1}^{n}(\bar{y}x_i) + \sum_{i=1}^{n}(-\hat{\beta}\bar{x}x_i) + \sum_{i=1}^{n}(\hat{\beta}x_i^2) = 0$$

Realize that $\bar{y}$, $\bar{x}$, and $\hat{\beta}$ are the same for all i and can be moved outside of the respective sums:

$$\sum_{i=1}^{n}(-y_i x_i) + \bar{y}\sum_{i=1}^{n}(x_i) + \hat{\beta}\bar{x}\sum_{i=1}^{n}(-x_i) + \hat{\beta}\sum_{i=1}^{n}(x_i^2) = 0$$

Multiply by $\frac{1}{n}$:

$$\frac{1}{n}\sum_{i=1}^{n}(-y_i x_i) + \bar{y}\frac{1}{n}\sum_{i=1}^{n}(x_i) + \hat{\beta}\bar{x}\frac{1}{n}\sum_{i=1}^{n}(-x_i) + \hat{\beta}\frac{1}{n}\sum_{i=1}^{n}(x_i^2) = 0$$

Now realize that what is left in the second and third term is average x. So:

$$\frac{1}{n}\sum_{i=1}^{n}(-y_i x_i) + \bar{y}\bar{x} + -\hat{\beta}\bar{x}^2 + \hat{\beta}\frac{1}{n}\sum_{i=1}^{n}(x_i^2) = 0$$

Now let's put back those averages into the sums:

$$-\frac{1}{n}\sum_{i=1}^{n}(y_i x_i - \bar{y}\bar{x}) + \hat{\beta}\frac{1}{n}\sum_{i=1}^{n}(x_i^2 - \bar{x}^2) = 0$$

And now the hard part. We can verify that

$$\frac{1}{n}\sum_{i=1}^{n}(y_i x_i - \bar{y}\bar{x}) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})(x_i - \bar{x})$$

and

$$\frac{1}{n}\sum_{i=1}^{n}(x_i^2 - \bar{x}^2) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Applying these we get that

$$-\frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})(x_i - \bar{x}) + \hat{\beta}\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = 0$$

So that

$$\hat{\beta}\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2=\frac{1}{n}\sum_{i=1}^{n}(y_i-\bar{y})(x_i-\bar{x})$$

Which gives us the OLS formula for $\hat{\beta}$:

$$\hat{\beta}=\frac{\frac{1}{n}\sum_{i=1}^{n}(y_i-\bar{y_i})(x_i-\bar{x_i})}{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2}$$

The first-order conditions are necessary conditions but not sufficient in themselves. In principle, they may identify a maximum as well as a minimum (or an inflection point). To verify that these indeed give a minimum, let's look at the second-order conditions. For a minimum, a sufficient condition is to have the matrix of the second-order conditions to be positive definite (its determinant should be positive). There are four second-order conditions: differentiating twice by α, differentiating twice by β, and differentiating once by α and once by β – this third one takes care of the remaining two conditions, because by symmetry it is the same as differentiating in the other order.

The second derivatives are

$$\frac{\partial^2}{\partial\alpha^2}\sum_{i=1}^{n}(y_i-\hat{\alpha}-\hat{\beta}x_i)^2=\frac{\partial}{\partial\alpha}\sum_{i=1}^{n}\Big[2(y_i-\hat{\alpha}-\hat{\beta}x_i)(-1)\Big]=2$$

$$\frac{\partial^2}{\partial\beta^2}\sum_{i=1}^{n}(y_i-\hat{\alpha}-\hat{\beta}x_i)^2=\frac{\partial}{\partial\beta}\sum_{i=1}^{n}\Big[2(y_i-\hat{\alpha}-\hat{\beta}x_i)(-x_i)\Big]=\sum_{i=1}^{n}2x_i^2$$

$$\frac{\partial^2}{\partial\alpha\partial\beta}\sum_{i=1}^{n}(y_i-\hat{\alpha}-\hat{\beta}x_i)^2=\frac{\partial}{\partial\beta}\sum_{i=1}^{n}\Big[2(y_i-\hat{\alpha}-\hat{\beta}x_i)(-1)\Big]=2\sum_{i=1}^{n}x_i=2\sum_{i=1}^{n}\bar{x}$$

The determinant of the matrix of the second-order conditions is positive:

$$det=4\sum_{i=1}^{n}x_i^2-4\sum_{i=1}^{n}\bar{x}\sum_{i=1}^{n}\bar{x}=4\sum_{i=1}^{n}(x_i^2-\bar{x})=4nVar[x]>0$$

So $\hat{\alpha}$ and $\hat{\beta}$ do indeed give minimum values.

7.U2 UNDER THE HOOD: MORE ON RESIDUALS AND PREDICTED VALUES WITH OLS

In this subsection we show that with OLS, the sum and the average of the residuals are zero, and average predicted *y* equals average *y*.

Recall that the regression residual for observation i in the data is

$$y_i - \hat{\alpha} - \hat{\beta} x_i$$

The sum of residuals is

$$\sum_{i=1}^{n} (y_i - \hat{\alpha} - \hat{\beta} x_i)$$

Plug in the formula for the intercept $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$ to get

$$\sum_{i=1}^{n} \left(y_i - (\bar{y} - \hat{\beta}\bar{x}) - \hat{\beta} x_i \right) = \sum_{i=1}^{n} \left(y_i - \bar{y} + \hat{\beta}\bar{x} - \hat{\beta} x_i \right)$$

Rewrite this by putting the sum in front of each term:

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \bar{y} + \sum_{i=1}^{n} \hat{\beta}\bar{x} - \sum_{i=1}^{n} \hat{\beta} x_i$$

Realize that $\sum_{i=1}^{n} y_i = n\bar{y}$, also $\sum_{i=1}^{n} \bar{y} = n\bar{y}$, take $\hat{\beta}$ outside the sum, and use $\sum_{i=1}^{n} x_i = n\bar{x}$ and $\sum_{i=1}^{n} \bar{x} = n\bar{x}$ to get

$$n\bar{y} - n\bar{y} + \hat{\beta} n\bar{x} - \hat{\beta} n\bar{x} = 0$$

So the sum of residuals is zero.

If the sum is zero then the average is zero, too: it's the sum divided by n.

Now let's show that, for a linear regression estimated by OLS, the average of the predicted dependent variable in the data is equal to the average of the dependent variable itself: $\bar{\hat{y}} = \bar{y}$.

Let's start with the OLS predicted value for observation i:

$$\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$$

Its average in the data is

$$\frac{1}{n} \sum_{i=1}^{n} \hat{y}_i = \frac{1}{n} \sum_{i=1}^{n} \hat{\alpha} + \frac{1}{n} \sum_{i=1}^{n} (\hat{\beta} x_i)$$

Plug in the formula for $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$:

$$\frac{1}{n} \sum_{i=1}^{n} (\bar{y} - \hat{\beta}\bar{x}) + \frac{1}{n} \sum_{i=1}^{n} (\hat{\beta} x_i)$$

Expand this and move $\hat{\beta}$ outside the sum:

$$\frac{1}{n}\sum_{i=1}^{n}\bar{y} - \hat{\beta}\frac{1}{n}\sum_{i=1}^{n}\bar{x} + \hat{\beta}\frac{1}{n}\sum_{i=1}^{n}x_i$$

Realize that $\frac{1}{n}\sum_{i=1}^{n}\bar{y} = \bar{y}$ and $\frac{1}{n}\sum_{i=1}^{n}\bar{x} = \bar{x}$ so that

$$\bar{y} - \hat{\beta}\bar{x} + \hat{\beta}\bar{x} = \bar{y}$$

So $\bar{\hat{y}} = \bar{y}$.

8 Complicated Patterns and Messy Data

How to uncover nonlinear patterns with linear regression, and how to deal with influential observations and measurement error in regression analysis

Motivation

Life expectancy at birth shows how long residents of a country live; it is a summary measure of their health. Residents of richer countries tend to live longer, but you want to know the strength of that pattern. You also want to identify countries where people live especially long for the income level of their country, to start thinking about what may cause their exceptional health. You download cross-country data on life expectancy and GDP per capita, and you want to uncover the pattern of association between them. How would you do that in a way that accommodates potentially nonlinear patterns and, at the same time, produces results that you can interpret?

Continuing with your investigation to find underpriced hotels, you want to examine the association between hotel price and hotel quality. Your data includes averages of customer ratings, which may be a good measure of hotel quality. However, the number of customers whose ratings are used to calculate this measure varies across hotels, from a few dozen ratings to many hundreds. Arguably, average ratings that are based on the ratings of a few customers are a less reliable measure of quality. Should you worry about this for your analysis? In particular, would a regression of price on average ratings be different, and less informative, if the average of ratings is based on too few customers? If yes, what can you do about it?

Linear regression gives a meaningful approximation to the patterns of association, but real-life data can be messy, and the patterns may be nonlinear. What those mean for regression analysis and what we can do about them is important to understand. There are several tools that we can apply to make linear regression approximate nonlinear patterns of association, but whether we want to do so depends on the goal of the analysis. The fact that real-life data tends to be messy, with errors and extreme values, poses other challenges for regression analysis.

The first part of this chapter covers how linear regression analysis can accommodate nonlinear patterns. We discuss transforming either or both the dependent variable and the explanatory variable, such as taking the log; piecewise linear spline of the explanatory variable; and quadratic and higher-order polynomials of the explanatory variable. We discuss whether and when to apply each technique, and emphasize the correct interpretation of the coefficients of these regressions.The second half of the chapter discusses potential issues with regression analysis using influential observations and variables that are measured with error. We outline potential consequences of

these issues and, when needed, ways to deal with them. The chapter closes by discussing whether and how to use weights in regression analysis.

There are three case studies in this chapter. The first one, **Finding a good deal among hotels with nonlinear function**, is a continuation of the study in the previous chapters: hotel prices in a European city and the distance of the hotel to the city center, using the `hotels-vienna` dataset. It illustrates the use of logarithms in linear regression. The case study **Hotel ratings and measurement error** uses the same data to illustrate the effect of measurement error in ratings on the regression of hotel price on ratings. The case study **How is life expectancy related to the average income of a country?** uses the `worldbank-lifeexpectancy` dataset on all countries to characterize the shape and strength of the association between life expectancy and GDP. This case study illustrates the transformations of variables (logs, per capita measures) and nonlinear functional forms.

Learning outcomes

After working through this chapter, you should be able to:

- identify situations that call for uncovering nonlinear patterns in regression analysis;
- estimate linear regressions using transformed variables, piecewise linear spline of the explanatory variable, or polynomials of the explanatory variable;
- interpret the parameters of such regressions and visualize the uncovered patterns;
- identify influential observations in linear regression and decide on whether and how to deal with them;
- assess the consequences of measurement error in variables of linear regression;
- identify the need for weights in linear regression analysis and estimate models using weights if necessary.

8.1 When and Why Care about the Shape of the Association between y and x?

The previous chapter introduced linear regression as the major tool to uncover the average association between y and x. The best way to think about linear regression is a linear approximation of a conditional mean of unknown shape: $y^E = f(x)$. Sometimes the pattern is close to linear. But often it isn't. In such cases we may or may not want to get that nonlinear pattern right. Typically, we want to get the regression to better characterize the nonlinear pattern:

- when we want to make a prediction or analyze residuals;
- when we want to go beyond the average pattern of association;
- when all we care about is the average pattern of association, but the linear regression gives a bad approximation to that.

In contrast, we may ignore nonlinear patterns if all we care about is the average pattern of association, and the linear regression gives a reasonable approximation to that average pattern.

This is an important point. How closely we want our regression to approximate a nonlinear pattern is a matter of choice. This choice should be guided by the benefits and the costs of the alternatives. In general, non-monotonic or highly nonlinear patterns may be more important to capture, but such patterns often are more difficult to include in regression. And, even if we choose to accommodate

a nonlinear pattern in a regression, we should aim at a simple solution. Simpler models are easier to interpret and communicate, and they tend to be more robust across datasets that represent the same population or general pattern.

If we want to capture a potentially complicated pattern, non-parametric regression seems to be the best option. Unfortunately, non-parametric regressions do not provide statistics or other numbers that summarize the most important aspects of a pattern. They are also difficult to generalize for multiple regressions – something that we'll show may be important.

Instead, when they want to capture nonlinear patterns of association in regressions, data analysts use relatively simple techniques to approximate those patterns. The next sections discuss several approaches for doing so.

Review Box 8.1 When and why care about the shape of the association between *y* and *x*?

- Whether the shape of a regression is linear or nonlinear matters if:
  - our goal is prediction;
  - our goal is uncovering the patterns of association and we want to uncover more than the average association;
  - all we want to uncover is the average association, but that average is badly estimated due to nonlinearities (most often due to strong skewness or extreme values).
- Potential nonlinear shapes don't matter if all we want to uncover is the average association, and we can get a good estimate of that average.

8.2 Taking Relative Differences or Log

Some frequent nonlinear patterns are better approximated by linear regressions if y or x, or both, are transformed by taking **relative differences**, or **percentage differences**. In a time series of GDP it makes sense to transform the variable to have year-to-year growth rates. In a time series of the prices of a company stock, it makes sense to transform the price variable into percentage change from the previous time period. This transformation is possible when there is a previous observation that could serve as the natural base for comparison. The case study in Chapter 12, Section 12. A1 will show an example. But doing so is impossible in cross-sectional data where there is no previous observation that could serve as a natural base for comparison. Taking the **natural logarithm** of a variable is often a good solution in such cases. When transformed by taking the natural logarithm, differences in variable values approximate relative differences.

The natural logarithm of a variable x is denoted as $\ln(x)$ or $\ln x$. Sometimes the notation $\log(x)$ is also used, but we prefer ln to log as log may refer to logarithm of another base, such as 10, while ln is reserved for the natural log. While we keep the notation ln in formulae, we shall often refer to the natural logarithm simply as **log** in the text. When we want to differentiate variables that are not in logs, we often say that they are in **level**, or levels.

Examining log differences works because differences in natural logs approximate percentage differences. For example, if x_2 is 10% larger than x_1, that is, $x_2 = 1.1 \times x_1$, then $\ln x_2 - \ln x_1 \approx 0.1$. Or, if x_2 is 10% smaller than x_1, that is, $x_2 = 0.9 \times x_1$, then $\ln x_2 - \ln x_1 \approx -0.1$.

Log differences are approximations of percentage differences that work well for small differences of 30% or less. For larger positive differences, the log difference is smaller (for example, a log difference of +1.0 corresponds to a +170% difference). For larger negative differences, the log difference is larger in absolute value (for example, a log difference of −1.0 corresponds to a −63% difference). The Under the Hood section 8.U1 provides more detail on why **log approximation** works and how bad it becomes with large relative differences.

To use precise language, we also call differences in logs **log point** differences. For example, if $\ln x_2 - \ln x_1 = 0.1$, we say that x_2 is 10 log points greater than x_1. If $\ln x_2 - \ln x_1 = -0.1$, we say that x_2 is 10 log points less than x_1. Most of the time the rationale for talking about log points is to avoid the difficulties of the approximation, especially with larger log differences. But, however precise this language, looking at log differences is most useful when they approximate relative differences well. As a result we'll talk about relative differences, or percentage differences, whenever we can, instead of using the more precise, but less useful, language of log point differences.

Measuring differences in relative terms is often a good choice with many variables analyzed in business, economics, and policy, including prices, incomes, sales, or measures of size. First, relative comparisons are free from measures of the variables that are often different across time and space and are arbitrary to begin with (counting in dollars or thousand dollars). Second, differences in relative terms are likely to be more stable across time and space than absolute differences, partly because causes behind these variables, such as trends in time or regional differences, tend to be more stable in relative terms. Third, many economic decisions are based on differences in relative terms, such as GDP growth or profit margins.

There is a related, statistical reason to take relative differences or logs. The distribution of variables in relative terms, including after taking the log, is often less skewed than distributions of levels (where level refers to the values as they are, as opposed to their logs). The distribution of many important economic variables is skewed with a long right tail and are reasonably well approximated by a **log-normal distribution** (see Chapter 3, Section 3.9). Examples include hourly wages or hotel prices. Thus, the natural log of such variables are close to normally distributed with all of the associated nice statistical properties such as symmetry and low frequency of extreme values.

Having variables with such nice properties is a good thing in linear regression analysis. Recall from Chapter 7 that the slope in a linear regression is an approximation of the average slope of an underlying conditional expectation that may be nonlinear. It turns out that the closer the distribution of variables is to normal, the better the approximation. Thus, even if we are interested in average associations only, taking relative differences, or logs, is often a good choice when the variables are skewed.

Taking relative differences or logs keeps some aspects of the variables the same but changes other aspects. Moreover, these transformations are not always feasible. Let us review the key features of the log transformation.

First, relative differences and log differences between two observations are not affected by the original units of measurement. Thus, relative, or log, differences in prices, wages, incomes, and other variables with a money dimension are the same whatever monetary unit is applied. Note that all of this is true for units of measurements that are multiples of each other, such as currencies or measures of distance. A 10% difference or a 10 log points difference is the same whether the underlying measurement is in dollars, million dollars, or euros. The same is not true for additive transformations $(x+c)$ – units that differ by an additive factor, potentially on top of a multiplying one. Thus, for example, a 10% difference in Fahrenheit is different from a 10% difference in Celsius. Indeed, it makes most sense to take relative differences or logs of ratio variables only (see Chapter 2, Section 2.1).

Second, when we take the log of a variable, we do a monotonic transformation: the ordering of observations in terms of a transformed variable is the same as their ordering in terms of the original variable. Thus, the observation with the smallest value also has the smallest log value, and observations with larger values also have larger log values.

Third, taking relative, or log differences is tricky with averages. In general, the average of many relative, or log differences, is not equal to the relative, or log difference, of average variables. A related fact is that the average of the log value is not the same as the log of the average value: $\overline{\ln(x)} \neq \ln(\bar{x})$. Technically this is because the log is a nonlinear function. That nonlinearity does not affect ordering and thus quantiles, but it matters for averages. This affects all statistics that are based on some kind of average. Importantly, it also affects prediction from regression analysis with log y variables. Regressions with $\ln y$ predict the expected value of $\ln y$. Most often, we are interested in predicting y itself. But getting $\hat{y}$ from $\widehat{\ln y}$ is not straightforward, and we shall discuss this in detail in Chapter 14, Section 14.3.

Summing up: taking relative differences, or logs, of variables with money or size dimensions is usually a good idea, and it may make sense to do so with other variables too. There are two related reasons to do so. The substantive reason is to consider differences in relative terms that are free from arbitrary units of measurement and represent how decision makers think about them. The statistical reason is to transform variables that are skewed with a long right tail to variables that are more symmetric and closer to being normally distributed. However, logs and relative differences are problematic when values are zero or negative. We explore that issue in the next section.

8.3 Log Transformation and Non-positive Values

An important constraint on taking relative differences or logs is that they are possible for positive values only. And that can be a problem, because we may encounter variables that we want to analyze in relative terms yet they have non-positive values. Examples of variables that may have zero values as well as positive values include a company's sales, earnings, expenditures. The relative change of sales from last year is impossible to calculate if sales were zero last year. It is also impossible to take the log of that zero number. Similarly problematic are variables that can take negative values, too, such as corporate profits or a family's net assets. The relative change in profits from last year is impossible to take if last year's profit was negative, and taking the log of that negative number is also impossible.

In such cases, relative difference or log cannot be taken of all of the observations. Yet, sometimes, we would like to analyze relative differences with such variables, too. Is there anything we can do in such cases? Let us briefly review some specific cases and potential solutions. We focus on taking logs here, but the discussion applies to taking relative differences in time series data, too, with appropriate modifications.

First, the reason for zero, or negative, values may matter. In particular, the variable may have only positive values in principle, but the data has zeros because of rounding. If the reason is rounding, the zeros are not really zeros but small positive values. In this case the problem is that we don't know what those positive values are.

When zeros are the results of rounding, it's good practice to replace those apparently zero values with what may be the average non-rounded value for them, such as 0.25 (if values below 0.5 are rounded to zero and the distribution of the variable is likely close to uniform in that region). This is in fact the most straightforward case of variables with positive and zero values. In what follows, we consider all other cases, when the non-positive values are real zeros or negative values.

Second, the magnitude of the problem matters. If only 1 percent of the observations have zero value, we may simply drop them from the data. If, in contrast, 60 percent, or 90 percent, have zero value, we have a very different problem: whether a value is zero or positive is important information, so much so that we may convert this variable into a binary one, showing whether it's positive or zero. It's the cases in-between that we'll focus on below: when a non-negligible minority of the variables have non-positive value.

Third, the available options are different for the y and x variables in the regression, and whether the non-positive values are all zero, or there are negative values, too.

Let's start with an x variable that is positive for most observations and zero for a non-negligible minority of the observations. In this case it is good practice to create a binary variable denoting whether $x = 0$. Then we make $\ln x = 0$ when $x = 0$, and enter both the binary variable and this modified $\ln x$ in the regression (see later in this chapter and in Chapter 10 of using more variables on the right-hand side of the regression). The interpretation of the first coefficient is expected y when $x = 0$ versus $x > 0$, and the interpretation of the second coefficient is the usual one with $\ln x$, given that $x > 0$ (see the next section for the interpretation of the coefficients of regressions with logs).

We can follow the same logic with an x variable that is positive for some observations but zero or negative for a non-negligible minority of the observations. Here we can define the binary variable denoting whether $x \leq 0$. Alternatively, we can create two binary variables, one for whether $x = 0$, and one for whether $x < 0$. Either way, we replace $\ln x = 0$ for all $x \leq 0$ observations, enter the binary variable, or variables, together with $\ln x$, and interpret their coefficients accordingly.

Unfortunately, the same options are not available for the y variable in the regression. But all is not lost here, either.

Data analysts often use involved nonlinear models that can deal with this case, but those are beyond the scope of our textbook. Another possibility is transforming y in a way that allows for examining relative differences and is feasible with non-positive values. One such transformation is $y/(y - \bar{y})$. This is a relative difference, with some of the nice features of logs, including being a monotonic transformation and yielding the same result regardless of the unit of measurement. Its interpretation is different from a relative or log difference, because the measurement scale of this transformed variable is different.

In summary, when some of the values are non-positive, various alternative options exist to make relative comparisons. Which option to use depends on the source of non-positive values, the prevalence of such values, and whether the variable is on the left-hand side of a regression or on the right-hand side. Moreover, not all options are equally good. Thus, data analysts can't simply use a one-fits-all tool but have to think and make a conscious decision.

Review Box 8.2 Taking relative differences and logs

- For many variables, differences in relative terms (e.g., percent difference) make more sense for the analysis.
- We can take relative change in time series data where the previous period's value is a natural basis for comparison.
- Taking the natural log of a variable approximates relative differences; this option is available in cross-sectional data, too.
- The difference in natural logs approximates the relative difference. If x_2 is D% larger than x_1, and $d = D/100$ so that $x_2 = (1 + d) \times x_1$, then $\ln x_2 - \ln x_1 \approx d$.

- This approximation works well for small relative differences of 30% or less (either positive or negative).
- Relative differences and logs can be taken of positive nonzero variables only. For non-positive values, various alternative options exist to make relative comparisons.

8.4 Interpreting Log Values in a Regression

When y, or x, or both are relative changes in time series data, the interpretation of the regression coefficients is straightforward, and we'll discuss it in Chapter 12. When, instead, the variables are log transformations, things are less straightforward. There are three possible simple regressions with log variables: both y and x are in log, only y is in log, only x is in log. In one sense, the interpretation of the coefficients of these regressions is the same as usual, using the language of log points (see the previous section). However, the fact that one or both of the variables is in logs makes another, more meaningful interpretation possible. The three kinds of regressions are as follows.

log–log regression: $(\ln y)^E = \alpha + \beta \ln x$. α here is average $\ln y$ when $\ln x$ is zero. This is rarely meaningful as average $\ln y$ is difficult to interpret. In contrast, the slope of a log–log regression has a nice interpretation. y is β percent higher on average for observations with one percent higher x. Note that the usual interpretation is still true, here with log points as units of measurement: y is β log points higher, on average, for observations one log point higher x. β in this regression is sometimes interpreted as an **elasticity**: the expected percent difference in y related to a one percent difference in x.

log–level regression: $(\ln y)^E = \alpha + \beta x$. α here is average $\ln y$ when x is zero – rarely meaningful again. The slope of a log–level regression has a nice interpretation although it is a bit complicated. y is $\beta \times 100\%$ higher on average for observations having one unit higher x. ($\ln y$ is β units higher, on average, for observations with one unit higher x, which translates into β times one hundred percent higher y.)

level–log regression: $y^E = \alpha + \beta \ln x$. α here is average y when $\ln x$ is zero, which happens when x is one. The slope, again, is a little complicated. y is $\beta/100$ units higher on average for observations having one percent higher x. (y is β units higher, on average, for observations with one unit higher $\ln x$; for observations with one hundredth of a unit higher $\ln x$, which is also one percent higher value for x, y is $\beta/100$ units higher.)

Review Box 8.3 Interpreting the slope of simple regressions with log variables

- log–log regression: $(\ln y)^E = \alpha + \beta \ln x$
  β: y is β percent higher on average for observations having one percent higher x.
- log–level regression: $(\ln y)^E = \alpha + \beta x$
  β: y is $\beta \times 100$ percent higher on average for observations having one unit higher x.
- level–log regression: $y^E = \alpha + \beta \ln x$
  β: y is $\beta/100$ units higher on average for observations having one percent higher x.

8.A1 CASE STUDY – Finding a Good Deal among Hotels with Nonlinear Function

Regression with logs

Recall our study that analyzes hotel prices to find a good deal for a couple that wants to spend a night in a European city. The analysis aims to identify hotels that are underpriced relative to their distance to the city center. Capturing the shape of the pattern is important here. If the pattern is well captured by the regression, the lowest prices are well captured by the smallest (most negative) regression residuals. If, however, the pattern is poorly captured, the calculated residuals will be off, too.

The case study uses data downloaded from a price comparison website. We narrowed the data to hotels that are within eight miles of the city center. In the previous chapter, we estimated a linear regression of $price^E = \alpha + \beta \times distance$, and listed the hotels with the lowest residuals to identify hotels with the lowest price for their distance (Table 7.1).

Our analysis in the previous chapter uncovered problems with the linear regression $price^E = \alpha + \beta \times distance$. Looking back at Figure 7.5, we see that the regression line does not fully reproduce the pattern suggested by the non-parametric regression (Figure 7.3): the same difference in distance is associated with larger price differences closer to the city center. That implies that the true shape of the $y^E = f(x)$ relationship is nonlinear: it is steeper close to the city center, flatter farther away, and may have positive slope for the farthest hotels in the data.

But does that nonlinearity matter? Our question is based on prediction: we want to identify hotels with the lowest price compared to what's predicted for them given their distance. So yes, this nonlinear pattern does matter here. Thus, we need to find a better regression that can accommodate it.

The distribution of hotel prices is skewed with a long right tail (see, e.g., the histogram in Figure 3.4 in Chapter 3, Section 3.A2). This gives a statistical reason to transform it to measure differences in relative terms instead of dollars. This is cross-sectional data so we can't take percent differences from a natural basis. Instead, we can make the regression approximate relative price differences by taking the natural log of the hotel price variable. Let's call this transformed variable ln *price*. Price is always positive so we can take the log for every observation. The range of ln *price* is [3.9, 5.9], mean is 4.6, the median is also 4.6. The distribution of ln *price* is rather symmetric, unlike the distribution of price. Recall that the average price is 110 US dollars. The natural log of 110 is 4.7, slightly more than the average of ln *price*.

We compare two simple linear regressions, one with price as the dependent variable, measured in dollars, and another with the natural logarithm of price as the dependent variable. Figures 8.1a and 8.2b show the scatterplots with the regression lines.

The R-squared of the log–level regression is higher, suggesting a better fit. However, we are comparing the R-squared of two regressions here that have different dependent variables. That comparison is unfair as it compares fit in different units. Thus, we usually avoid comparing R-squares of regressions with different dependent variables.

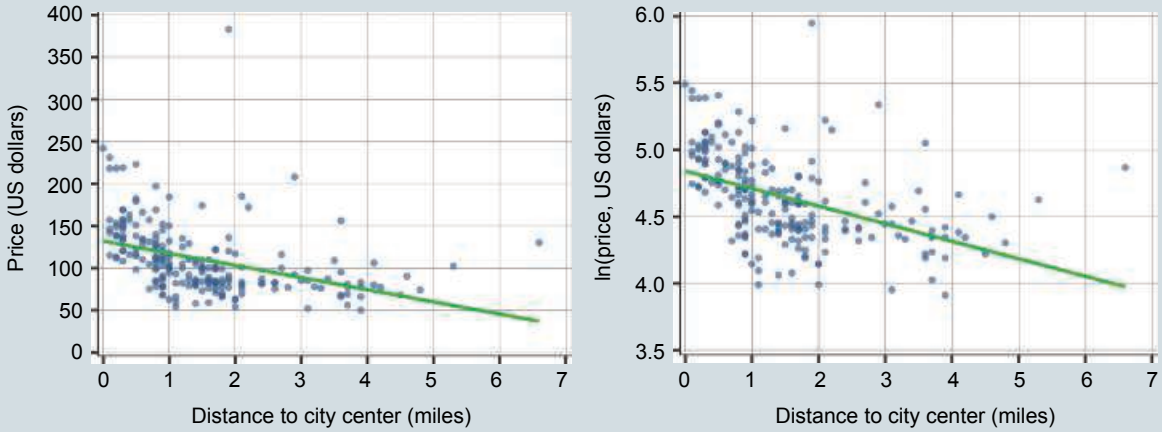


Figure 8.1 Regressions of hotel price and distance to the city center

Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. N=207.

But, setting measures of fit aside, we have a good reason to prefer the log–level regression: it gives a better approximation to the average slope of the pattern. That is because the distribution of log price is closer to normal than the distribution of price itself. We can see the consequence in Figures 8.1a and 8.1b: the scatterplot is more symmetrically distributed around the regression line in Figure 8.1b.

We may also have a substantive reason to examine relative differences, but that depends on the goal of the analysis. Here we are after a good deal on a single night. So, perhaps, absolute price differences are meaningful, as they tell us how much money we can save. However, if we were to generalize the findings to, say, another time, relative differences would be better. Percentage differences in price may remain valid if inflation and seasonal fluctuations affect prices proportionately.

Note, however, that the log–level regression still doesn't appear to capture the nonlinear pattern. Even just eyeballing Figure 8.1b suggests that the negative log price–distance association is steeper at lower levels of distance. As a consequence, we still have the hotels with the most negative residuals at around 1 mile, just as earlier with the level–level regression (Table 7.1). But, as in that case, this may very well be an artifact of our regression line not fitting the nonlinear pattern. As we shall see in the following sections, we can do a much better job of fitting to the nonlinear pattern by an extension of this log–level regression.

For comparison purposes, we can add two additional simple linear regressions, with the natural logarithm of distance as the explanatory variable. Of the 207 hotels in the data there is one with zero distance. Note that distance is rounded to 0.1 mile, and it is measured from the center of the main square. This hotel is on the main square, but not at its center, so this zero is in fact a rounded number that is less than 0.1. We'll replace its value by 0.05. Figures 8.2a and 8.2b show the scatterplots with the regression lines.

Recall that the results of the level–level regression are: intercept 133, slope −14, R-squared 0.157. The slope of the level–level regression implies that hotels that are 1 mile farther away from

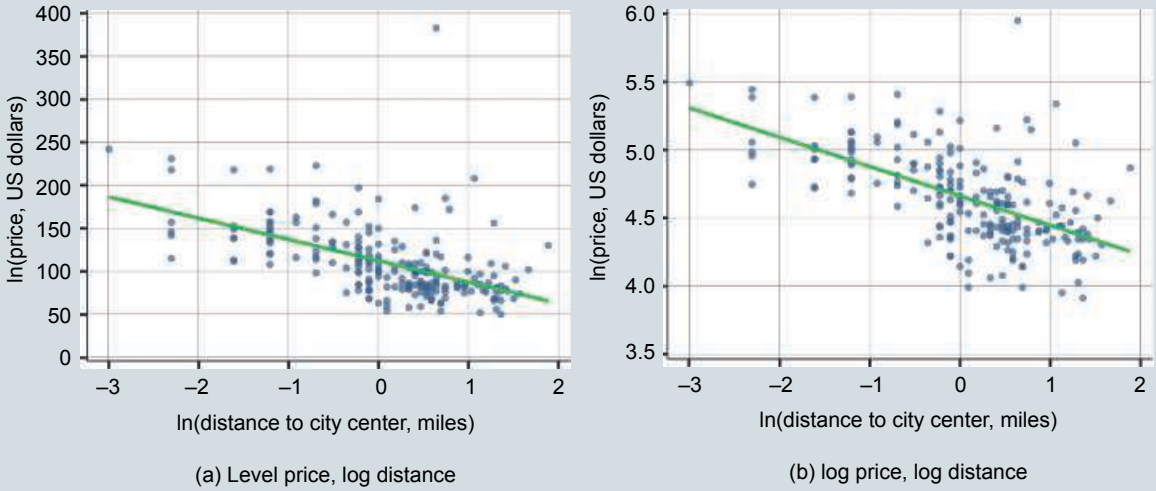


Figure 8.2 Regressions of hotel price and distance to the city center

Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. N=207.

the city center are 14 dollars less expensive, on average. The results of the log–level regression are: intercept 4.84, slope −0.13, R-squared 0.205. The slope of the log–level regression implies that hotels that are 1 mile farther away from the city center are 13 percent less expensive, on average.

All results are summarized in Table 8.1. Let us now interpret the additional two regressions with log distance as explanatory variable. The results of the level–log regression are: intercept 112.42 ≈ 112, slope −24.77 ≈ −25, R-squared 0.280. The intercept implies that hotels that are 1 mile from the center (for which ln *dist* = 0) cost 112 dollars, on average. The slope coefficient implies that hotels that are 10 percent farther away from the city center are 2.5 dollars less expensive, on average. Finally, the results of the log–log regression are: intercept 4.66, slope −0.225, R-squared 0.334. We don't interpret the intercept because the left-hand-side variable is in log. The slope coefficient implies that hotels that are 1 percent farther away from the city center are 0.22 percent less expensive, on average.

Table 8.1 Hotel price and distance regressions

VARIABLES	(1) price	(2) ln(price)	(3) price	(4) ln(price)
Distance to city center, miles	−14.41	−0.13		
ln(distance to city center)			−24.77	-0.22
Constant	132.02	4.84	112.42	4.66
Observations	207	207	207	207
R-squared	0.157	0.205	0.280	0.334

Source: `hotels-vienna` dataset. Prices in US dollars, distance in miles.

Which regression should we prefer from these four? The fit, in terms of the R-squared, is best for the log–log regression. Nevertheless, our choice is the log–level regression. The reasons for having

log price are its better interpretation and skewed distribution. The reason for having distance in level and not log is better interpretation. Our question (how a visitor should pick a hotel) implies thinking about time or money spent on traveling, which are likely linear in terms of distance.

8.5 Other Transformations of Variables

Sometimes we need transformations of y or x for the linear regression to make more sense and/or give a better approximation to the underlying nonlinear conditional expectation. In this section we cover one of many such transformations: ratios of variables – most notably, per capita measures of aggregate variables.

It is typically useful to use **per capita measures** rather than total measures of aggregate variables for cross-country comparisons. Take, for example, the pattern of association between CO_2 emission and GDP across countries. If we have both variables as totals, we have an issue: size. Small countries have low levels of total CO_2 emission and low levels of total GDP at the same time. Large countries have large levels of both. Thus, the pattern of association between totals mixes in size differences. In fact, even if there were no substantive association between CO_2 emission and GDP, we would have a strong positive association of the totals. But all this would show is that, for example, China is large and Singapore is small. Per capita measures both for GDP and CO_2 emission solve this issue. Similar arguments work for many sum-like variables in aggregate data, such as imports, exports, and number of internet users.

Per capita measures are the most frequent normalization of totals; they divide by population. Sometimes other variables are used to normalize totals, such as area, GDP, or labor force. All of these produce ratios, either in their own units (e.g., percent of area covered by forests, or debt over GDP) or dividing different units (e.g., GDP per capita).

We can take the log of a ratio just like any other variable to express differences in relative terms. Technically, we could consider it as a difference of two variables, because the log of a ratio equals the difference of the two logs: $\ln(GDP/Pop) = \ln(GDP) - \ln(Pop)$. Usually, however, it makes more intuitive sense to treat it as a single variable that approximates percentage differences in something that happens to be a ratio. For example, a difference of 0.1 in $\ln(GDP/Pop)$ between two countries is a 10 percent difference in average income produced in the two countries for each of their residents.

8.B1 CASE STUDY – How is Life Expectancy Related to the Average Income of a Country?

Per capita measure and logs

Our second case study examines the pattern of association between how long people live in a country and how rich people in that country are, on average. Uncovering that pattern may be interesting for many reasons. One may be to identify countries where people live longer than what we would expect based on their average income, or countries where people live shorter lives.

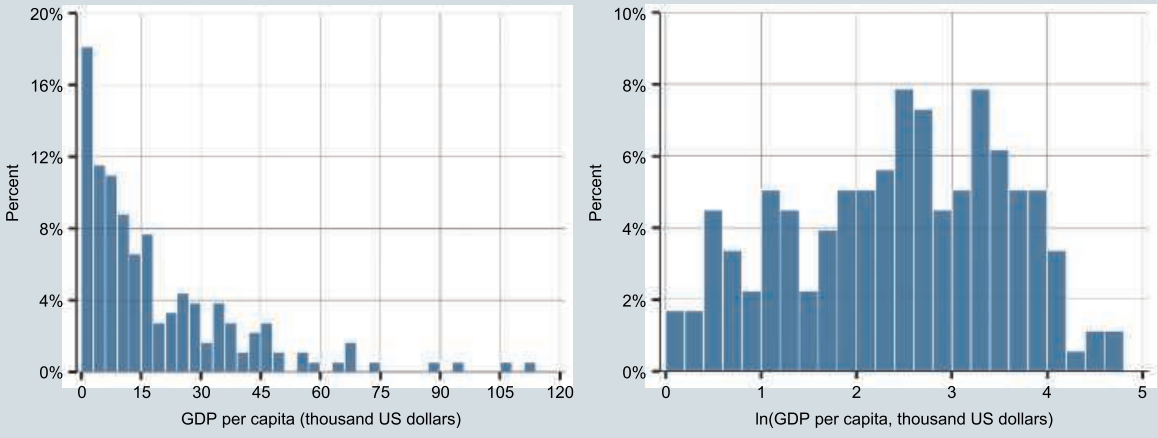
That would require analyzing regression residuals, similarly to our hotel price case study. For this purpose, getting a good approximation of the $y^E = f(x)$ function is important.

Life expectancy at birth in a given year is a measure of how long people live in a country on average. It is the average age at which people die in the given year. If newborn children were to live to the same age as the current population, this would be the expected length of their lives – hence the name of the variable. GDP per capita is a commonly used measure of average income in a country. It is a flow variable, measuring the total income generated within a country over a year. It is also a per capita measure, dividing by country population, the transformation we discussed in the previous section. At the end of this section we will illustrate why dividing by population is important.

We use data from the publicly available World Development Indicators website, maintained by the World Bank. This is a panel data, and we only look at data referring to 2017. There are 217 countries in this data table, but GDP and life expectancy is available for only 182 of them. Most of the 35 countries with missing values are very small, but a few are medium-sized countries with missing GDP data that are affected by conflict or a collapse of the state (Syria or Venezuela) or do not collect reliable data to assess their GDP (Cuba, North Korea).

Average life expectancy across the 182 countries is 72 years, with a range of 52 to 85. Thus, there is substantial variation in the average length of life across countries. Countries with the lowest life expectancy are from sub-Saharan Africa, including Chad and Nigeria. Countries with the highest life expectancy are a mixture of East Asian countries (Hong Kong, Japan) and Western European ones (Switzerland, Spain).

The distribution of GDP per capita is skewed with a long right tail; Figure 8.3a shows its histogram. There are four countries with extremely high GDP per capita values: Luxembourg, Singapore, Macao, and Qatar. Average GDP per capita is 19 000 US dollars in this data; the median is 12 000 US dollars, which also shows the skewness of the distribution. The standard deviation is 20 000 US dollars, suggesting substantial spread.



(a) Histogram of GDP per capita

(b) Histogram of ln(GDP per capita)

Figure 8.3 The distribution of GDP per capita

Note: Histograms. GDP measured at Purchasing Power Parity.
Source: `worldbank-lifeexpectancy` dataset. World Development Indicators 2017. N=182.

When we take the natural log of GDP per capita, the distribution is close to symmetric (Figure 8.3b). There are four countries with less than one thousand dollars of GDP per capita (Burundi, the Central African Republic, the Democratic Republic of Congo, and Niger); their log GDP per capita value is less than zero. The average of log GDP per capita is 2.4 in this data; its median is 2.5, suggesting a fairly symmetric distribution. The standard deviation is 1.2.

Figure 8.4 shows the results of a linear regression with life expectancy as the *y* variable, and GDP per capita as the *x* variable. This regression shows a positive slope, but the scatterplot implies that there are strong nonlinearities in the association that the linear regression doesn't capture. In particular, the slope appears a lot steeper at lower levels of GDP per capita and a lot flatter at higher levels. Thus, the linear regression does not capture the shape of the pattern, and, because of the skewed distribution of the *x* variable, it is not a very good approximation of the average of that nonlinear pattern either. The R-squared of this regression is 0.44.

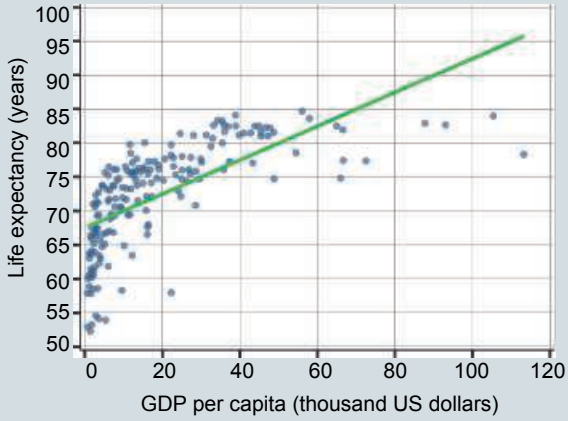


Figure 8.4 Life expectancy and GDP per capita

Note: Regression line and scatterplot. GDP measured at Purchasing Power Parity.
Source: `worldbank-lifeexpectancy` dataset. World Development Indicators 2017. N=182.

Figure 8.5a shows the same regression with the natural log of GDP per capita as the *x* variable. In contrast with the regression without taking logs, the scatterplot with log GDP per capita suggests a remarkably linear pattern – except maybe at the highest levels, a question we'll return to later. Indeed, for most of the range of log GDP per capita, the linear fit appears to be a good approximation of the conditional expectation. The R-squared is higher than without taking logs, 0.68.

Before we move on to quantify and interpret the results of this regression, let's spend a little time on the details of visualizing regressions with variables in logs. One way to visualize them is what we did for Figure 8.5a: measure the units of the log variable on the appropriate axis (here the *x* axis). That's the usual choice of visualization for expert audiences. However, those numbers don't mean much for most people. Thus, an alternative way to visualize the same regression is to put natural units as axis labels instead of the log values. Figure 8.5b shows this solution. It's the exact same figure, with different units for the *x* labels. To alert users of the graph that the labels are in natural units but the regression is in logs, the axis title adds "**ln scale**."

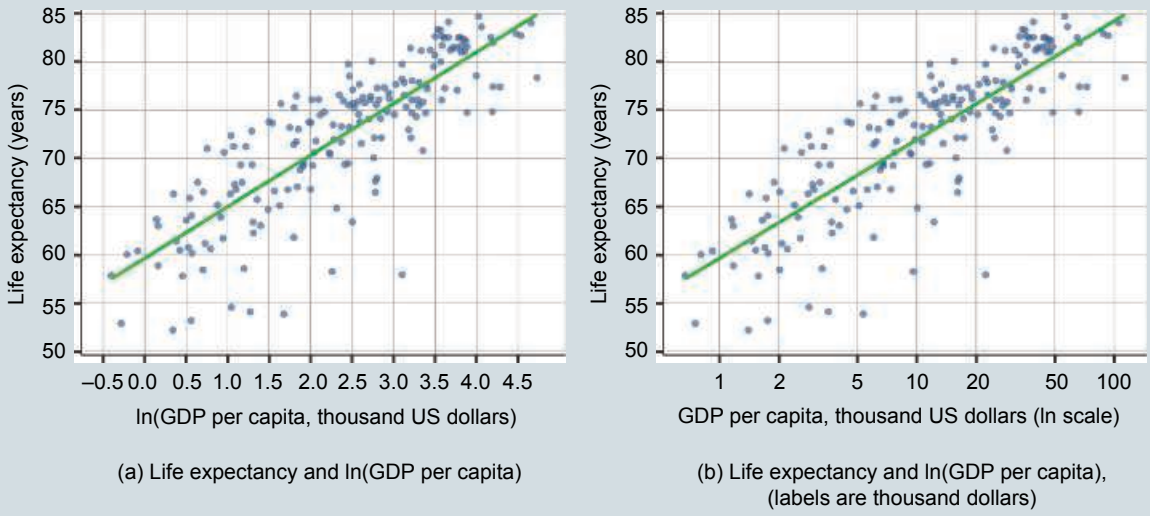


Figure 8.5 Life expectancy and GDP per capita

Note: Life expectancy regressed on the natural log of GDP per capita. Two visualizations of the same regression, with different labels on the *x* axis. Scatterplots and regression lines. GDP measured at Purchasing Power Parity.
Source: `worldbank-lifeexpectancy` dataset. World Development Indicators 2017. N=182.

We can conclude, therefore, that the association between life expectancy and average GDP is best captured with a regression that has log GDP per capita as the *x* variable. The slope of this regression is 5.3. This is a level–log regression, so the interpretation of the slope is a bit tricky. It shows that countries with a 1 percent higher GDP per capita have life expectancy higher by 0.053 years, on average. Note that 1-percent differences in GDP per capita are very small. Recall that the standard deviation of log GDP per capita is more than 1 so that typical cross-country differences are in the order of hundred percents not one percents. At the same time, the log approximation becomes quite bad at such high levels. As a middle ground, let's interpret the slope coefficient by looking at 10-percent differences. Countries with a 10 percent higher GDP per capita have a half (0.53) year higher life expectancy on average.

One purpose of the exercise is identifying countries where people live especially long or especially short lives for their average income. With the level–log linear regression, the countries with the shortest lives given their average income include Equatorial Guinea, Nigeria, and Cote d'Ivoire, where the average age at death is 18, 15, and 12 years shorter than what we could expect given their GDP per capita. The countries with the longest lives given their income include Vietnam, Nicaragua, and Lebanon where the average age at death is a little more than seven years longer than what we could expect given their GDP per capita. Other interesting results include that lives are more than two years shorter than expected in the USA, two years longer than expected in China, and five years longer than expected in Japan. Looking at the richest countries, lives are six years shorter than expected in Qatar, 0.5 years shorter than expected in Macao and Singapore, and one year shorter than expected in Luxembourg.

In fact, the eight countries with the highest levels of GDP per capita have lower life expectancy than predicted by the linear regression. These countries are a mix of city-states (Macao, Singapore), oil-rich countries (Brunei, United Arab Emirates, Qatar), and well-off small Western European

countries (Ireland, Luxembourg). This may mean that these particular countries have specific reasons for falling behind what's expected, maybe including mismeasurement of income due to GDP accounted for that does not enrich local people. But, maybe, this result shows that we just can't expect life expectancy to be proportionately higher with higher (log) GDP per capita among countries with the highest levels of GDP per capita. We'll try to do some more analysis to shed light on this question.

Before turning to investigate functional forms, let's examine what would have happened if, instead of per capita GDP, we had used total GDP for comparison. Total GDP varies even more than GDP per capita, from 0.2 billion to 20 trillion, reflecting variation both in size (number of people) and average income (GDP per capita). The distribution of total GDP is also very skewed with a long right tail: the mean is 630 billion and the median is 74 billion. There are two countries with extreme values, China and the USA.

Figures 8.6a and 8.6b show two regression lines, each with the corresponding scatterplot. Like earlier, the dependent variable is life expectancy in each regression. Different from what we had earlier, the explanatory variable is total GDP and its natural log, not GDP per capita and its natural log.

Figure 8.6a shows that something is obviously wrong with the regression on total GDP. Most of the observations are very close to each other at the bottom of the distribution of total GDP, and the regression line fits the few observations with very high values. The R-squared of this regression is 0.02, which is very low, due mainly to the fact that there is a lot of variation around the regression line at lower levels of total GDP. This picture illustrates why it's a good idea to include the scatterplot together with the regression line, as that in itself can reveal serious problems with the variables in the regression.

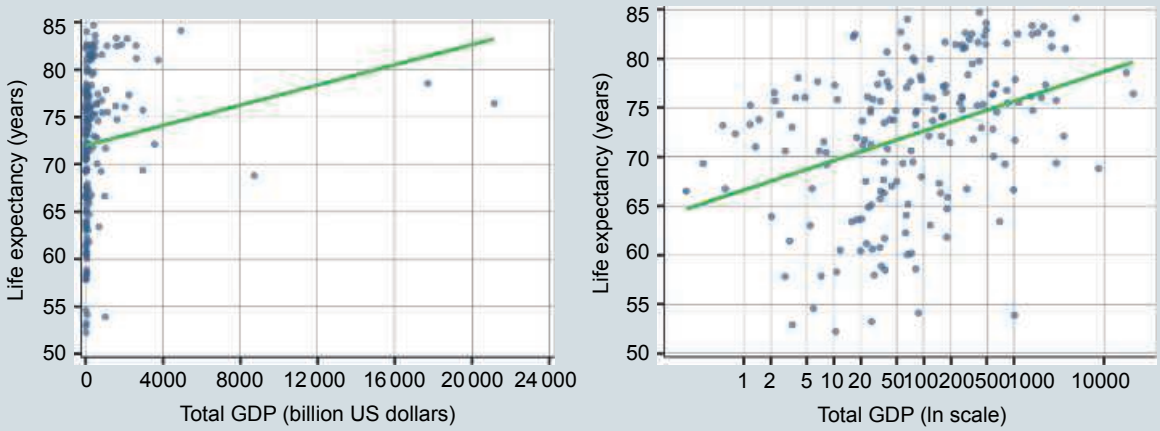


Figure 8.6 Life expectancy and total GDP

Note: Life expectancy regressed on total GDP. Level (panel a) and natural log (panel b). Scatterplot and regression line. GDP measured at Purchasing Power Parity.

Source: `worldbank-lifeexpectancy` dataset. World Development Indicators 2017. N=182.

More disturbing is the fact that Figure 8.6b doesn't show that something is necessarily wrong. In fact, it shows a linear regression that fits the scatterplot rather well. The R-squared of this regression is 0.14, which is low, but not as low as it was before taking logs of total GDP. However, this regression is misleading, too. It mixes two associations: the association of life expectancy with average income (GDP per capita) and with country size (population). Since our question is about the association with average income, we should get rid of the second one. Having GDP per capita as our *x* variable ensures that; having its log value ensures that we get a much better approximation of the functional form of the association.

8.6 Regression with a Piecewise Linear Spline

Data analysts use two main tools to approximate nonlinear patterns in the framework of linear regression besides transforming variables: piecewise linear splines and polynomials. Both tools involve specifying a nonlinear functional form for f in $y^E = f(x)$.

Although they allow for nonlinearities in the functional form, both tools specify regressions that are estimated within the framework of linear regression. This may sound strange, but recall from Chapter 7, Section 7.4 that a linear regression is linear in its coefficients. Technically, these two tools have more than one right-hand-side variable, as a multivariate linear function. Thus, technically, they are multiple regressions, which we'll cover in Chapter 10. Substantively, however, they have one right-hand-side variable, only entered in a more complicated way.

A **piecewise linear spline** allows the slope to differ for different sections (values) of the explanatory variable. It is a set of connected line segments instead of a single line with a single slope. The points at which the different line segments are connected are called knots. With two line segments we have one knot. With three line segments we have two knots. With m line segments we have $m - 1$ knots.

Importantly, piecewise linear splines are not the same thing as separate linear regressions for different ranges of the explanatory variable. Both methods allow for a different slope for the different intervals of the explanatory variable. However, piecewise linear spline regressions produce connected line segments while separate regressions produce line segments that are typically not connected.

It is the job of the analyst to specify the number and the place of the knots of the piecewise linear spline – i.e., at what values of the explanatory variable *x* the knots are. Then, the regression uncovers the slopes of the different line segments. Whether their slopes are different indeed is determined by the data. The decision of where to allow for knots is not an easy one. We may start with a nonparametric regression and allow for knots where the shape appears to change. We should aim for many observations within each *x* segment, and, in any event, simplicity is a virtue, so the fewer line segments the better.

The formula for a piecewise linear spline with two segments connected at knot k is

$$y^E = \alpha_1 + \beta_1 x[\text{if } x < k] + (\alpha_2 + \beta_2 x)[\text{if } x \geq k] \tag{8.1}$$

The interpretation of the coefficients of this regression are straightforward generalizations of the coefficients of a simple linear regression:

α_1: Average *y* for observations with value zero for *x*.

β_1: Among observations with *x* values less than k, *y* is β_1 units higher, on average, for observations with one unit higher *x* value.

β_2: Among observations with x values greater than k, y is β_2 units higher, on average, for observations with one unit higher x value.

Note that when we estimate a spline, α_2 is automatically calculated as $\alpha_2 = \alpha_1 + \beta_1 k$.

The formula for a piecewise linear spline with m line segments connected at knots k_1, k_2, ..., k_{m-1} is

$$y^E = \alpha_1 + \beta_1 x[\text{if } x < k_1] + (\alpha_2 + \beta_2 x)[\text{if } k_1 \leq x \leq k_2] + \cdots + (\alpha_m + \beta_m x)[\text{if } x \geq k_{m-1}] \quad (8.2)$$

The interpretations of the coefficients in this regression are similar: each line segment has its own slope coefficient. Once again $\alpha_2, \ldots, \alpha_m$ are calculated automatically. Software will publish the estimated α_1 and $\beta_1, \beta_2, \ldots, \beta_m$ coefficients.

Specifying a regression with piecewise linear splines is relatively straightforward in most statistical software as they have built-in routines to do that. The analyst just needs to define the knots.

Note that there is an alternative way to present the results of a regression with a piecewise linear spline: selecting a line segment as a reference and showing the slope of each of the other line segments in terms of its difference from the reference slope. The interpretation of the coefficients is different, and the underlying formula for this representation is different, too. However, taken together, the information content of the coefficients is the same. Usually, statistical software allows for presenting the results in both ways, and they have one, and not always the same one, as their default setting. Data analysts need to choose the version they prefer and understand the default settings.

Review Box 8.4 Regression with piecewise linear spline of x

- A piecewise linear spline regression results in connected line segments, each line segment corresponding to a specific interval of x.
- The formula for a piecewise linear spline regression with m line segments (and $m - 1$ knots in-between) is

  $$y^E = \alpha_1 + \beta_1 x[\text{if } x < k_1] + (\alpha_2 + \beta_2 x)[\text{if } k_1 \leq x \leq k_2] + \cdots + (\alpha_m + \beta_m x)[\text{if } x \geq k_{m-1}]$$

  - α_1: Average y when x is zero.
  - β_1: Among observations with x values less than k_1, y is β_1 units higher, on average, for observations with one unit higher x value.
  - β_2: Among observations with x values between k_1 and k_2, y is β_2 units higher, on average, for observations with one unit higher x value.
  - β_m: Among observations with x values greater than k_{m-1}, y is β_m units higher, on average, for observations with one unit higher x value.

8.7 Regression with Polynomial

An alternative to piecewise linear splines are **polynomials**, having squared, cubed, or higher-order terms of the explanatory variable together with the linear term. The simplest and most widely used is the **quadratic function** of x that adds a square term:

$$y^E = \alpha + \beta_1 x + \beta_2 x^2 \quad (8.3)$$

Graphically, a quadratic function of x replaces the regression line with a regression parabola. A cubic functional form adds a third power term: $y^E = \alpha + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$. Higher-order polynomials have more terms: 4th term, 5th term, and so on, allowing for even more complicated curves. The highest power in the polynomial is called its order. The order of the regression equation determines the number of potential points at which the curve may twist (turning steeper, flatter, up, or down).

Compared to piecewise linear splines, quadratic or higher-order polynomial specifications have both advantages and disadvantages. Polynomials do not require the analyst to specify where the pattern may change. The only thing to decide is the order of the polynomial; where and how the curve would twist is determined by the data. That comes at a cost: polynomials are rigid functional forms. Quadratic polynomials result in a parabola so that the pattern is bound to be U-shaped: negative slope for low levels of x and positive slope for high levels (or an upside-down U with positive slope first and negative slope next). Moreover, the pattern is bound to be steeper farther away from the middle of the parabola. Higher-order polynomials specify curves with fixed features too.

In practice, the limitation of a fixed curve may or may not be important. For example, while a quadratic always results in a parabola if extended to the entire x line, it is possible that the range of x in the data shows only a part of the parabola. Therefore it is possible that with a quadratic model, the curve that best fits the data may be increasing over the entire range of x in the data, or decreasing over the entire range. The eventually unavoidable turn would occur outside the range of the data, which, in most cases, has no practical consequence for the analyzed patterns.

For example, the pattern of association between the (log) wage of employees and their age is often well captured by a quadratic. This quadratic is increasing over most of the age range in most economies. But it's a nonlinear increase, being a lot steeper at younger ages and flat at old age. We usually don't want to extrapolate that pattern to very old ages with no employees so it doesn't matter that the quadratic would imply a steeper and steeper decline in wages.

Another disadvantage of having polynomials in the regression is that their coefficients are hard to interpret. In a regression with a quadratic function of x $y^E = \alpha + \beta_1 x + \beta_2 x^2$, α is average y when $x = 0$, but β_1 has no interpretation, and β_2 only shows whether the parabola is U-shaped, in other words, convex (if $\beta_2 > 0$) or inverted U-shaped, or concave (if $\beta_2 < 0$).

Sometimes we are interested in the slope of the curve. With a quadratic or a higher-order polynomial the slope is different for different values of x, and getting the slope requires work. In the regression with a quadratic $y^E = \alpha + \beta_1 x + \beta_2 x^2$, the slope is $\beta_1 + 2\beta_2 x$ (the first derivative of the quadratic function). We can compare two observations, denoted by j and k, that are different in x, by one unit so that $x_k = x_j + 1$. y is higher by $\beta_1 + 2\beta_2 x_j$ units for observation k than for observation j.

Review Box 8.5 Regression with quadratic or higher-order polynomial in x

$$y^E = \alpha + \beta_1 x + \beta_2 x^2$$

- The result is a parabola that is either U-shaped (convex) or upside-down U-shaped (concave).
- Interpretation of the coefficients:
  - α: average y when x is zero;
  - β_1: no meaningful interpretation;
  - β_2: signals whether the relationship is convex (if β_2 is positive) or concave (if β_2 is negative).

- Regression with higher-order polynomial of the explanatory variable:

$$y^E = a + \beta_1 x + \beta_2 x^2 + \cdots + \beta_k x^k$$

8.8 Choosing a Functional Form in a Regression

The choice of functional form for the regression analysis is rarely straightforward. The most important guiding principle should be the ultimate goal of the analysis. The importance of functional form is different if the question is prediction, uncovering an average effect, or learning how an effect varies. Statistical tools can help make this choice but should never substitute for thinking about the substantive goals of the analysis.

To recap, we may want to ignore nonlinear patterns if all we care about is an average association and the linear regression gives a reasonable approximation to that average association. If that approximation is not good, we may want to transform variables, for example by taking logs, but only if the interpretation of the transformed variables makes more sense for the substantive question of the analysis. However, if our goal is prediction, identifying observations with large residuals, or learning about how an association varies with x, nonlinear patterns are important. In such cases we should uncover the shape of the pattern. As simplicity is a virtue for many reasons, it's often best to approximate such a shape within the framework of the linear regression.

As we have seen, we may have more options to uncover nonlinear patterns and incorporate them in linear regression analysis: transforming variables, piecewise linear spline, quadratic or other polynomials. Statistics can help choosing the best, or a good enough, method, but it never substitutes for thinking. In the subsequent chapters of the book (for example in Chapters 13 and 14) we'll cover some more systematic methods and more systematic ways of thinking for choosing the best regression to answer the question of the analysis. These are sometimes referred to as methods for model selection.

Even before learning those more systematic methods, we can often make good enough choices. Here is a roadmap that we think is useful to follow. We'll apply this roadmap for our case study below.

- Start by deciding whether you care about nonlinear patterns.
  - We usually are fine with a linear approximation if our focus is on an average association.
  - Even then we often transform variables if that makes for a better interpretation of the results and makes linear regression better approximate the average association.
  - In contrast, we usually want our regression to accommodate a nonlinear pattern if our focus is:
    - on prediction;
    - the analysis of residuals;
    - about how an association varies with x.
- If we want to uncover and include a potentially nonlinear pattern in the regression analysis we should:
  - uncover the most important features of the pattern of association by examining a scatterplot or a graph produced by a non-parametric regression such as lowess;
  - choose one or more ways to incorporate those features into a linear regression (transformed variables, piecewise linear spline, quadratic, and so on);
  - compare the results across various regression approaches that appear to be good choices.

In social sciences, the last point is referred to as **robustness checks**. Doing robustness checks basically means running several different regressions and comparing results. In other words, it amounts to checking the extent to which the most important conclusions of the analysis are robust to our choices. Here the choice is of functional form. This is an important step because, most often, all we can aspire to is a good enough approximation. Thus, there is rarely a single best choice. Robustness checks focus our attention on what's really important in our analysis: whether and how much our choices during the analysis affect the answer to the ultimate question of the analysis.

8.B2 CASE STUDY – How is Life Expectancy Related to the Average Income of a Country?

Piecewise linear spline and polynomials

Our reason for analyzing the pattern of association between life expectancy and GDP per capita is to identify countries that do especially well on this health measure relative to how much income they generate. This calls for an analysis of the residuals. Thus, getting the shape of the pattern is important here.

We want to have a regression that fits the pattern of association between life expectancy and log GDP per capita. The scatterplot reveals that the association may be flatter for countries with the highest levels of GDP per capita. Recall from Figure 8.4b that life expectancy is below the simple linear regression line for all of the eight highest-GDP countries, with GDP per capita over 50 000 dollars.

One choice is thus a piecewise linear spline regression with a knot at GDP per capita of 50 000 dollars, which translates to 3.9 in natural logs. It produces the results shown in Figure 8.7a.

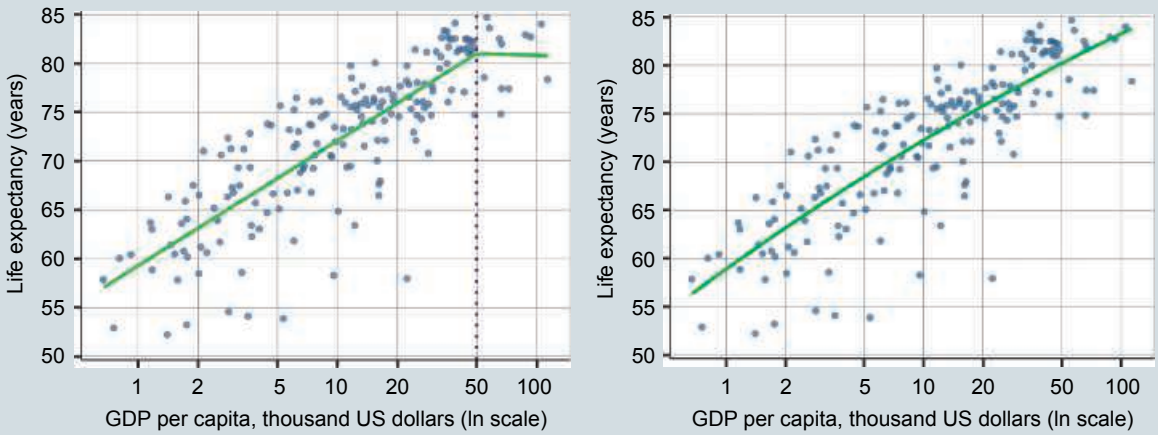


Figure 8.7 Life expectancy and GDP per capita: scatterplot and nonlinear regression

Note: Piecewise linear spline with one knot at 50 000 dollars per capita. GDP measured at Purchasing Power Parity.
Source: `worldbank-lifeexpectancy` dataset. World Development Indicators 2017. N=182.

The slope of the first, long, line segment is 5.5, just a little steeper than with a single log GDP per capita right-hand-side variable. The slope of the other line segment is −0.3. Comparing countries with GDP per capita below 50 thousand dollars, life expectancy is higher, on average, by approximately 0.55 years in countries with 10% higher GDP per capita. But, comparing countries with GDP per capita above 50 thousand dollars, life expectancy is lower, on average, by approximately 0.03 years in countries with 10% higher GDP per capita. This is a very small difference, and we can safely treat it as if it were zero: no difference in life expectancy by GDP per capita above 50 thousand dollars.

The R-squared is 0.69 here. The R-squared of the previous linear regression, with a single regression line, was 0.68. This is an improvement, but only a small one. A linear regression that allows for slopes to differ for different values of the explanatory variable must produce at least as good a fit as a linear regression that does not allow for different slopes. However, the improvement on the fit is very small in this case. That should not be surprising as the new regression provides a substantially better fit for only a few observations.

Let's see if an alternative way to capture the nonlinear pattern leads to a similar conclusion. One such way may be to fit a quadratic. Figure 8.7b shows that the quadratic fit settles for slight nonlinearity. The pattern is a positive association through the range of observed ln GDP per capita, with less positive association at higher levels than at lower levels of GDP per capita. The model fit is once again just marginally improved with R-squared of 0.69.

Note that a quadratic function always fits a parabola (Section 8.7). This time it is an upside-down parabola. But we don't see it on the graph. Why? That's because the estimated segment of the parabola is very close to linear. The turning point of the parabola would be at levels of GDP per capita way above what's observed in the data.

The two slope parameters are 6.3 and −0.22, with no clear interpretation except the second, negative, number showing that the parabola is concave (upside-down). The slope of the curve varies with the value of the x variable: it is $6.3 - 0.44 \times \ln(GDPpercapita)$. A quick calculation shows that this slope would be zero for ln GDP per capita around 14.3, corresponding to GDP per capita over 1 600 000 measured in thousand dollars, which is USD 1.6 trillion. That would be the turning point above which the implied pattern would be negative. Of course, that value is stratospherically high, so the main message of the quadratic is that the pattern is always positive, only it is a tiny bit flatter at higher levels of income. Whether that aspect of the pattern is stable, whether it reflects anything causal, and whether it's generalizable from this data are questions that we shall address in the subsequent chapters.

Looking at specific countries with unusually low or unusually high life expectancy given their level of income, we see very similar groups as for the models without the spline or quadratic on the right-hand side. The countries with the shortest lives given their income include Equatorial Guinea, Nigeria, and Cote d'Ivoire, and the countries with the longest lives given their income include Vietnam, Nicaragua, and Lebanon again. The ranking of the top and bottom lists are similar across these various regressions, although the magnitudes of the residuals differ slightly. The fact that these results don't change much is not surprising, as neither the piecewise linear spline nor the quadratic specification lead to strong nonlinear shapes in the estimated regression line.

What can we conclude from this case study? We'll estimate one more kind of regression later in Section 8.B3, but its results will not change the conclusions we can draw. The cross-sectional relationship between life expectancy and (log) GDP per capita is strong positive throughout almost

the entire range of observable GDP per capita. For most of the range, a 10 percent difference in GDP per capita is associated with a higher life expectancy by one half of a year on average. Life expectancy in the richest countries is lower than the average pattern would suggest. That can be viewed in various ways; maybe, the association at very high income levels is flat, or maybe it is attributable to country-specific factors.

Given the pattern of association between life expectancy and GDP per capita, we identified countries with unexpectedly low life expectancy and unexpectedly high life expectancy. The former include the sub-Saharan African countries of Equatorial Guinea, Nigeria, and Cote d'Ivoire. The latter group includes, perhaps surprisingly, countries such as Nicaragua and Lebanon next to some less surprising ones such as Vietnam.

In terms of methods, this case study illustrated that we may want to transform a variable, here the *x* variable, in more than one way. We took GDP per capita instead of total GDP in order to focus on the association with the level of income without the association with size. And we took the log of GDP per capita in order to have a more symmetric distribution of the *x* variable, which gave a substantially better fit of the regression. It also made the regression very close to linear. We have also learned how to apply the piecewise linear spline and the quadratic specifications. In this example, those more complicated specifications didn't lead to very different conclusions because the pattern of association turned out to be fairly linear overall. Finally, we have illustrated that we can identify countries with unusually high and unusually low life expectancy in a way that is robust to how we specify the regression.

8.9 Extreme Values and Influential Observations

After the overall functional form of the regression, let's spend some time on a potential issue that may have consequences for the shape, as well as the slope, of the regression. This issue arises when patterns are driven by a few observations. Typically, these observations have unusually large or small values for *y* and *x*. For example, nonlinear patterns may be largely due to a few observations with low or high values of *x*. Such extreme values are rare, by definition. Whether to do something with them, and what to do, are important questions in data analysis – unfortunately, without an easy answer.

As we discussed in Chapter 3, **extreme values** are simply values of a variable that are very small or very large and are somewhat separated from the rest of the values. For example, distributions with long right tails have extreme large values.

However, not all extreme values matter for regression analysis. Observations with extreme values that influence the results of regression analysis are called **influential observations**. Influential observations tilt regression lines: the slope of the regression is different when we include them in the data from the slope when we exclude them from the data. Extreme values with little effect on the results of regression analysis are not really an issue: they just don't make a difference. It is influential observations that matter.

Influential observations are often called "outliers." We prefer not using this term because it suggests that such observations do not belong in the analysis. Instead of making such a judgment right away, data analysts need to think about what to do with those influential observations.

Whether influential observations belong in the analysis is an important decision that should be based on thinking. Two issues need to be considered: (1) why the values are extreme for the influential observations, and (2) what the question of the analysis is.

Influential observations should be excluded from the data if they are due to errors. Spotting whether extreme values are due to errors is an important goal of exploratory data analysis, as we discussed in Chapter 3. Observations with erroneous extreme values are best discarded at the exploratory phase of data analysis, even before they could show up as influential observations for regression analysis.

If not due to errors, extreme values belong to the distribution of variables. As a general rule, we should keep such observations in the data even if they prove to be influential observations.

An exception to this rule is when we decide that we don't want our analysis to say anything about certain kinds of observations. That is usually fine if the decision is made based on the value of the *x* variable. Indeed, that is the decision we made when we discarded hotels from our data with less than 3 stars or more than 4 stars or that were more than 8 miles away from the city center. In contrast, it is usually not fine to discard observations based on the *y* variable. That's because we want our analysis to explain variation in *y*. If we were to discard some *y* values we would "explain" some of the variation in *y* without an analysis by saying that those values are the result of a fundamentally different pattern.

Indeed, observations with extreme values may be the most valuable observations. For example, the pattern of association between prices and sales is difficult to uncover unless there is enough difference in prices. When comparing markets with very similar prices, the few instances of large price difference may be the only source of variation that can help uncover the pattern.

Moreover, it turns out that observations that are influential in some kinds of regressions are not influential in others. Taking the log of variables with long right tails often takes care of extremely large values and thus influential observations. Similarly, expressing variables as ratios (GDP per capita, stock price per earnings, and so on) makes otherwise extreme values not extreme. Dropping such extreme values before transforming them would be a mistake.

Sometimes, our decision is clear (drop if error), sometimes it is debatable (is it part of the pattern we are interested in?). Hence, on some occasions, there is no clearly superior decision. Importantly, we should be transparent in what we decide and document every step of data manipulation. When no obvious single best solution exists, it is good practice to carry out robustness checks, just like when choosing a functional form (Section 8.8): examine whether and how key results change if we make different decisions.

Review Box 8.6 Extreme values and influential observations

- Observations with extreme values may or may not influence the results of the regression analysis; if they do they are called influential observations.
- It is not good practice to drop observations with extreme values without thinking.

8.10 Measurement Error in Variables

Measurement error in variables may have severe consequences for regression analysis beyond producing extreme values and potentially influential observations. Such errors may arise due to technical or fundamental reasons.

Technical sources of measurement error include reporting errors when the source of variables is a survey; recording errors in data recorded by humans as well as machines; and coding errors that are introduced during transforming raw data into analysis data, or data cleaning. Unfortunately, both humans and machines tend to make errors when they have the opportunity. Another technical source

of measurement error is frequent in aggregate variables that are calculated from a sample of observations (average wage in firms, total sales summed over stores). As we have seen in Chapter 5, the number of observations used to produce averages matters: with fewer observations, those numbers may fluctuate more for statistical reasons.

Fundamental sources of error occur when the variable we would like to include in the analysis is not contained in the data; instead, we use another variable, or variables, that are related. Such fundamentally unobserved variables are called **latent variables**, a concept we introduced earlier in Chapter 4, Section 4.7. Observed variables that we use in analyses instead of latent variables are called **proxy variables**. As we discussed in Chapter 4, we often have more than one observed variable to measure a latent concept. When we combine those, most often into an average score, this generated variable can be thought of as a proxy variable, too.

Proxy variables are erroneous measures of latent variables. When we want to measure the relationship of earnings and skills, skills is a fundamentally latent variable. Skills may be measured by scores on a test. But scores are not the same as skills; they can be thought of as erroneous measures of skills. When we want to measure the relationship of firm productivity and worker turnover, productivity is a latent variable. Sales per staff member may be used as a proxy variable for firm productivity. It would be an erroneous measure: productivity is output conditional on many inputs besides staff numbers.

Regardless of whether being due to fundamental or technical reasons, measurement errors in variables affect the results of regression analysis. The effect depends on properties and magnitude of the errors, as well as whether they affect a dependent variable or an explanatory variable in a regression. Consider simple linear regressions of the form $y^E = \alpha + \beta x$. The questions to consider are (1) whether the error leads to a bias in the slope coefficient – i.e., whether it is expected to lead to a different slope coefficient from what it would be without an error; (2) whether the error has other consequences, such as worse fit. As we shall see, the answer to the first question may be very different depending on the properties of the error and whether it affects y or x. As for the second question, measurement error almost always results in worse fit of the regression.

8.11 Classical Measurement Error

We first consider the simplest kind of error: **classical measurement error**. Classical measurement error is an error that is zero on average and is independent of all other relevant variables, including the error-free variable. To be more specific, classical measurement error, denoted by e, is added to the error-free value of a variable y^* or x^*, resulting in an observed variable y or x. Classical measurement error is also called noise.

Technical errors are often close to being classical errors. These include recording errors, such as errors in surveys (wrong values entered) or in automatic measurements (noise in recording). Variables in aggregate data are often estimates from samples, and such estimates are often well represented as variables measured with classical error (estimations have error reflecting the fact that other, equally representative samples may result in different values, as discussed in Chapter 5).

An example is using county-level data to analyze how the local unemployment rate may affect local support for a political party. Typically, both the unemployment rate and the party support variables are estimated from individual-level surveys from samples of respondents. Thus, each variable contains error: the true unemployment rate plus error, and true party support rate plus error.

If measurement error is classical, the average of the error is zero, so the average of the observed variable is the same as the average of the error-free variable, which would be: $\bar{x} = \overline{x^*}$. At the same

time classical measurement error makes the spread of the measured variable larger than what the spread of the error-free variable would be. In fact, the variance of an error-ridden variable is the sum of the variance of the error-free variable and the variance of the error.

For example, if x is measured with classical error e:

$$Var[x] = Var[x^*] + Var[e] \tag{8.4}$$

The variance of the sum of two variables is the sum of the variance of the two variables plus twice their covariance; the covariance here is zero because classical measurement error is independent of the error-free values.

The importance of measurement error is captured by the **noise-to-signal ratio**: the ratio of error variance to what the variance of the error-free variable would be. When a variable x includes classical measurement error so that $x = x^* + e$, the noise-to-signal ratio is $Var[e]/Var[x^*]$. When the noise-to-signal ratio is small, we may safely ignore the problem. This occurs when we are confident that coding errors are not important or when our data has an aggregate variable estimated from very large samples. When the noise-to-signal ratio is substantial, we may be better off assessing its consequences.

In our example of county-level estimates, we can expect a larger error, and thus a larger noise-to-signal ratio, when the original sample is smaller. The original sample here is the sample of individuals who participated in a labor force survey to measure the unemployment rate, and the sample of individuals who participated in a political poll to measure party support. The sizes of the two samples may be different in each county.

What are the consequences of classical measurement error for regression analysis? We focus on linear regressions of the form $y^E = \alpha + \beta x$, and examine what happens to the fit of the regression and the slope coefficients when either y or x is measured with classical error. It turns out that the consequences are different for the two variables.

Classical measurement error leads to worse fit of a regression when the error affects the dependent variable, $y = y^* + e$, but it does not affect estimates of the regression coefficients. It's quite intuitive to see why the fit is worse. Regressions fit the variation in observed y. But when y is measured with classical error, part of that variation is just noise, independent of everything. Regressions do not fit noise: they find patterns of association between variables, and noise due to classical measurement error is not associated with anything relevant. It also makes sense that regressions are expected to produce the same slope (and intercept) when the dependent variable is measured with classical error as if they were free from error. That is because classical error in y averages to zero for all values of x, so when the regression looks at average y for different values of x, it is expected to find the same value as for average y^*. (See Under the Hood section 8.U2, for a more technical derivation.)

In contrast, when the explanatory variable is measured with classical measurement error, the regression coefficients are off. That is less intuitive at first, but it also makes sense. The problem is due to the fact that the error conflates comparisons between different x values. When comparing observations, their typical distance in measured x is inflated by noise: part of it is error-free x^*, but part of it is error e. Thus, if a unit difference in x^* would go with an average difference of, say, β^* in y, a unit difference in measured x would go with a smaller average difference in y. Technically, this is all due to the fact that the noise inflates the variance ($Var[x] = Var[x^*] + Var[e]$), and this variance is in the denominator of the slope coefficient estimate ($\hat{\beta} = Cov[y, x]/Var[x]$).

Notice the difference between the role of error in the dependent variable and the explanatory variable. Noise-induced extra variance in the dependent variable leads to worse fit but does not affect the coefficients. Noise-induced extra variance in the explanatory variable affects the coefficients.

In our county-level example, we would run a regression in which y is the percent of supporters of the political party in the county and x is the unemployment rate in the county. Suppose that the estimated slope is $\hat{\beta} = 0.1$. That means that in counties with one percentage point higher unemployment rate, support for the party tends to be higher by one tenth of a percentage point. But there is measurement error in both y and x here. The former has no effect on the estimated slope; the latter makes the estimated slope smaller than it truly is. If the noise-to-signal ratio is small, the estimated slope is close to what it would be without the error. With large noise-to-signal ratio, the difference is larger. If, for example, the samples are smaller in rural counties than in urban ones, then we would see a smaller slope in rural counties even if the true slope were the same.

The effect of classical measurement error in the explanatory variable is called the **attenuation bias**. The slope of a simple linear regression is flatter if the explanatory variable is measured with classical error. The larger the noise-to-signal ratio, the larger the attenuation. A flatter line means smaller slope coefficient in absolute value. If the slope with error-free x^* was positive, we would expect the slope with the error-ridden x to be positive, too, but smaller. If the slope with error-free x^* was negative, we would expect the slope with the error-ridden x to be negative, too, but less negative. An additional consequence of this extra flatness is that the intercept is biased in the other direction: if the slope is biased in the negative direction, the intercept is biased in a positive direction, and vice versa. (See Under the Hood section 8.U2, for a more technical derivation.) We'll see these features in our hotels case study (Section 8.C1).

Review Box 8.7 Classical measurement error in simple linear regression

- Classical measurement error is pure noise, independent of everything of interest.
- If y is measured with classical error: $y = y^* + e$
  - the fit of the regression is worse;
  - but the slope of the regression is expected to be the same as it would be with an error-free y^*.
- If x is measured with classical error: $x = x^* + e$
  - this leads to an estimated slope coefficient that is closer to zero (attenuation bias).
- Often, errors in variables are not classical. Such errors affect regression coefficients in different ways.

8.C1 CASE STUDY – Hotel Ratings and Measurement Error

Regression and measurement error in average customer ratings

Measurement error is not something we may know – if we did, we would correct it and it would not be a problem anymore. However, we can explore our data to get a better understanding of its magnitude. To do that we would need a variable in the dataset that is a good indicator of the magnitude of the measurement error, and use that to map out the impact of measurement error on regression estimates.

Going back to the case study of hotel prices, let's investigate another association: price and customer rating. The price comparison website publishes the averages of ratings customers gave to each hotel. Ratings vary between 1 and 5, with 5 being excellent. The website collects such ratings and shows their average value for each hotel. Customer ratings are a measure of quality: hotels with higher ratings provide an experience that was valued higher by their customers. Thus a regression of hotel price on rating may show an association between price and quality. Large negative residuals from this regression may help identify hotels that are underpriced relative to their quality.

However, the measure of customer rating is an average calculated from individual evaluations. That is a noisy measure of hotel quality, for at least two reasons.

First, a fundamental reason. Ratings may be influenced by factors other than quality. For example, they may be influenced by the price itself as customers may give higher ratings to cheaper hotels of the same quality. That is an error that is certainly not classical as it is related to an important variable: price. We can't investigate this source here, in part because we don't have the statistical tools, but also because it's a fundamentally difficult question.

Second, a technical reason. Average ratings are based on a number of individual ratings, and few ratings provide noisier averages than many ratings. That is simply due to larger variation of estimates from small samples than from large samples, as covered in Chapter 5. This is typical classical measurement error: observed average customer ratings are a noisy measure because they are averages from a sample of all customers. The noise is more important if fewer customers provide ratings that go into the average measure. That is, the noise-to-signal ratio is larger, the fewer the ratings.

The data includes the number of ratings that were used to calculate average customer ratings. If classical measurement error plays a role, it should play a larger role for hotels with few ratings than for hotels with many ratings. To investigate this possibility, we divided our data into three subsamples of roughly equal size: hotels with few ratings, hotels with a medium number of ratings, and hotels with many ratings.

We labeled hotels as having few ratings if they had less than 100 ratings (77 hotels, with 57 ratings each on average). We labeled hotels as having many ratings if they had more than 200 ratings (72 hotels, with 417 ratings each on average). In-between are hotels with a medium amount of ratings (58 hotels, with 147 ratings each on average). Average customer rating is rather similar across these three groups (3.94, 4.03, and 4.20). But the standard deviation of the average customer ratings across hotels is a lot larger among hotels with few ratings (0.42 versus 0.26). Thus, there is more spread across hotels in terms of average customer ratings when we compare hotels in which those ratings were given by fewer customers. That may be due to many reasons, but it is also in line with more noise among hotels with fewer ratings.

We regressed log hotel price on average ratings separately for hotels with few ratings (less than 100) and hotels with many ratings (more than 200). If there is classical measurement error in average ratings, the error should be more prevalent among hotels with few ratings, and so the regression line should be flatter there.

That is indeed what we find. Figure 8.8 shows two regression lines, one for hotels with few ratings ($<$ 100) and one for hotels with many ratings ($>$ 200). The first slope coefficient is 0.35; the second one is 0.55 (we don't show the regression line for hotels in-between; its slope is 0.45). The corresponding intercepts are ordered in the opposite way: 3.14 for the first regression and

2.39 for the second. The figure visualizes those differences: flatter, less positive slope and higher intercept among hotels with few ratings.

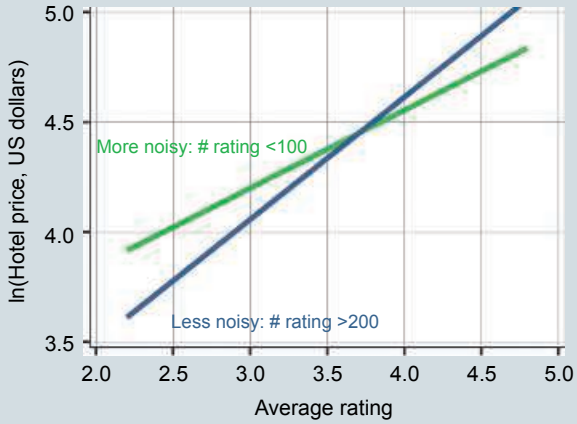


Figure 8.8 Log hotel price and average customer ratings: samples by noisiness

Note: Log hotel price and average customer ratings. Hotels with noisier measure of ratings (number of ratings < 100) versus hotels with less noisy measure (number of ratings > 200).
Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. N=207.

The take-away from this analysis is that there appears to be substantial measurement error in average customer ratings among hotels where that average rating is based on a few customers' reports. Thus, we can expect a regression with average customer ratings on the right-hand side to produce an attenuated slope. Should we do anything about that? And if yes, what?

If we are interested in the effect of ratings on prices, this is clearly an issue. In such an analysis we may want to discard hotels with less than a minimum number of reviews (maybe 10 or 20 or 50 or 100). This is hard, as there is no clear threshold here, so we may try various thresholds. Another approach would be to keep all data but acknowledge that the slope estimate is attenuated due to measurement error. In fact, we could assess the magnitude of the error and adjust the estimates with it. That would require more sophisticated analysis that is beyond the scope of this textbook.

8.12 Non-classical Measurement Error and General Advice

Not all errors in variables are classical in nature. In fact, most errors in variables in real data are not classical. For example, when asked about their earnings or income, survey respondents tend to report values that vary less than their earnings, or income measured in tax records or other administrative sources. Survey responses here are erroneous measures of the values from administrative sources, but they are actually less dispersed, not more dispersed as classical errors should be.

Errors due to misreporting or omission of elements that should be included can lead to biased estimates, too. The consequences of such errors for regression analysis are typically different from the consequences of classical error. Discussing them is beyond the scope of this textbook. Nevertheless, when we suspect substantive errors in variables, it pays to understand the sources and features of error and its consequences for the analysis.

Most variables in economic and social data are measured with noise. So what is the practical consequence of knowing the potential bias? The first take-away is that we should have a good idea about the magnitude of the problem. Noisy measures may be a problem if the noise-to-signal ratio is high. Second, if the ratio is high, we need to have a good idea of the source of the error. If that source makes the measurement error classical, we know what to expect: error in x is a bigger problem for inference than error in y. If non-classical error, we need to think more and, possibly, ask for more advice.

Ultimately, the potentially severe consequences of errors in variables highlight the importance of data quality. A variable that is better measured is more useful for analysis. By "better" we mean both high validity (intended content close to actual content) and high reliability (repeated measurement would lead to similar values). For example, the quality of management of firms is measured with higher validity if based on a scoring system (as in the World Management Survey, see Chapter 1) than on what the firms' managers would say. And answers to difficult survey questions on, for example, household savings or the value of one's home, are more reliable the more background information respondents have available to them.

Thus, assessing the magnitude and properties of measurement error in the most important variables is important. Measurement error may have substantial, and sometimes surprising, consequences. When possible, it pays to work with variables with less error. When error in variables is unavoidable, we need to document it and think about its potential effects.

8.13 Using Weights in Regression Analysis

In the last section of this chapter we discuss whether, when, and how to use **weights in regression analysis**. Using weights makes some observations count more than others in the analysis. We consider two cases for using weights: samples that represent populations, and aggregate data that represent smaller units.

Datasets that are samples of a population often have weights to tell us the number of population items each observation represents (see Chapter 1 for more detail). Such weights may compensate for unequal sampling probabilities, or they may be meant to compensate for some of the biases in the sample due to differences in coverage or response rates.

Using weights in regression analysis is one way to compensate for unequal sampling and biased coverage. Weights in regressions $y^E = f(x)$ produce weighted average values for y^E instead of an unweighted average. In weighted averages, observations with higher weights count more than observations with lower weights.

In practice, weights that compensate for unequal sampling or unequal coverage rarely make a big difference in regression results, especially in multiple regression analysis (to be covered in Chapter 10). Nevertheless, it is good practice to do the analysis both with and without such weights and show whether the results are robust. If they are not, that is if results with and without these kinds of weights differ a lot, there usually is some problem in the background. It is advised to dig deeper in such cases and perhaps incorporate variables that may explain the discrepancies in multiple regression analysis (see later in Chapter 10).

The second potential case for weighting arises with aggregate data. Observations in aggregate data may be units of different size (countries, firms, families). Analyzing aggregate data without weights treats each aggregate unit equally. Analyzing aggregate data with weights makes larger units more important.

Whether to use the size of each observation as weight in analysis of aggregate data should be guided by the substantive question of the analysis. Without weights, a regression using aggregate data compares the aggregate units (e.g., countries). With weights, a regression using aggregated data is best interpreted as approximate comparisons of the smaller units that make up the aggregate ones (e.g., people living in various countries).

For example, we may not want to use weights when interested in how certain policies affect aggregate economic measures (say, the effect of tax policy on income inequality). In this case each country is as important an observation as another country. In contrast, we may want to use weights if we use aggregate data to uncover how certain conditions of people are related to their outcomes (say, the association between vaccination rate and disease prevalence). Here we use aggregate measures but the outcome (disease) makes sense for individuals.

Sometimes whether to use weights or not to use weights is a difficult question. Without a clear best answer we should carry out robustness checks, as usual. If the most important results are similar, we're fine – weights just don't matter. If, however, they turn out to be different, we need to understand the difference and make a choice accordingly.

Review Box 8.8 Using weights in regression analysis

- Different observations may have different weights for two reasons:
  - to compensate for unequal representation of the population (e.g., due to unequal sampling probabilities);
  - to denote different size of larger units when those larger units are the observations in the data (e.g., population of countries).
- It is good practice to use weights in regression analysis if those weights compensate for unequal representation. (This rarely matters, so it can be part of robustness checks.)
- Weights of size should be used if we want to uncover the patterns of association for the individuals who make up the larger units (e.g., people in countries), but not if we want to uncover the patterns of association between the larger units (e.g., countries).

8.B3 CASE STUDY – How is Life Expectancy Related to the Average Income of a Country?

Weights in the regression

Let's go back to cross-country association between GDP per capita and life expectancy. Recall that, when life expectancy (in years) is regressed on log GDP per capita, the slope coefficient is 5.3. This shows that life expectancy is around half a year higher on average in countries with 10% higher GDP per capita.

The same linear regression using population as weight gives a slope of 5.8. This shows that people who live in countries with 10% higher GDP per capita live, on average, 0.6 years longer. This interpretation is similar to the previous one but it is not quite the same. Whereas the unweighted regression compares countries, the regression weighted by population compares people living in

different countries. Thus people in larger countries are compared more often to other people than those in smaller countries.

Figures 8.9a and 8.9b show the two regression lines with the corresponding scatterplots. The scatterplot for the weighted regression shows the size of each country: the area of the circles is proportionate to their population.

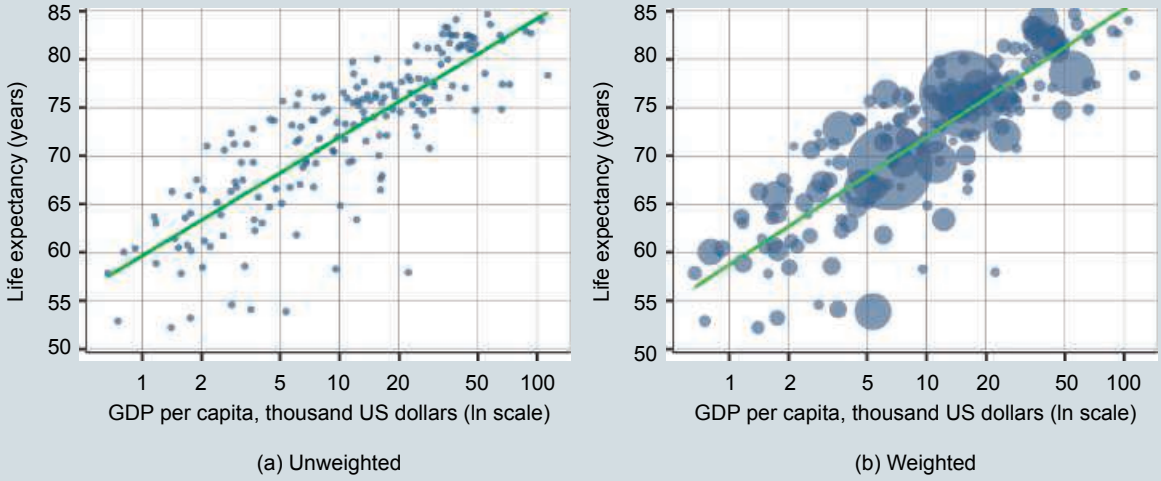


Figure 8.9 Life expectancy and log GDP per capita: unweighted and weighted regressions

Note: Life expectancy regressed on log GDP per capita. Regression without weights (panel a); regression weighted by population (panel b). GDP measured at Purchasing Power Parity.
Source: `worldbank-lifeexpectancy` dataset. World Development Indicators 2017. N=182.

The differences in principle notwithstanding, the actual slope estimates turn out to be similar. The R-squared values are similar, too, with 0.68 in the unweighted regression and 0.66 in the weighted regression. The two regressions show similar results because larger countries do not tilt the regression line much. For example, India and China are in the middle of the GDP per capita distribution so they do not affect the slope much when they are counted with their huge weights. As the weighted regression produces results that are similar to the unweighted regression, the conclusions that we drew at the end of Section 8.B2 remain the same.

8.14 Main Takeaways

Nonlinear functional forms may or may not be important for regression analysis.

- They are usually important for prediction.
- They are usually less important for causal analysis.
- When important, we have multiple options, such as taking logs or specifying linear splines or polynomials such as a quadratic.

Influential observations and other extreme values are usually best analyzed with the rest of the data; we should discard them only if we have a good reason.

PRACTICE QUESTIONS

1. When and why may it make sense to take the natural log of a variable to be used in a regression?
2. The slope parameter of a linear regression $(\ln y)^E = \alpha + \beta lnx$ is 0.5. Interpret this number.
3. The slope parameter of a linear regression $(\ln y)^E = \alpha + \beta x$ is 0.2. Interpret this number.
4. The slope parameter of a linear regression $y^E = \alpha + \beta lnx$ is 5. Interpret this number.
5. What kind of pattern could be captured by a regression with a single explanatory variable entered as a piecewise linear spline with one knot? Give an example.
6. The slope parameters of a linear regression $y^E = \alpha_1 + \beta_1 x[if\, x < 5] + (\alpha_2 + \beta_2 x)[if\, x > 5)$ are $\beta_1 = 2,\ \beta_2 = 1$. Interpret these numbers.
7. The slope parameters of a quadratic regression $y^E = \alpha + \beta x + \gamma x^2$ are $\beta = 2,\ \gamma = -0.1$. Interpret these numbers. Is the relationship positive between y and x?
8. A piecewise linear spline regression of y on x shows a steeper positive slope in the lower 90% of the range of the x variable and a flatter, but still positive slope, for the remaining 10%. What shape do you expect from a quadratic fitted on the same data? What are the advantages and disadvantages of these two specifications?
9. Observations with extreme values are also called outliers and are often discarded from the data before analysis. Is this good practice? If yes, why? If not, why not and what would be a better approach?
10. When interviewed for a household survey, people who drink large amounts of alcohol often tend to understate the frequency and quantity of their alcohol consumption. Can you think of the frequency of alcohol consumption variable in such data as a variable measured with error? Is it a classical error?
11. When interviewed for a household survey, people give rough and approximate answers to the question of how much time they spend commuting to work. However, those with longer commuting times are not more likely to understate or overstate their time than those with shorter times. Can you think of the commuting variable in such data as one measured with error? Is it a classical error?
12. You are interested in whether more educated people are different in terms of their commuting time to work than less educated people. Time to commute to work is measured with classical error in the data. What is the effect of this measurement error on the slope coefficient in a regression of commuting time on education?
13. You are interested in whether people who need to commute a lot to work earn more. Time to commute to work is measured with classical error in the data. The slope coefficient of a regression of log earnings on commuting time (in hours) is 0.01. Interpret this coefficient. Would it be smaller, larger, or the same if commuting time were measured without any error?
14. In a dataset on firms, the percentage of employees leaving a firm within a year is regressed on the average number of paid holidays employees receive. The slope coefficient is -0.1 in a regression without weights; it is -0.2 in a regression weighted by the number of employees. Interpret each number.

15. The amount of credit card debt is regressed on (log) household income. Compare the slope coefficient you expect if income is measured with substantial classical error to the slope coefficient you expect if income is measured without any error.

DATA EXERCISES

Easier and/or shorter exercises are denoted by [*]; harder and/or longer exercises are denoted by [**].

1. Continue the hotel price vs. distance to the city center regression analysis with log dependent variable but allowing for a piecewise linear spline in the explanatory variable, then a quadratic, and then a cubic. Compare the five best deals from the four specifications (linear, linear spline, quadratic, cubic). What do you conclude from this analysis? [*]
2. Re-do the hotel vs. distance to the city center regression analysis in a log–log specification. How do you interpret the coefficients of the simple linear regression? Then estimate corresponding regressions allowing for a piecewise linear spline in the explanatory variable, then a quadratic. Compare the five best deals from these specifications (linear, linear spline, quadratic). What do you conclude from this analysis? [*]
3. Re-do the cross-country regression analysis of life expectancy and ln GDP per capita for a different year. Use weights if you think that's a good choice, and use a functional form that allows for nonlinearity if you think that's a good choice. Compare your results to what's in our case study. [*]
4. Your task is to understand the pattern of association between how easy it is to do business in a country and how much income the country generates. Download data from the World Bank on GDP per capita in PPP (y) and the time required to start a business (x) for a recent year. Besides an average association, you are also asked to uncover if the association is linear or it is different at higher numbers of days. Estimate a regression that would measure such a potentially nonlinear pattern of association, interpret its coefficients. and draw conclusions. (During your analysis you may want to consider taking logs, inspecting and dealing with influential observations, experimenting with functional forms, using weights, and so on.) [**]
5. Download data on used cars of a specific brand and type, and analyze price vs. age of cars in order to find a good deal. Estimate a regression that fits the pattern of association well enough. Use the results from the best linear regression to list candidate cars for the best deal using the residuals of the regression. (During your analysis you may want to consider taking logs, inspecting and dealing with influential observations, experimenting with functional forms, and so on.) [**]

REFERENCES AND FURTHER READING

The influential study Preston (1975) analyzed the relationship between per capita GDP and life expectancy and the changes of this association with time.

8.U1 UNDER THE HOOD: DETAILS OF THE LOG APPROXIMATION

In this subsection we dig deeper into why and how log differences approximate relative, or percentage differences. Understanding these details helps appreciate the approximate nature of log differences and why they are better with small differences than larger differences.

For a number d that is small enough (say, 0.1 or –0.1), it is true that $\ln(1+d) \approx d$, then $(x_1 - x_0)/x_0 = d \approx \ln(1+d) = \ln(1+(x_1-x_0)/x_0) = \ln(1+x_1/x_0-1) = \ln(x_1/x_0) = \ln(x_1) - \ln(x_0)$.

The exact difference, using the exponential function is
$(x_1 - x_0)/x_0 = x_1/x_0 - 1 = \exp(\ln(x_1/x_0)) - 1 = \exp(\ln(x_1) - \ln(x_0)) - 1$.

Table 8.2 shows selected exact values for $(x_1 - x_0)/x_0$ and $\ln(x_1) - \ln(x_0)$.

Table 8.2 Tabulating log approximation

in the negative range		in the positive range		in the positive range, continued	
$(x_1 - x_0)/x_0$	$\ln(x_1) - \ln(x_0)$	$(x_1 - x_0)/x_0$	$\ln(x_1) - \ln(x_0)$	$(x_1 - x_0)/x_0$	$\ln(x_1) - \ln(x_0)$
−1.00	*n.a.*	0.00	0.00	1.00	0.69
−0.95	−3.00	0.05	0.05	1.05	0.72
−0.90	−2.30	0.10	0.10	1.10	0.74
−0.85	−1.90	0.15	0.14	1.15	0.77
−0.80	−1.61	0.20	0.18	1.20	0.79
−0.75	−1.39	0.25	0.22	1.25	0.81
−0.70	−1.20	0.30	0.26	1.30	0.83
−0.65	−1.05	0.35	0.30	1.35	0.85
−0.60	−0.92	0.40	0.34	1.40	0.88
−0.55	−0.80	0.45	0.37	1.45	0.90
−0.50	−0.69	0.50	0.41	1.50	0.92
−0.45	−0.60	0.55	0.44	1.55	0.94
−0.40	−0.51	0.60	0.47	1.60	0.96
−0.35	−0.43	0.65	0.50	1.65	0.97
−0.30	−0.36	0.70	0.53	1.70	0.99
−0.25	−0.29	0.75	0.56	1.75	1.01
−0.20	−0.22	0.80	0.59	1.80	1.03
−0.15	−0.16	0.85	0.62	1.85	1.05
−0.10	−0.11	0.90	0.64	1.90	1.06
−0.05	−0.05	0.95	0.67	1.95	1.08
0.00	0.00	1.00	0.69	2.00	1.10

8.U2 UNDER THE HOOD: DERIVING THE CONSEQUENCES OF CLASSICAL MEASUREMENT ERROR

In this subsection we formally derive the consequences of classical measurement error in variables for the OLS estimates of the slope and intercept parameters of the simple linear regression.

Classical measurement error in the dependent variable does not affect the regression coefficients. To see this, compare the slope of a simple linear regression with an error-free dependent variable y^* to the slope of the same regression in which the dependent variable y is measured with classical error:

$$y = y^* + e \tag{8.5}$$
$$y^{*E} = \alpha^* + \beta^* x \tag{8.6}$$
$$y^E = \alpha + \beta x \tag{8.7}$$

The slope coefficient in the first regression with error-free dependent variable is

$$\beta^* = \frac{Cov[y^*, x]}{Var[x]} \tag{8.8}$$

The coefficient in the second regression with a dependent variable measured with classical error is

$$\beta = \frac{Cov[y, x]}{Var[x]} \tag{8.9}$$

The two are equal because the measurement error e is uncorrelated with all relevant variables, including x, so that $Cov[e, x] = 0$.

$$\beta = \frac{Cov[y, x]}{Var[x]} = \frac{Cov[(y^* + e), x]}{Var[x]} = \frac{Cov[y^*, x] + Cov[e, x]}{Var[x]} = \frac{Cov[y^*, x]}{Var[x]} = \beta^* \tag{8.10}$$

Classical measurement error in the left-hand-side variable makes the slope coefficient unchanged because the expected value of the error-ridden y is the same as the expected value of the error-free y.

In contrast, classical measurement error in a right-hand-side variable makes the regression coefficient estimates different. Here we need to compare the slope of a simple linear regression with an error-free right-hand-side variable x^* to the slope of the same regression in which the right-hand-side variable x is measured with classical error:

$$x = x^* + e \tag{8.11}$$
$$y^E = \alpha^* + \beta^* x^* \tag{8.12}$$
$$y^E = \alpha + \beta x \tag{8.13}$$

The slope coefficient in the first regression with error-free dependent variable is

$$\beta^* = \frac{Cov[y, x^*]}{Var[x^*]} \tag{8.14}$$

The coefficient in the second regression with a dependent variable measured with classical error is

$$\beta = \frac{Cov[y, x]}{Var[x]} \tag{8.15}$$

These two are different in the presence of classical measurement error. The slope coefficient in the regression with an error-ridden right-hand-side variable is smaller in absolute value than the slope coefficient in the corresponding regression with an error-free right-hand-side variable:

$$\beta = \beta^* \frac{Var[x^*]}{Var[x^*] + Var[e]} \tag{8.16}$$

The derivation follows:

$$\begin{aligned}\beta &= \frac{Cov[y,x]}{Var[x]} = \frac{Cov[y,(x^*+e)]}{Var[x^*+e]} = \frac{Cov[y,x^*]+Cov[y,e]}{Var[x^*]+Var[e]} = \frac{Cov[y,x^*]}{Var[x^*]+Var[e]} \\ &= \frac{Cov[y,x^*]}{Var[x^*]}\frac{Var[x^*]}{Var[x^*]+Var[e]} \\ &= \beta^* \frac{Var[x^*]}{Var[x^*]+Var[e]}\end{aligned}$$

Technically, attenuation bias occurs because the error inflates the variance in the right-hand-side variable and the slope coefficient normalizes the (unaffected and thus unbiased) covariance with this inflated variance.

If the slope is attenuated towards zero, the intercept is tilted away from zero. Recall that the intercept equals average y minus average x multiplied by the slope coefficient. Without measurement error,

$$\alpha^* = \bar{y} - \beta^* \overline{x^*} \tag{8.17}$$

With measurement error,

$$\alpha = \bar{y} - \beta \bar{x} \tag{8.18}$$

Classical measurement error leaves expected values unchanged so we can expect

$$\bar{x} = \overline{x^*} \tag{8.19}$$

The regression line on error-free variables with slope β^* and the regression line on erroneous variables with slope β go through the same $(\bar{x}, \bar{y})$ point. Classical measurement error in x makes β attenuated relative to β^*. That means a flatter regression line. With x and y positive only (and thus the $(\bar{x}, \bar{y})$ point in the upper right quadrant of the coordinate system) and a positive slope, an attenuated slope leads to an upward biased intercept. Depending on where the point is and whether the slope is negative or positive, the intercept with an error-ridden x can be higher or lower than the intercept with an error-free x.

We can derive that the difference in the two intercepts depends on the sign of the slope as well as the sign of average x:

$$\begin{aligned}\alpha &= \bar{y} - \beta\bar{x} = \bar{y} - \beta\overline{x^*} = \bar{y} - \beta\overline{x^*} + \beta^*\overline{x^*} - \beta^*\overline{x^*} = \alpha^* + (\beta^* - \beta)\overline{x^*} \\ &= \alpha^* + \left(\beta^* - \beta^* \frac{Var[x^*]}{Var[x^*]+Var[e]}\right)\overline{x^*} = \alpha^* + \beta^* \frac{Var[e]}{Var[x^*]+Var[e]}\overline{x^*}\end{aligned}$$

9 Generalizing Results of a Regression

How to construct confidence intervals, prediction intervals, and tests in linear regression analysis to uncover general patterns represented by the data, and how to think about external validity

Motivation

You want to find out if the earnings of women and men tend to be different in your country in the occupation you are considering: market analysts. Analyzing data from a random sample of market analysts in the USA, you find that women earn less, by 11 percent on average. How much gender difference can you expect among all market analysts in the USA? In particular, is there a difference in the population, or is it just a chance event true in your data? And can you generalize these results to the future or to other countries?

Using the same data, you uncover that earnings tend to be higher for older market analysts than younger ones, but the association is not linear. In particular, the age differences are large and positive between older versus younger employees if both are below age 30; they are smaller but still positive if both are between 30 and 40; they are zero or slightly negative if comparing employees over age 40. How can you tell whether, and by how much, the nonlinear association you estimated in your sample generalizes to the population of all market analysts? How could you visualize the shape of the expected association in the population based on the shape you estimated in your data? And, can you generalize these results to learn what earnings trajectory you could expect in the future among market analysts?

Most often, we want to generalize the results of a regression from the data we are analyzing to a decision situation we care about. We introduced the relevant conceptual framework in Chapters 5 and 6 for simpler statistics. The same framework applies to regression coefficients, and we can build on the methods we introduced there. In particular, we can use methods to quantify the uncertainty brought about by generalizing to the general pattern represented by the data (statistical inference), and we can have guidelines to assess whether the general pattern represented by the data is likely close to the general pattern behind the situation we care about (external validity).

We start by describing the two steps of generalization in the context of regression analysis: statistical inference and external validity. Then we turn to quantifying uncertainty brought about by generalizing to the general pattern represented by our data. We discuss how to estimate the standard errors and confidence intervals of the estimates of the regression coefficients, how to estimate prediction intervals, and how to test hypotheses about regression coefficients. We introduce ways

to visualize the confidence interval and the prediction interval together with the regression line, and we introduce the standard way to present the results of regression analysis in tables.

The first, and main, case study in this chapter, **Estimating gender and age differences in earnings**, uncovers earnings differences by gender and age among people employed as market analysts, using standard survey data from the United States. Learning such differences may help young adults predict what kinds of differences in earnings they can expect. This case study uses the `cps-earnings` dataset. The second case study, **How stable is the hotel price–distance to center relationship?**, is a short illustration of potential issues with external validity. It uses the `hotels-europe` dataset to explore how the coefficients of the price–distance regression in one city for one day compare to the estimates on other days and in other cities.

Learning outcomes

After working through this chapter, you should be able to:

- estimate standard errors of regression coefficients in an appropriate way;
- construct and interpret confidence intervals of regression coefficients;
- construct and interpret prediction intervals;
- carry out and interpret tests of hypotheses about regression coefficients;
- present results of regression analysis in standard and easily interpretable ways;
- think about the external validity of a regression analysis.

9.1 Generalizing Linear Regression Coefficients

With the help of a simple linear regression $y^E = \alpha + \beta x$, we can uncover $\hat{\beta}$, the average difference of y corresponding to unit differences in x in the data. The regression coefficients calculated within the data allow for constructing predicted values. $\hat{y}_i$ is our best guess for expected y for an observation i with value x_i – based on the regression coefficients that characterize patterns in the data.

Sometimes all we care about are patterns, predicted values, or residuals in the data we have. An example is our case study of finding a good deal among hotels in the data. Most often, though, we are interested in patterns and predicted values in situations that are not contained in the data we can analyze. That's true for all other case studies in this textbook. There the question is whether and to what extent predictions and patterns uncovered in the data generalize to a situation we care about. This is the question of generalization that we covered in Chapter 5 earlier, applied to regression analysis.

As we argued there, answering this question is best broken down into two steps. One step is **statistical inference**: generalizing from the data to the population, or general pattern, it represents. Our best tool here is the confidence interval (Chapter 5, Section 5.4), but sometimes we carry out more formal hypothesis testing (Chapter 6). The other step is about **external validity**. We have to define the population, or general pattern, we are interested in and compare it to the population, or general pattern, that the data represents. If we conclude that external validity is high, the confidence interval is the end of the story. If external validity is not that high, then the confidence interval we estimated does not capture all uncertainty, and we may consider wider ranges – how much wider is hard to tell, unfortunately. If external validity is very low, we may not learn anything meaningful from

our data for the situation we care about. It's perhaps best to see if our case is that bad early in the analysis so we don't do the entire analysis for nothing.

Applying these principles to the results of regression analysis is straightforward conceptually, but it involves some new techniques. Typically, there are two statistics we want to generalize from regression analysis: the slope coefficient β that summarizes the pattern of differences in average y corresponding to differences in x and the predicted value of $\hat{y}$ for specific values of x. Often, we want to test hypotheses about the coefficients. All of these questions are discussed in this chapter.

9.2 Statistical Inference: CI and SE of Regression Coefficients

With a simple linear regression $y^E = \alpha + \beta x$, the estimate $\hat{\beta}$ shows the average difference in y corresponding to one unit difference in x in the data. The question of statistical inference is the true value of β. The **true value** means its value in the population, or general pattern, represented by the data. The **confidence interval of the regression coefficient** can answer that question with the caveats we discussed in Chapter 5, Section 5.5. The 95% CI of β can tell us where its true value is with 95% likelihood. Its 99% CI can tell us where its true value is with 99% likelihood. And so on, with confidence intervals for different percentage likelihood.

Estimating the 95% CI of the slope coefficient β of a linear regression is similar to estimating a 95% CI of any other statistic. We take the value of the statistic estimated from the data and measure twice its standard error around it. To be precise, the 95% CI value is 1.96. However, as all of this is an approximation, we prefer the round number 2. For the slope of the linear regression, the 95% CI is

$$CI(\hat{\beta})_{95\%} = [\hat{\beta} - 2SE(\hat{\beta}), \hat{\beta} + 2SE(\hat{\beta})] \tag{9.1}$$

As always, the CI is narrower the smaller the SE, in which case we also say that the estimate of the regression coefficient is **more precise**. A larger SE and thus a less precise estimate leads to a wider CI.

The standard error (SE) of the slope coefficient is conceptually the same as the SE of any statistic. It measures the spread of the values of the statistic across hypothetical repeated samples drawn from the same population, or general pattern, that our data represents – as discussed in detail in Chapter 5, Section 5.2. Just as with the standard error of an average, there are two ways to get the standard error of the slope of a linear regression: bootstrap and formula.

Recall that the bootstrap is an intuitive method to get standard errors. It generates samples that represent the same population, or general pattern, as our original data and have the same size, too. Each bootstrap sample is a random sample from the original data with replacement. After estimating the slope coefficient $\hat{\beta}$ in each sample, we can look at its distribution. The standard deviation of this distribution is the bootstrap estimate of the standard error (SE) of the slope coefficient. The simple bootstrap procedure we discussed in Chapter 5 works here as well.

There are various versions of the SE formula for the slope of the linear regression, and they are not equally good. Here we show what is called the **simple SE formula**. The simple SE formula assumes that variables are independent across observations, which is likely true in most cross-sectional data. However, the formula assumes something else, too, called homoskedasticity, which may not be a good assumption in many cases. We'll discuss this second assumption later. Thus, this formula is not correct in many applications. So the simple SE formula turns out to be not a good formula in general. We discuss it, nevertheless, because it is the simplest formula that contains the most important elements

of other, more complicated SE formulae for a slope coefficient. The simple SE formula of the slope is

$$SE(\hat{\beta}) = \frac{Std[e]}{\sqrt{n}Std[x]} \tag{9.2}$$

The formula contains three elements:

1. $Std[e]$, the standard deviation of the regression residual ($e = y - \hat{\alpha} - \hat{\beta}x$);
2. $Std[x]$, the standard deviation of the explanatory variable;
3. $\sqrt{n}$, the square root of the number of observations in the data.

A smaller standard error translates into a narrower confidence interval, which pins down the true value of the slope coefficient with more precision. The simple SE formula shows that the SE is smaller, the smaller the standard deviation of the residual, the larger the standard deviation of the explanatory variable, and the more observations there are in the data. There is a corresponding formula for the intercept; see Under the Hood section 9.U1 for the details.

As we indicated, the simple SE formula is not correct in general. It relies on the assumption that the fit of the regression line is the same across the entire range of the x variable, an assumption called **homoskedasticity**. In general, though, the fit may differ at different values of x, in which case the spread of actual y around the regression line is different for different values of x, a phenomenon called **heteroskedasticity**.

There is a SE formula that is correct whether the situation is one of homoskedasticity or of heteroskedasticity. That is called the **robust SE formula**, also known as the "White" or "Huber" formula (for the two statisticians who uncovered it independently), or the heteroskedasticity-robust SE formula. This formula shares the same properties as the simple formula in that it is smaller the smaller $Std[e]$, the larger $Std[x]$, and the larger $\sqrt{n}$.

Most statistical software can calculate both the simple and the robust SE of the regression coefficients. Typically, the default choice of statistical software is to show the simple SE estimate. However, this is not the best default. The robust SE is correct both under heteroskedasticiy and homoskedasticity; the simple SE is correct only under homoskedasticity. (In small samples, the robust SE may be worse.) Therefore, in reasonably large samples, it is good practice to consider the robust SE as a default choice. Note that, most often, this choice does not make much of a difference, as the simple and robust SE estimates tend to be very similar. But sometimes it does, and then the robust SE tends to be the better choice.

Review Box 9.1 Statistical inference for regression coefficients

- Consider the simple linear regression: $y^E = \alpha + \beta x$ with coefficient estimates in the data $\hat{\alpha}$ and $\hat{\beta}$.
- We are interested in the value of β in the population, or general pattern, of interest.
- The question is best broken down into two parts:
  1. Statistical inference: what is the value of β in the general pattern represented by the data if we know its estimated value $\hat{\beta}$?
  2. External validity: how close is the general pattern of interest to the general pattern represented by the data?

- The most useful tool of statistical inference is the confidence interval (CI). The 95% CI around $\hat{\beta}$ gives a range in which we can expect β to lie with a 95% confidence in the general pattern represented by our data.
- The 95% CI around $\hat{\beta}$ is $\hat{\beta} \pm 2SE(\hat{\beta})$.
- We can estimate $SE(\hat{\beta})$ by bootstrap or a formula.
- The simple SE formula for the slope estimate is

$$SE(\hat{\beta}) = \frac{Std[e]}{\sqrt{n}Std[x]}$$

- Accordingly, the SE of the slope estimate is smaller the better the fit of the regression, the larger the sample, and the larger the spread of the x variable.
- In practice, we should use the robust SE formula, not the simple SE formula.

9.A1 CASE STUDY – Estimating Gender and Age Differences in Earnings

Question, data, confidence interval

Do women earn the same as or less than men in jobs that many students consider taking after their studies in data analysis? And how much more do older employees earn than younger employees? Answering these questions helps set the expectations of students about their future salaries. It also helps answer a bigger question: are there systematic differences in earnings by gender and age of employees in the same occupation? And, if yes, why?

The case study in this chapter examines patterns of earnings by gender and age. We will first uncover patterns in the data. But our question relates to a situation outside our data: what to expect in the future? This generalization involves both of the usual steps: do statistical inference to learn what the average difference may be in the population represented by the data, and assess external validity to understand how that average difference may look like in the future situation we care about.

In this chapter we focus on employees in a particular occupation category: "market research analysts and marketing specialists". We use the `cps-earnings` dataset. It is part of the Current Population Survey (CPS) of the USA, a major survey that provides federal statistics on employment, unemployment, and earnings and allows for in-depth research on many related topics. The so-called morg sample of the CPS has earnings data. The CPS is a sample, although a large one. It is a random sample aimed at representing the adult population of the USA. The data we use is from 2014.

The 2014 CPS earnings data has 317 056 observations, each observation being an individual of age 16 through 85. We kept all respondents between age 16 and 64 who were employed but not self-employed, reported a wage that was more than zero, and reported working more than zero hours in the previous week.

Out of the 317 056 in the 2014 survey, there are 254 904 working age individuals, 175 661 of whom were in employment. 367 individuals reported zero wage or zero hours worked, and 26 345

had missing or zero earnings and/or worked hours in the data. That's 15% of all observations. It turns out that whether earnings or hours are missing (or zero) is weakly associated with some demographic variables (gender, race), but it's not associated with education. Thus, while this is not a completely random phenomenon, it doesn't appear to lead to a strong selection bias, so we decided to simply exclude the observations missing data for our analysis. The size of this sample is 149 316 individuals.

In this sample, 281 individuals are in the "market research analyst and marketing specialist" occupation, 61% of whom are women. The dependent variable of the analysis is the natural logarithm of earnings per hour, which we denote as $\ln w$. We calculated this by dividing weekly earnings by usual hours worked. 63% of the sample reports 40 hours per week, and an additional 17% reports 41–50 hours. It turns out that weekly earnings are capped in the CPS: if its value is above a threshold level, the data shows the threshold level for earnings and not the true higher value. This is due to confidentiality reasons, to prevent potential identification of very high-earning respondents. The threshold value was USD 2 285 in 2014, and it affects only 5% of the sample of market analysts. We'll ignore this issue for our analysis and take the capped values as actual earnings. The distribution of earnings per hour is distributed with a long right tail; the distribution of its natural log is more symmetric and quite close to normal.

We regressed log earnings per hour on a binary variable that is one if the individual is female and zero if male:

$$(\ln w)^E = \alpha + \beta \times female \tag{9.3}$$

This is a log–level regression. The slope shows average differences in relative wages by gender in the data: women earn $\beta \times 100$ percent more than men, on average. The regression estimate is -0.11: female market analysts earn 11 percent less, on average, than male market analysts in this data.

Based on these estimates, what can university students expect about the gender gap in pay if they choose to become market analysts?

First, statistical inference. Our data is a random sample of all market analysts working in the USA in 2014. The CPS is a high-quality sample with careful random sampling and high response rates in general. Earnings or hours are missing for 15% of the respondents, but that's not strongly related to variables that matter for earnings. Thus, we treat it as missing at random, and so we treat the part of the data with valid earnings and hours measures as a random sample of the original sample. We can therefore use the standard tools of statistical inference to estimate the standard error and then the confidence interval for the estimated slope coefficient of -0.11. We computed three alternatives for the standard error and the corresponding confidence interval. First by bootstrap, second using the simple standard error formula, and third using the robust standard error formula. Of these three, the second, the simple formula, may be wrong, while the bootstrap and the robust SE should both be fine.

Second, external validity. Students are interested in the future not the past. Whether gender differences will follow the same general pattern in the future as in the recent past is impossible to know. However, a more thorough analysis may shed light on this question. For example, we may analyze gender differences in earnings in many years in the past. Such analysis would reveal that gender differences in earnings have been relatively stable over those years. Therefore, we may assume that they will be similar in the future as well, at least in the near future. Students outside

the USA should also ask if gender differences in the USA are close to what they would be in their countries. Again, some research may help assess this question. We may not have access to average earnings by gender and occupation to make direct comparisons among market analysts across countries, but average earnings by gender among all employees are available in most countries (typically, estimated from surveys). If, for example, earnings differences by gender are similar, in general, in our country to those surveyed in the USA, we may assume that they are likely similar among market analysts, too.

Figure 9.1 shows the histogram of the slope estimates from the 1000 bootstrap samples we took, with a vertical line denoting −0.11. The bootstrap SE is 0.06, and the 95% CI is [−0.23,0.01]. The simple SE and the robust SE estimates are also 0.06, implying the same 95% CI, [−0.23,0.01] (there are minor differences in the third digit). Thus, the two estimates that should be similar (bootstrap and robust SE) are similar indeed. The simple SE is similar, too; while it may be off in principle, it turns out just fine here.

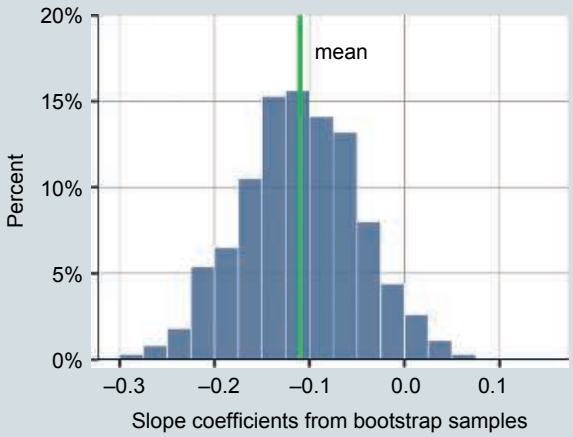


Figure 9.1 Bootstrap distribution of the average female–male wage difference among market analysts

Note: Histogram of 1000 bootstrap sample estimates of the slope in $(\ln w)^E = \alpha + \beta \times female$.
Source: `cps-earnings` dataset. USA, 2014. Employees in "market research analyst and marketing specialist" occupations, N=281.

Interpreting the 95% CI, we can say that in 2014 in the USA (the population represented by the data), we can be 95% confident that the average difference between hourly earnings of female market analysts versus male market analysts was −23% to +1%. This confidence interval includes zero. Thus we can't rule out with 95% confidence that their average earnings are the same. More likely, though, female market analysts earn less in the population as a large part of the CI is negative.

Note that in the case study, our confidence interval is rather wide. As we discussed, the sample size affects the standard error, and in the case study we had 281 observations only. If, instead of market research, we considered computer science occupations with 4740 observations, our slope coefficient estimate would be −0.15 with a smaller standard error of 0.018 and narrower 95% CI

of [−0.185, −0.115]. As a data exercise you will be asked to reproduce the whole case study for an occupation other than market research analysts.

9.3 Intervals for Predicted Values

One of the two main goals of regression analysis is prediction. There the task is predicting the value of y for observations outside the data, for which only the value of x is known. To predict y with the help of x using linear regression, we need to know the coefficients α and β. To be more precise, we need to know their values in the general pattern that is relevant for the situation where we want to predict y. This is the usual problem of generalizing statistics from a dataset, with its two steps: statistical inference and external validity.

Using our data we can estimate the regression coefficients. Then, with their help, we can calculate the predicted value of y for a particular observation with known x. For an observation j with known value x_j this is $\hat{y}_j = \hat{\alpha} + \hat{\beta}x_j$. The question we ask here is how we can quantify and present the uncertainty corresponding to this specific predicted value, $\hat{y}_j$. The answer is producing an interval around it, with some level of confidence.

It turns out that there are two kinds of intervals to consider for predicted values. The first one is the **confidence interval (CI) of the predicted value**, also called the **CI of the regression line**. The second one is called the **prediction interval** for an observation, y_j. Let's start with the first one.

The predicted value $\hat{y}$ combines the estimated values $\hat{\alpha}$ and $\hat{\beta}$. Thus the CI of the predicted value combines the CI for $\hat{\alpha}$ and the CI for $\hat{\beta}$. It answers the question of where we can expect y^E to lie if we know the value of x and we have estimates of coefficients $\hat{\alpha}$ and $\hat{\beta}$ from the data. Similarly to the CI formulae discussed earlier, the 95% CI of the predicted value is the value estimated from the sample plus and minus its standard error. We call this estimate the **standard error of the predicted value** y and denote it by $SE(\hat{y}_j)$:

$$95\%CI(\hat{y}_j) = \hat{y}_j \pm 2SE(\hat{y}_j) \tag{9.4}$$

$SE(\hat{y}_j)$ can be estimated using bootstrap or an appropriate formula. As usual, we need the robust SE formula in practice. However, as usual, that robust formula is quite complicated and difficult to understand. Therefore, we continue to show the simple formula, because it contains the most important ingredients of the more complicated robust formula and thus helps understand the intuition behind it. The simple formula for the SE of the predicted value from a simple linear regression is

$$SE(\hat{y}_j) = Std[e]\sqrt{\frac{1}{n} + \frac{(x_j - \bar{x})^2}{nVar[x]}} \tag{9.5}$$

One way of making sense of what's in this formula is to recognize that it has elements from the standard error of the slope estimate: $Std[e]$, n, and $Std[x]$ ($Std[x] = \sqrt{Var[x]}$). Thus, $SE(\hat{y}_j)$ is smaller, the smaller $Std[e]$, the larger n, and the larger $Std[x]$. These are the same things that make $SE(\hat{\beta})$ small. Thus, not surprisingly, the standard error of the predicted value is small if the standard error of $\hat{\beta}$ is small. There are two additional elements here, under the square root. The factor $\frac{1}{n}$ within the square root amplifies the role of the sample size: the larger n is, the smaller the SE. This extra element

comes from the SE formula of the intercept $\hat{a}$. The second element is new: $(x_j - \bar{x})^2$. It says that the standard error of predicted y_j depends on the corresponding value x_j. The closer that x value is to $\bar{x}$, the smaller the SE of the predicted value.

To summarize: the SE of the predicted value for a particular observation is small if the SEs of the coefficient estimates are small and the particular observation has an x value close to its average. The second part means that predictions for observations with more extreme x values are bound to have larger standard errors and thus wider confidence intervals.

The CI of the predicted value may be estimated for all kinds of regressions, not just for simple linear regression. That includes regressions with piecewise linear spline, quadratic or other polynomial, as well as non-parametric regressions including lowess. The general idea is the same: it is an interval that tells us where to expect average y given the value of x in the population, or general pattern, represented by the data. The way it is computed is different for different kinds of regressions, but it is always true that the CI is narrower, the smaller the $Std[e]$, the larger the n, and the larger the $Std[x]$.

The best way to use the CI of the predicted value is visualizing it, together with the regression line (or curve). The CI of the predicted value shows up as a band around the regression line, telling us where else the regression line could be besides where it is in the data, with 95% (or some other) chance, in the population, or general pattern represented by the data.

A frequent use of the CI of the predicted value is choosing the functional form for the $y^E = f(x)$ regression. We may compare two regressions that apply a different functional form for f, say a quadratic and a piecewise linear spline. Visualizing the estimated regressions together with their 95% confidence intervals is very informative. This visualization may show that the confidence intervals overlap over the entire range of x. That would imply that the quadratic is indistinguishable from the piecewise linear spline with a 95% chance in the population, or general pattern, represented by the data. Thus, the quadratic and the piecewise linear spline are very similar in capturing the unknown f function. Conversely, we may find that the two CIs don't overlap for a substantial part of the range. That would imply that the quadratic and the piecewise linear splines are different approximations for the pattern in the population, or general pattern, represented by the data.

Let's turn to the second kind of interval now, the one around predicted values: the prediction interval for y_j. This interval answers a different question: where to expect the particular y_j value if we know the corresponding x_j value and the estimates of the regression coefficients. Note the difference. The CI of the predicted value is about $\hat{y}_j$: the expected value if we know x_j. The prediction interval is about y_j itself: the actual value of y_j if we know x_j.

The prediction interval for y_j starts from the CI for $\hat{y}_j$ and adds the extra uncertainty due to the fact that the actual y_j will be somewhere around $\hat{y}_j$. To distinguish it from the confidence interval around $\hat{y}_j$, we denote the **prediction interval** for y_j as $PI(\hat{y}_j)$ and the corresponding standard error as the **standard prediction error**, $SPE(\hat{y}_j)$.

The formula for the 95% prediction interval is

$$95\%PI(\hat{y}_j) = \hat{y}_j \pm 2SPE(\hat{y}_j) \tag{9.6}$$

The simple formula for $SPE(\hat{y}_j)$ is instructive to look at for the same reason as the simple formulae earlier: it only contains the most important ingredients.

$$SPE(\hat{y}_j) = Std[e]\sqrt{1 + \frac{1}{n} + \frac{(x_j - \bar{x})^2}{nVar[x]}} \tag{9.7}$$

The formula has one, seemingly minor, additional element to $SE(\hat{y}_j)$: the plus one under the square root. It is as if we added $Std[e]$ to the previous formula. This additional part summarizes the additional uncertainty here: the actual y_j value is expected to be spread around its average value. The magnitude of this spread is best estimated by the standard deviation of the residual. In very large samples, the standard error for average y is very small. In contrast, no matter how large the sample, we can always expect actual y values to be spread around their average values. In the formula, all elements get very small if n gets large, except for this new element. We shall discuss the prediction error in more detail in Chapter 13.

Note that the additional assumption of homoskedasticity is explicit in the simple formula: it assumes that the spread of actual y around the regression line is $Std[e]$, the same everywhere. If this assumption fails, and the fit of the regression is not the same at different x values, the additional standard deviation will be different, too. (Appropriate robust formulae are difficult to obtain and standard statistical software do not contain such a formula in a simple fashion.) Thus, when the fit of the regression is very different across the range of x, we do not have a general formula to capture the prediction interval.

Review Box 9.2 Intervals for predicted values

Consider the simple linear regression estimates $\hat{y} = \hat{\alpha} + \hat{\beta}x$:

- The confidence interval (CI) of the predicted value (CI of the regression line)
  - tells us about where y^E is in the general pattern represented by the data if we know the value of x and we have estimates of coefficients $\hat{\alpha}$ and $\hat{\beta}$;
  - is narrower, the narrower the CIs of the regression coefficient (their SEs are small);
  - is narrower, the closer the value of x is to its average $\bar{x}$.
- The prediction interval
  - tells us about where the actual value of y_j is in the general pattern represented by the data if we know the value of x_j and we have estimates of coefficients $\hat{\alpha}$ and $\hat{\beta}$;
  - adds uncertainty to the CI of the predicted value: the actual y values are expected to be spread around the regression line.

9.A2 CASE STUDY – Estimating Gender and Age Differences in Earnings

Visualizing confidence and prediction intervals

Let's now turn to the pattern of association between earnings and age in our sample of 281 market analysts. Learning this age pattern may help early-career market analysts predict how they can expect their earnings to change throughout their career. How much we can learn about this question from uncovering the age–average earnings pattern in the data depends

on the usual issues plus an additional one here. The usual steps of generalization are statistical inference and external validity (the data is from the USA in 2014; students are interested in the future, possibly in another country). The additional issue is what we can learn from cross-sectional differences in age about what to expect about changes in earnings as individuals advance in age.

For illustration purposes, let us start by showing the confidence interval and the prediction interval for the simple linear model. Figure 9.2a shows the confidence interval. It's fairly narrow overall, but wider farther away from the sample mean. Figure 9.2b depicts the prediction interval, it is rather wide showing how hard indeed it is to make predictions for single observations.

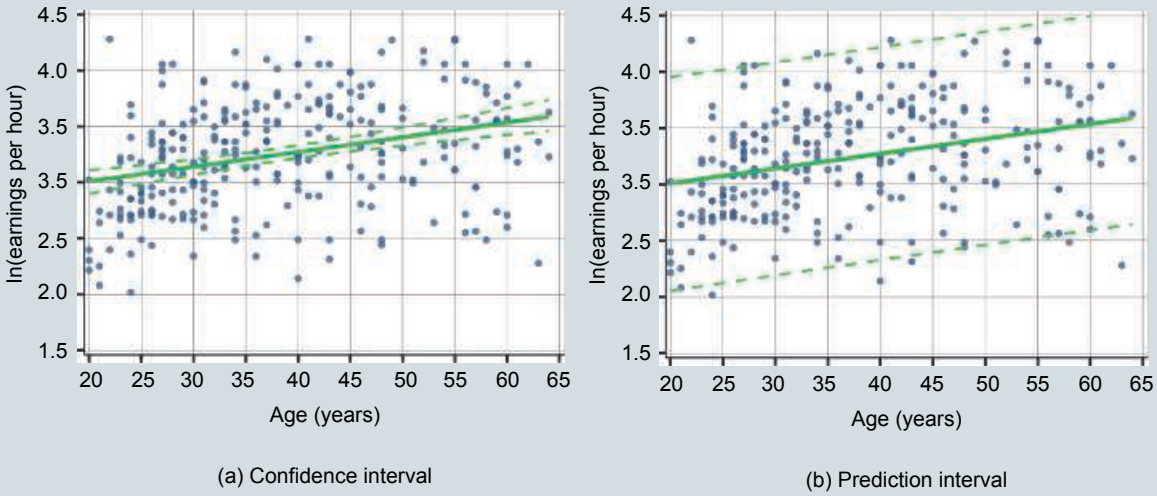
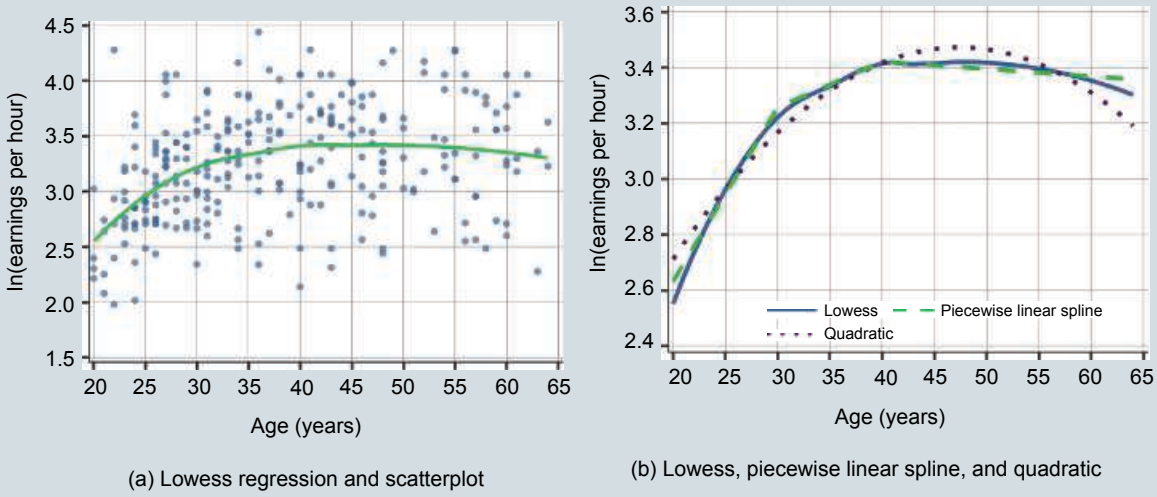


Figure 9.2 Log hourly wage and age: regression line, confidence interval, prediction interval.

Note: Market analysts. 95% confidence and prediction intervals.
Source: `cps-earnings` dataset. USA 2014. Employees in "market research and marketing specialist" occupations, N=281.

Using the same data on 281 market analysts, let's now turn to uncovering the actual shape of the $(\ln w)^E = f(age)$ pattern. To begin with, we estimated a lowess regression; the result is visualized in Figure 9.3a, together with the scatterplot. According to the pattern uncovered by lowess, earnings are substantially higher for older ages among young adults. Average log earnings is about 0.4 log unit (or 50%) higher for 25-year-old market analysts than for 20-year-old market analysts. The age differences are positive but get smaller and smaller as we compare older people: around 0.2 log units between age 25 and 30, 0.1 between age 30 and 35. There is little if any difference by age between 40 and 60, and those older than 60 earn less on average. The pattern is, thus, increasing in young age, flat in middle age, and slightly decreasing in old age, and concave throughout. It is an asymmetric upside-down U curve that is steeper in its positive part and a lot flatter in its negative part.

Figure 9.3b shows the same lowess curve as in Figure 9.3a together with two parametric regressions approximating it: a piecewise linear spline with knots at ages 30 and 40 and a quadratic model. To better see differences, we applied a different scale on the *y* axis.



(a) Lowess regression and scatterplot

(b) Lowess, piecewise linear spline, and quadratic

Figure 9.3 Log hourly wage and age: regressions that capture nonlinearity

Source: `cps-earnings` dataset. USA 2014. Employees in "market research analyst and marketing specialist" occupations, N=281.

Both the piecewise linear spline and the quadratic capture the increasing-then-decreasing concave pattern shown by the lowess, but they do that differently. The piecewise linear spline does not capture the smooth transition from steep to flat, and it shows a little higher average earnings than the lowess around age 30. But it does a good job of capturing the average slopes between the knots (not surprisingly as this is what splines are for). Moreover, it captures the very flat part above age 40. The quadratic captures the smooth change in slopes, and it is close to the lowess curve below age 35. But it is a little too steep between 35 and 45, and it is way too steep in the declining range above age 50. This is the result of the rigid upside-down U functional form of the quadratic: it has to be symmetric and thus as steep in the negative part as in the positive part, whereas the lowess suggests an asymmetric curve.

But how seriously should we take these differences across the regressions? That depends on whether we expect these regression lines or curves to be different in the population represented by the data. The best way to answer that question is comparing the CI of the predicted values, also known as the CI of the regression line.

Figure 9.4 shows the same three regressions as the previous one, but it adds the 95% confidence bounds to the piecewise linear regression and the quadratic regression. These confidence bounds show up with the same color and line pattern on the graph as the corresponding regression, only with a thin line.

The graph is a bit hard to read: it has three regression lines plus two CI bounds for two of the regressions. But its two main messages are not that hard to see. First, and most importantly, the lowess regression line is within the CI bounds of both the piecewise linear spline and the quadratic regression. Second, the quadratic is within the CI bounds of the piecewise linear spline, and, vice versa, the piecewise linear spline is within the CI bounds of the quadratic. (Well, almost: the kink of the piecewise linear spline at age 30 is just above the upper CI bound of the quadratic, but that one blip we ignore as the rest of the regression is very much inside.) Moreover, the confidence bounds around the quadratic and the piecewise linear spline overlap over the entire age range in the data.

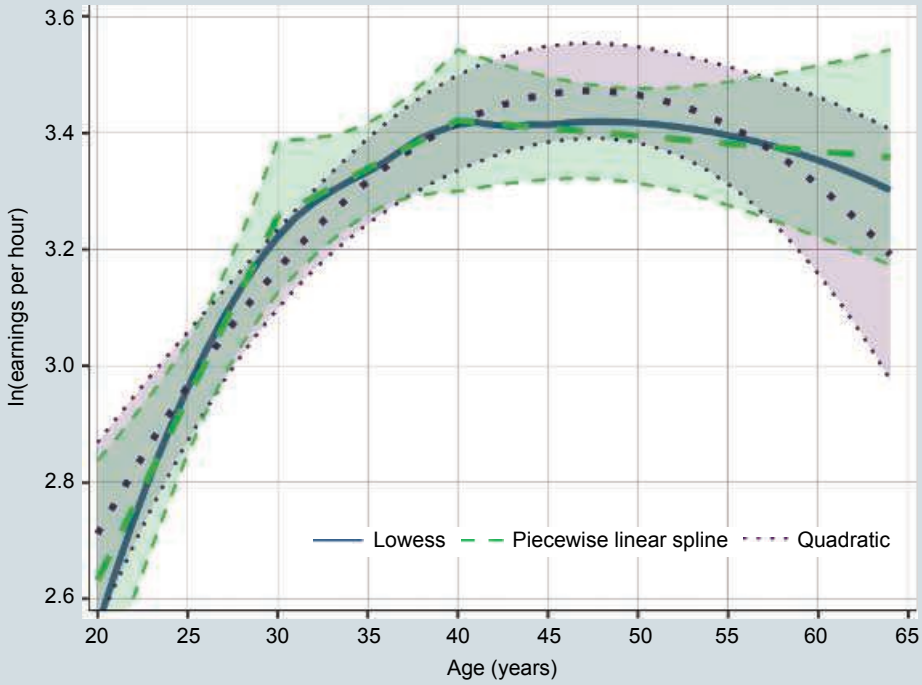


Figure 9.4 Average log earnings and age: regressions with CI

Note: Lowess regression; piecewise linear spline in age and its 95% CI; quadratic in age and its 95% CI. The CI are shown as dashed curves of the same color as their respective regressions.
Source: `cps-earnings` dataset. USA, 2014. Employees in "market research analyst and marketing specialist" occupations, N=281.

What should we conclude from these figures? The most important conclusion is that both the piecewise linear spline and the quadratic are pretty good at capturing the general pattern in the population represented by our data. If we wanted to represent the pattern in a parametric regression, both are equally good choices. They are, as is sometimes said, "statistically indistinguishable" from each other.

That conclusion may be due to the fact that we have relatively few observations: 281 people. The confidence bounds are wide. Many other kinds of patterns may fit into them. But all such patterns should be steep in younger ages and flat above age 40. These are the most important characteristics of the pattern anyway. With this much data, we simply cannot pin down the pattern more precisely.

Returning to the original question: what can students expect about how their earnings may change through their career if they choose to be market analysts? High external validity requires that the general patterns that determine age differences in earnings in 2014 remain stable for the coming decades. But even that is not sufficient to conclude that these cross-sectional age differences show how earnings are expected to change as individuals age; for that we need to assume that those cross-sectional differences show the effects of aging only, as opposed to mixing in how the content of education or initial labor market conditions differed for different birth cohorts of people. These are big ifs, and thus external validity is unlikely to be very high. For that reason it makes more sense to take away the qualitative findings as opposed to the numerical estimates

themselves. Even if external validity is not super high, we can expect earnings of market analysts to rise fast until age 30 and then more slowly until age 40 or 45. But above age 45 they aren't expected to increase as much, or at all, in this profession.

Analyzing additional data won't resolve all the issues with external validity, but it may help a little. We could, for example, look for panel data in which we can examine how earnings change as people age, instead of looking at the cross-section. It turns out that most analyses of such data reveal similar patterns between age and average earnings: concave shape with steep upward part and a mostly flat part afterwards. Of course, no data from the past can tell us if the future is going to be similar or very different. But, with those caveats, the results of this simple analysis are informative.

9.4 Testing Hypotheses about Regression Coefficients

Recall from Chapter 6 that testing hypotheses means making a statement about a general pattern and deciding whether we can reject it using information in the data. Such statements are called hypotheses, and in statistical testing hypotheses are about values of statistics. We specify a **null hypothesis** H_0 ("the null") and an **alternative hypothesis** H_A ("the alternative").

The logic remains the same when testing hypotheses about regression coefficients. An important question in regression analysis is whether y and x are mean-dependent. In the context of the linear regression $y^E = \alpha + \beta x$, this question is whether the true value of β is zero, which means its value in the population, or general pattern represented by the data. The null and alternative hypotheses are

$$H_0: \beta_{true} = 0 \tag{9.8}$$

$$H_A: \beta_{true} \neq 0 \tag{9.9}$$

The test is carried out with the help of the **t-statistic**. As we discussed in Chapter 6, Section 6.3, the t-statistic is best viewed as a measure of distance: how far the value of the statistic in the data ($\hat{\beta}$) is from its value hypothesized by the null (zero). The distance is measured in terms of the standard error (SE). A value 0 of the t-statistic means that the value of $\hat{\beta}$ is exactly what's in the null hypothesis (zero distance). A t-statistic of 1 means that the value of $\hat{\beta}$ is exactly one SE larger than the value in the null hypothesis. A t-statistic of -1 means that the value of $\hat{\beta}$ is exactly one SE smaller than the value in the null hypothesis. And so on.

With the null $H_0: \beta_{true} = 0$ and the alternative $H_A: \beta_{true} \neq 0$, the t-statistic is

$$t = \frac{\hat{\beta}}{SE(\hat{\beta})} \tag{9.10}$$

Another often investigated null states that the slope is of a particular value c, $H_0: \beta_{true} = c$ and the alternative $H_A: \beta_{true} \neq c$. Simply rewriting these hypotheses to $H_0: \beta_{true} - c = 0$ and the alternative $H_A: \beta_{true} - c \neq 0$, we can derive the t-statistic to be

$$t = \frac{\hat{\beta} - c}{SE(\hat{\beta})} \tag{9.11}$$

The t-statistic for the intercept coefficient is analogous.

To make a decision we apply the usual **critical value** to the t-statistic: ±2. We reject the null if the t-statistic is larger than 2 or smaller than −2, and we don't reject the null if it's in-between.

Alternatively, we can make a decision with the help of the appropriate **p-value**. This saves us from calculating t-statistics and thinking about critical values. As we have discussed in Chapter 6, Section 6.5, the p-value is the smallest significance level at which we can reject the null. Thus, for example, a p-value of 0.001 means that we are safe to reject the null if we can tolerate a 0.001 (0.1%) or larger chance of a false positive. We have to simply look at the p-value and decide if it is larger or smaller than the level of significance that we set for ourselves in advance. Following common practice, we set a 5% level of significance, and then we reject the null if the p-value is less than 0.05.

Most often, the hypothesis we test is H_0: $\beta_{true} = 0$. This amounts to the hypothesis that average y does not differ according to the value of x, which also means that y and x are uncorrelated. If this null is rejected at the 5% level of significance, x is said to be **statistically significant** at 5% in the regression. Statistical software shows the result of this among the default set of regression results. Sometimes we put a star next to the coefficients (or their SE) that are significant at 5%, and an additional star if significant at 1%.

As we explained in Chapter 6, Section 6.6, the general practice is to set a 5% critical value to avoid making biased decisions. But there may be very legitimate reasons to deviate from this rule, in either way, to require a more lenient criterion (higher level of significance) or a more conservative criterion (lower level of significance).

We may want to set a more lenient criterion if the purpose of our analysis is exploration or a **proof of concept**. Examples include when we use cross-country data to explore how various policies may affect CO_2 emission, or when we use time series data from our company to see if changes in marketing mix may affect sales. In many such instances, the size of the data is small, and there may be a great amount of noise in measurement. Thus, we are unlikely to get precise estimates, and we may not reject the null of no association at the 5% level (and the 95% CI of the slope coefficient would contain zero). But then we would risk missing something potentially important. If, instead, we adopt a more lenient criterion of a 10% or 15% level of significance, we are more likely to find a positive result and thus avoid a false negative. Of course we should acknowledge that our level of significance is higher than is conventional, with all of its potential consequences.

In contrast, we may want to have a more conservative criterion if we are looking for a **proof beyond reasonable doubt**. For example, if a firm is sued for violating gender equality rules, the judge would need convincing evidence that wages are indeed different from one another. In this scenario, we typically aim for a conservative critical value of 1%. If multiple testing is taking place, we should take an ever lower critical value. Once again, being explicit about the details of statistical inference and our assumptions is critical.

Review Box 9.3 Steps of testing hypotheses about regression coefficients

- Select the statistic to test. Most often it is β, the slope coefficient.
- State the null and alternative hypotheses. They are about the "true" value: the value of the coefficient in the population, or general pattern, represented by the data. Often: H_0: $\beta_{true} = 0$, H_A: $\beta_{true} \neq 0$.
- Calculate or read the appropriate t-statistic. With H_0: $\beta_{true} = 0$ it is $\frac{\hat{\beta}}{SE(\hat{\beta})}$.
- Choose a critical value (or values). Reject the null if the absolute value of the t-statistic is outside the range defined by the critical value(s); don't reject otherwise. Often, we choose ±2 for critical

values, corresponding to a 5% level of significance. But this choice may depend on the purpose of the work.
- Instead of the last two steps, we may look at the appropriate p-value. Reject the null if the p-value is smaller than the level of significance (say, 5%); don't reject if it is larger.
- Think external validity. (Compare the general pattern you are interested in to the general pattern your data represents.)

9.5 Testing More Complex Hypotheses

Some hypotheses are more complex than H_0: $\beta_{true} = 0$, H_A: $\beta_{true} \neq 0$. We consider three cases.

The first case is testing against a **one-sided alternative**. We do that when we are interested in whether the slope is negative or positive. The corresponding hypotheses here would be H_0: $\beta_{true} \geq 0$, H_A: $\beta_{true} < 0$. As discussed in Chapter 6, Section 6.7, the question that we are truly interested in should be in the alternative (e.g., whether women earn less on average), and the equality sign should be in the null. In practice, testing these hypotheses is similar to testing H_0: $\beta_{true} = 0$, H_A: $\beta_{true} \neq 0$, with two modifications. First, don't reject the null if the coefficient falls in the region that is specified in the null hypothesis (here $\hat{\beta} > 0$). Second, divide the p-value of the H_0: $\beta_{true} = 0$, H_A: $\beta_{true} \neq 0$ test by two and apply it for the decision if the coefficient falls in the direction of the alternative (here $\hat{\beta} \leq 0$).

The second case is hypotheses that involve more than one coefficient. An important example is whether two slope coefficients are the same. That question can be asked about slope coefficients in the same regression. For example, we may want to know if there is a difference in slopes between two parts of a piecewise linear spline. Alternatively, the question may be asked about slope coefficients in different regressions estimated on different data. Whether the slopes of, say, earnings on age are the same in different occupations may be asked after estimating the slopes in two regressions on two samples.

The intuition and principles of testing hypotheses that involve more than one regression coefficient are the same as the principles of testing hypotheses about a single regression coefficient. We have to set null and alternative hypotheses in such a way that the null has the form of "something equals zero." For example, when interested in whether $\beta_{1true} = \beta_{2true}$, the corresponding null should be

$$H_0: \beta_{1true} - \beta_{2true} = 0 \tag{9.12}$$

$$H_A: \beta_{1true} - \beta_{2true} \neq 0 \tag{9.13}$$

Then we have to estimate the SE of the estimated version of the statistic in the null: $\hat{\beta}_1 - \hat{\beta}_2$. With the SE in hand, we can calculate the appropriate t-statistic, say $\frac{\hat{\beta}_1 - \hat{\beta}_2}{SE[\hat{\beta}_1 - \hat{\beta}_2]}$. With the appropriate SE estimates, all follows as usual: reject the null if the t-statistic is outside the range specified by appropriate critical values, say ± 2. Of course, if we have the appropriate p-value, the decision is even easier. Estimating the appropriate SE may be tricky, but sophisticated statistical software can do that for us and can also report appropriate p-values.

In the third case the null is more than one statement; these are called **joint hypotheses**. One example of joint hypotheses is that all slope coefficients are zero in a piecewise linear spline regression.

With two such slopes, these hypotheses have the form

$$H_0: \beta_{1true} = 0 \ \& \ \beta_{2true} = 0 \tag{9.14}$$

$$H_A: \beta_{1true} \neq 0 \ or \ \beta_{2true} \neq 0 \tag{9.15}$$

Testing joint hypotheses has the same overall principle as before: estimate the test statistic, and compare the test statistic against the critical value that corresponds to a pre-specified level of significance. In practice, though, all this is more complicated for testing joint hypotheses. Luckily, we do not carry out the test this way. Instead, we make use of the p-value that statistical software produces. We estimate the regression with the appropriate standard errors, specify the null hypothesis to test, and ask the software to return the appropriate p-value.

9.A3 CASE STUDY – Estimating Gender and Age Differences in Earnings

Testing hypotheses about regression coefficients

We return to our question of whether women earn the same, on average, as men among market analysts, using our data on 281 market analysts from the USA in 2014 (the `cps-earnings` dataset). We regressed log hourly earnings on a binary variable that is one if the person is female and zero if male: $(\ln w)^E = \alpha + \beta \times female$. Recall that the coefficient estimate was -0.11, implying that women earn 11 percent less, on average, than men in this sample. The SE of the coefficient was 0.06, and the 95% CI was $[-0.23, +0.01]$ implying that we can be 95% confident that the difference in average earnings between women and men was between -23% and +1% among all market analysts in the USA in 2014. Notably, this interval contains zero. Thus, we cannot rule out with 95% confidence that average earnings are the same among women and men in this occupation (in the USA in 2014).

To formally test whether average earnings are the same by gender, we simply test if the coefficient on the binary variable is zero, against the alternative that it is not zero: $H_0: \beta_{true} = 0$ against $H_A: \beta_{true} \neq 0$. The regression output from software will typically show coefficient estimates together with their standard errors, as well as t-statistic and p-values for testing whether each coefficient is zero (against the alternative that it isn't zero). In our case, the corresponding t-statistic is -1.8. The critical values for a 5% significance level are ± 2, and -1.8 falls within the critical values not outside of them. Thus, we cannot reject the null of equal average earnings at a 5% level of significance. The same conclusion is implied by the p-value that the software presents: it is 0.07, which is greater than 0.05 that would correspond to a 5% level of significance.

After testing whether gender differences in earnings are zero in the population represented by our data, let's turn to a more complex hypothesis: whether the age–earnings pattern is the same between ages 20 and 30 versus 30 and 40. To answer this question, we consider the regression with piecewise linear spline in age. If the slopes of the first two splines are the same, the pattern is the same; if they are different, the pattern is different. This can be formulated as a hypothesis that involves two regression coefficients.

The results of the piecewise linear estimates give a slope coefficient of 0.06 for the first line segment (age 20 to 30) and 0.02 for the second line segment (age 30 to 40). These are different in the data. Do we have enough evidence to tell if they are different in the population as well?

One way to approach this question is by looking at the confidence intervals. The 95% CI around the first coefficient is [0.035,0.089]; the 95% CI around the second coefficient is [−0.003, +0.036]. The point estimates of the coefficients (0.06 and 0.02) are each outside of the CI of the other coefficient. At the same time, the two CIs overlap, if only barely. What should we conclude? Apparently, the CIs can't tell us that here.

We should carry out a formal test to decide. We asked the software to test whether the two coefficients are the same, after having estimated a regression with robust standard errors. The software returned a p-value of 0.03. That implies that we are safe to reject the null at a 5% level of significance. Thus we decide that the patterns are different in the population represented by the data. By doing so, we allow ourselves to be wrong with less than a 5% chance.

Unlike formal tests, confidence intervals are not always informative when testing hypotheses that compare regression coefficients. Checking whether each point estimate is within the other coefficient's confidence interval is too harsh: even if they fall outside, we may not want to reject the null that they are the same in the general pattern. Checking whether the confidence intervals overlap is too lenient: even if they overlap, we may want to reject the null that they are the same in the general pattern. We need to carry out formal tests in such cases.

Nevertheless, there are more straightforward situations when looking at the confidence intervals is enough. If the confidence intervals do not overlap, we can always conclude that the two coefficients are different. Conversely, if the point estimates are both within the CI of the other coefficients, we can always conclude that we cannot reject the null of equal coefficients. It is the occasions similar to our case study that are difficult: when the point estimates are outside each other's CI but the CIs overlap. In any case, carrying out formal tests is always a good solution.

9.6 Presenting Regression Results

The standard way to present the results of linear regression is in a table format, with coefficients and corresponding standard errors listed in one column. Usually, the standard errors are below the coefficient estimates, in parentheses or brackets. Typically, two lines at the bottom of the table show the number of observations and the R-squared of the regression. Column headings show the name of the dependent variable, or some information that differentiates the regressions in the same table.

In addition, stars are sometimes put as superscripts to the coefficients to indicate whether the coefficients are significant: that is, whether we can reject the hypothesis that each one is zero. Usually, one star is attached to coefficients that are significant at 5% (we can reject that they are zero at the 5% level), and two stars are attached to coefficients that are significant at 1%.

Some data analysts have strong views about whether and how such stars should be included in **regression tables**. Many argue that stars are misleading: they focus attention on whether a particular coefficient is zero or not in the population, even though that is not always the most important question. (In R, one package that creates regression tables with stars is called stargazer.) We do not

take a firm stand on this issue. Most software that produces standard regression tables includes stars by default; analysts should keep them or erase them according to their preference.

It is good practice to include notes at the bottom of regression tables that clarify things. If the table contains stars for levels of significance, then the notes should explain what they mean. The notes should also clarify what is shown in the parentheses: e.g., are they **robust standard error** estimates, or simple standard error estimates? Sometimes values of t-statistics are shown instead of standard errors (i.e., values of t-statistics that test whether the coefficient is zero). Yet other times the CI are shown instead of, or in addition to, the SE or the t-statistic. The information content of the two is the same since we can get one from the other. We prefer showing standard errors because they are easy to use to construct t-statistics for testing H_0: $\hat{\beta} = 0$ or H_0: $\hat{\beta} = c$ for any c value. It is also straightforward to use them to construct confidence intervals for any level of confidence (95%, 90%, 99%).

9.A4 CASE STUDY – Estimating Gender and Age Differences in Earnings

Presenting regression results

The standard regression table of the regression $(\ln w)^E = \alpha + \beta \times female$ from our data on 281 market analysts is presented in Table 9.1

Table 9.1 Wage and gender gap baseline regression

Variables	ln wage
Female	−0.11
	(0.062)
Constant	3.31**
	(0.049)
Observations	281
R-squared	0.012

Note: Robust standard error estimates in parentheses.
** p<0.01, * p<0.05.

Source: `cps-earnings` dataset. USA, 2014. Employees in "market research analyst and marketing specialist" occupations.

The leftmost column has the variable names, also indicated by its header. The column with numbers shows the results of the regression analysis. The header of the column contains the name of the dependent variable. Then we have the slope estimate of the variable of the regression ("female"), followed by its standard error in parentheses. The next pair of numbers is the intercept estimate and its standard error (variable name "**Constant**"). The bottom of the table shows the number of observations in the regression and the R-squared. Stars denote levels of significance at 5% and 1% levels. This table summarizes the results we discussed earlier: a −11% difference in average earnings in the data, with SE of 0.6 implying a confidence interval [−0.23, +0.01] and a test that we cannot reject the null of $\beta = 0$ at 5% or 1% because the p-value is greater than 0.05 (it's 0.07, not shown in the table).

Note that we set the table to show coefficient estimates up to two digits; standard error estimates are up to three digits. We show two digits for the coefficient estimates as more digits would convey a false impression of precision. Showing one more digit would tell us that the slope coefficient is −0.114 in the data. But that is not really informative. The confidence interval [−0.23, +0.01] swamps such differences. Even if the confidence interval was super narrow, it wouldn't make any meaningful difference to know that the average earnings difference is 11.4% not 11%. We show one more digit for the SE estimates because that third digit can be helpful when constructing confidence intervals of various confidence (95%, 90%, 99%).

This standard table format allows showing results of more than one regression at the same time. Table 9.2 shows results from three regressions using the same data, each specifying a different functional form to the $(\ln w)^E = f(age)$ regression. The first column is a linear specification, the second a quadratic, the third a piecewise linear spline with two knots, at ages 30 and 40. With smaller coefficients, we now use three digits.

Table 9.2 Wage and age – different specifications

Variables	(1) ln wage	(2) ln wage	(3) ln wage
age	0.014**	0.096**	
	(0.003)	(0.018)	
age squared		−0.001**	
		(0.000)	
age spline <30			0.062**
			(0.014)
age spline 30–40			0.017
			(0.010)
age spline 40<			−0.003
			(0.006)
Constant	2.732**	1.193**	1.383**
	(0.101)	(0.341)	(0.369)
Observations	281	281	281
R-squared	0.098	0.168	0.173

Note: Robust standard error estimates in parentheses. ** p<0.01, * p<0.05.
Source: `cps-earnings` dataset. USA, 2014. Employees in "market research analyst and marketing specialist" occupations.

The first column shows that a line fit on the scatterplot would have a slope of 0.014: market analysts that are one year older earn 1.4% more, on average, in this sample. We know that the pattern is nonlinear: increasing in a concave way up to middle age and flat, or even decreasing, above that. This average slope combines the steep slope at young age with the flat slope in middle and older ages. The average difference of 1.4% is what it is: the average difference. It may be informative for some purpose; it is not very informative for students if they want to uncover the

age pattern of earnings to learn about what to expect during their career. That's why we estimated two alternatives that capture the nonlinear pattern.

The second column shows the results of the quadratic specification. The coefficient of the linear term ("age") has no interpretation in itself; the coefficient of the quadratic term ("age square") is negative, indicating a concave pattern. This term is statistically significant, meaning that the concave pattern is very likely true in the population.

The third column shows the estimates of the piecewise linear spline with two knots, resulting in three line segments. The slope estimate of the first one ("age spline <30") is 0.06: among market analysts of age 20 to 30, those that are one year older earn 6% higher, on average. Its standard error implies a 95% confidence interval [0.03, 0.09]. It is significant at 1%: age and earnings are very likely correlated in the population (all market analysts in the USA in 2014 younger than 30). The slope estimate of the second line segment is 0.017, but its 95% CI includes zero [−0.003, 0.037]. The estimated slope of the third line segment ("age spline >40") is −0.003, practically zero. The slopes of the second and third line segments are not statistically different from zero: they may very well be zero in the population.

9.7 Data Analysis to Help Assess External Validity

As external validity is about generalizing beyond what our data represents, we can't assess it using our data. But analyzing other data may help. The most important question usually is β, the slope coefficient on x. As emphasized in Chapter 5, Section 5.8, the three common dimensions of generalization are time, space, and other groups.

If we want to generalize for our country but we have data from other countries only, we can estimate β from several countries and see if they differ substantially, or whether their differences have some systematic pattern. For example, we may want to know how much domestic prices change if the exchange rate changes substantially. We may not have enough data for the particular country we are interested in, but we may have estimates using data from several countries with various degrees of openness. Then, if the country we are interested in is among the more open ones, the coefficient estimates for such countries are more likely to be valid for this country.

Similarly, if we are worried about the fact that β may change in the future, we can try to estimate it for many years and see if it's stable or, if not, if its changes show a particular pattern. The same goes for groups: gender differences in earnings may be different for different occupations, and while we may not have enough data for market analysts, estimating the difference for some other occupations may shed light on how stable that difference is or, if not, whether there is a pattern in its variation.

9.B1 CASE STUDY – How Stable is the Hotel Price–Distance to Center Relationship?

External validity, other cities and time periods, beyond hotels

Let's go back to our example of hotel prices and distance to the city center. The goal of the analysis in the previous chapters was to find a good deal from among the hotels for the date contained in the data. External validity was not an issue there.

Here we ask a different question: whether we can infer something about the price–distance pattern for situations outside the data: different time, different city, or different kinds of hotels. Such a speculation may be relevant if we want to expand development services we offer for relatively low priced hotels. Or maybe we simply want to find a good deal in the future without estimating a new regression but taking the results of this regression and computing residuals accordingly.

Recall that our data is collected from a price comparison website. So far we have used the data we collected for Vienna for a particular autumn weekday. In the `hotels-europe` dataset, we have similar data for other dates and other European cities, too. We collected data for various dates. In the present analysis, we restricted our data to hotels with 3 to 4 stars for the analysis.

Let's start with generalizing to the future. Although all the price data were collected at a single point in time, we can use data on prices shown for a variety of future dates to assess how stable the price–distance pattern may be. The pattern we examine is linear regression of log price with a piecewise linear spline in distance with knot at 2 miles.

Table 9.3 shows results for four dates: November 2017 weekday (the date we used in our baseline example in Chapters 7 and 8); a November 2017 weekend; December 2017 holidays; June 2018 weekend. The coefficient estimates for the spline regressions show the slope separately for 0–2 miles and 2–7 miles.

Table 9.3 External validity – comparing dates

Variables	(1) 2017-NOV-weekday	(2) 2017-NOV-weekend	(3) 2017-DEC-holiday	(4) 2018-JUNE-weekend
Distance spline <2	-0.31**	-0.44**	-0.36**	-0.31**
	(0.038)	(0.052)	(0.041)	(0.037)
Distance spline 2–7	0.02	-0.00	0.07	0.04
	(0.033)	(0.036)	(0.050)	(0.039)
Constant	5.02**	5.51**	5.13**	5.16**
	(0.042)	(0.067)	(0.048)	(0.050)
Observations	207	125	189	181
R-squared	0.314	0.430	0.382	0.306

Note: Dependent variable is log price, distance is entered as spline with a knot at 2 miles. Robust standard error estimates in parentheses. ** p<0.01, * p<0.05.

Source: `hotels-europe` dataset. Vienna, reservation price for November and December 2017, June 2018.

Among hotels in the data that are within 2 miles from the city center, prices are 0.31 log units or 36% (exp(0.31)−1) cheaper, on average, for hotels that are 1 mile farther away from the city center both for the November weekday and the June weekend. The coefficient estimate is similar for December (−0.36) and June (−0.31). The coefficient is −0.44 for the November weekend, which suggests a stronger negative association with distance. The corresponding 95% confidence intervals overlap somewhat: they are [−0.39, −0.23] and [−0.54, −0.34], so they don't help in deciding whether price–distance patterns are different during the weekday and weekend in November. However, a test of equal coefficients produces a p value of 0.035, which tells us that we can reject that they are equal at a 5% level.

The second part of the piecewise linear spline has an estimated slope that is close to zero with CI including zero, for all dates. Thus, among hotels that are 2 miles or farther away from the city center, average prices are essentially the same regardless of how far away the hotels are. This holds for all dates.

The regression constants show the intercept estimates, which don't have a good interpretation because this is a log–level regression. But we can compare them across regressions, which tells us differences in expected hotel price in the very center of the city. The results suggest that prices are almost 50 log points (60%) higher during the November weekend than on the November weekday, and they are 15% higher during the June weekend.

So it seems that guests are more willing to pay extra to be close to the city center during the November weekend, and perhaps also during the December holidays. Why do we see these differences? We can only speculate. Perhaps the weekend makes distance from the city center more undesirable for tourists when the weather is cold.

Before we conclude from the comparisons across time, let's pause for an important technical issue. When comparing regression results across time, we should always be careful with the observations we actually compare. Here the question is whether the regressions for the different dates are estimated for the same hotels. Looking at the number of observations, we see that there must be some differences: there are more hotels for the November weekday in the data than for the November weekend. (Even if the number of observations was the same, the hotels themselves may be different.)

This difference could be important: hotels with available rooms at a weekend or during a holiday could be different in a non-random way. Variation in demand for hotel rooms makes available hotels vary by dates. This is an illustration of potential selection bias due to a non-random sampling, a concept we introduced in Chapter 1, Sections 1.7 and 1.8.

To compare apples to apples, we re-estimated the same three regressions with hotels that are observed at all four times. Table 9.4 shows the results.

Table 9.4 External validity – comparing dates 2

Variables	(1) 2017-NOV-weekday	(2) 2017-NOV-weekend	(3) 2017-DEC-holiday	(4) 2018-JUNE-weekend
Distance spline <2	−0.28**	−0.44**	−0.40**	−0.28**
	(0.058)	(0.055)	(0.045)	(0.053)
Distance spline 2–7	−0.03	−0.02	−0.01	−0.03
	(0.049)	(0.041)	(0.031)	(0.039)
Constant	5.02**	5.52**	5.19**	5.12**
	(0.068)	(0.069)	(0.067)	(0.078)
Observations	98	98	98	98
R-squared	0.291	0.434	0.609	0.332

Note: Hotels available for all four dates only. Dependent variable is log price, distance is entered as spline with a knot at 2 miles. Robust standard error estimates in parentheses. ** p<0.01, * p<0.05.

Source: `hotels-europe` dataset. Vienna, reservation price for November and December 2017, June 2018.

First note the number of observations: we now have the same 98 hotels for all four dates, and so selection is not an issue anymore. The results turn out be rather similar: weekend patterns in November and December look a bit different now with a stronger price slope (−0.44 and −0.40 versus −0.28 log units).

What can we conclude from these results about the external validity of our findings? While we can't tell how the price–distance pattern would look like in the future in Vienna, we saw some instability in the pattern from weekdays to weekends. If we want to generalize the pattern into some future situation, we now know that we need to pay attention to dates. More generally, the instability in the pattern we uncovered highlights the fact that the pattern may very well change in other future situations. As important as differences in the price–distance pattern were differences in the intercept. Those differences reflect the fact that prices at weekends and during the holiday season tend to be higher overall. This is important if we want to use coefficients from one day to predict prices on another day: such predictions may be off by a large amount.

External validity across space is another potential question we may be interested in. To show some examples, we estimated the same regression for the same date (November 2017 weekday) in two more cities, Amsterdam and Barcelona. Table 9.5 shows the results.

Table 9.5 External validity – comparing cities

Variables	(1) Vienna	(2) Amsterdam	(3) Barcelona
Distance spline <2	−0.31**	−0.27**	−0.06
	(0.038)	(0.040)	(0.034)
Distance spline 2–7	0.02	0.03	-0.05
	(0.033)	(0.037)	(0.058)
Constant	5.02**	5.24**	4.67**
	(0.042)	(0.041)	(0.041)
Observations	207	195	249
R-squared	0.314	0.236	0.023

Note: Dependent variable is log price, distance is entered as spline with a knot at 2 miles. Robust standard error estimates in parentheses. ** p<0.01, * p<0.05.

Source: `hotels-europe` dataset. November 2017, weekday.

The slope coefficients are quite similar from Vienna to Amsterdam. Among hotels within 2 miles from the city centers, those that are 1 mile further away from the center are, on average, 35 percent (around 30 log points) less expensive. However, Barcelona is very different: there seems to be no difference in prices by distance from the city center. Confidence intervals do not overlap. Prices in Barcelona decline little as we consider hotels further away from the center (around the famous Las Ramblas). This could come from different city structure, or a variety of other reasons.

In terms of space, external validity seems weak. Patterns found in Vienna may work out fine in some, but not all, other cities. Hence the external validity of Vienna results is much weaker regarding space than it is regarding time.

Finally, we can compare hotels with another type of accommodation, apartments, that were also advertised on the price comparison website we used to collect our data. The data are from the same city (Vienna) for the same date (November weekday). Similarly to hotels, we restricted apartments to those with 3 to 4 stars. Table 9.6 shows the results.

Table 9.6 External validity – accommodation types

Variables	(1) Hotels	(2) Apartments
Distance spline <2	-0.31**	-0.26**
	(0.035)	(0.069)
Distance spline 2–7	0.02	0.12
	(0.032)	(0.061)
Constant	5.02**	5.15**
	(0.044)	(0.091)
Observations	207	92
R-squared	0.314	0.134

Note: Dependent variable is log price, distance is entered as spline with a knot at 2 miles. Robust standard error estimates in parentheses. ** p<0.01, * p<0.05.

Source: `hotels-europe` dataset. Vienna, November 2017, weekday.

Again, the patterns are quite similar across the two groups, which here differ by accommodation type. Among hotels and apartments within 2 miles of the city center, those that are 1 mile further away tend to be around 30 percent (26 vs. 31 log points) less expensive. The 95% confidence intervals overlap to a great extent. Among apartments that are more than 2 miles away from the center, those that are 1 mile further away are not less expensive; in this data they are actually slightly more expensive, although the 95% confidence intervals contain zero.

Let us summarize how we now think about external validity. We started by showing a pattern for a particular lodging type, time, and city. We then continued and looked for evidence of the same pattern for other dates, locations, and types. We found some additional variation to what is suggested by statistical inference. In particular, external validity across dates and types seems fairly solid, but across space seems moderate: the patterns in our data may not transfer to other cities.

9.8 Main Takeaways

The confidence interval (CI) is the most useful tool for statistical inference of regression coefficients.

- The standard error of a slope coefficient estimate is smaller (its CI is narrower) the more observations we have, the larger the spread (standard deviation) of x, and the larger the R-squared (smaller residual standard deviation).
- We should use heteroskedasticity-robust standard errors unless we have a good reason to do otherwise.

PRACTICE QUESTIONS

1. The SE of $\hat{\beta}$ in the simple regression is larger, the smaller the standard deviation of x. Why?
2. The SE of $\hat{\beta}$ in the simple regression is larger, the smaller the R-squared of the regression. Why?
3. The 95% CI for $\hat{\beta}$ in the simple regression is narrower, the more observations we have in the data. Why?
4. Which one is narrower – the 95% CI for $\hat{\beta}$ or the 90% CI for $\hat{\beta}$? Why?
5. The 95% CI for $\hat{y}$ in the simple regression is narrower, the closer x is to its average. Why?
6. What additional uncertainty do we have when we want to predict y_j instead of $\hat{y}_j$ in the simple linear regression?
7. What's a false positive for H_0: $\beta = 0$ in the regression $y^E = \alpha + \beta x$? Give an example.
8. What's a false negative for H_0: $\beta = 0$ in the regression $y^E = \alpha + \beta x$? Give an example.
9. Which is a more conservative rule for rejecting H_0: $\beta = 0$: a 5% level of significance or a 1% level of significance? Why?
10. When exploring potential associations in the data, we often allow a more lenient criterion for rejecting H_0: $\beta = 0$ than when we want to prove beyond reasonable doubt that an association exists. Give an example and argue why the difference in the rules makes sense.
11. We estimate a regression of log hourly wage on a binary "female" variable on a random sample of employees in computer science occupations in the USA in 2014. The slope coefficient estimate is -0.15; its 95% CI is $[-0.18, -0.12]$. Interpret these results.
12. In the previous regression, the software reports a p-value of 0.000 next to the slope coefficient. What does this mean?
13. Interpret the coefficient estimate on the first piecewise linear spline and the confidence interval you create around it in Table 9.5 in our **How stable is the hotel price–distance to center relationship?** case study.
14. In Table 9.5 in our **How stable is the hotel price–distance to center relationship?** case study, the coefficient estimates on the first piecewise linear spline are different for Vienna and Amsterdam in the data. Can you reject the hypothesis that they are the same in the general pattern represented by the data?
15. In Table 9.5 in our **How stable is the hotel price–distance to center relationship?** case study, the coefficient estimates on the first piecewise linear spline are different for Vienna and Barcelona in the data. Can you reject the hypothesis that they are the same in the general pattern represented by the data?

DATA EXERCISES

Easier and/or shorter exercises are denoted by [*]; harder and/or longer exercises are denoted by [**].

1. Replicate the case study of patterns of earnings by gender and age using data on employees in the same occupation (market research analysts and marketing specialists) for another year. Use another CPS Morg file (`www.nber.org/data/morg.html`) to pick another year in the USA. Carry out the analysis and compare your results to what we found for the USA in 2014. [*]
2. Replicate the case study of patterns of earnings by gender and age using data on employees in the same occupation in another country. For a different country find and use survey data on

earnings, hours, and occupations that represent the population of interest. If possible, pick an occupation close to market research analysts and marketing specialists. Carry out the analysis and compare your results to what we found for the USA in 2014. [**]

3. Replicate the case study of patterns of earnings by gender and age using data on employees in another occupation. Use the same 2014 USA CPS Morg file, carry out the same sample selection steps, and create the same variables, but select another occupation group. Go on the USA Census website to see which codes refer to which occupations. Carry out the analysis and compare your results to what we found for market research analysts. [*]
4. Use the `hotels-europe` dataset and pick a European city (not Vienna). Select a sample that is appropriate for a couple searching for a good deal among 3-to-4-star hotels not very far from the center. Show a lowess regression for the pattern between log hotel price and ratings. Estimate a simple linear regression and alternative regressions that better fit the nonlinear pattern. Visualize the confidence interval around the regression lines together with the lowess regression, and decide which specification is the best approximation. [*]
5. Use the `wms-management-survey` dataset and pick a country. Estimate a linear regression with the management quality score and employment. Interpret the slope coefficient, create its 95% CI, and interpret that, too. Explore potential nonlinearities in the patterns of association by lowess. Estimate a regression that can capture those nonlinearities, and carry out a test to see if you can reject that the linear approximation was good enough for the population of firms represented by the data. [**]

REFERENCE AND FURTHER READING

For a recent review on the extent and potential causes of gender differences in earnings, see Bertrand (2018).

9.U1 UNDER THE HOOD: THE SIMPLE SE FORMULA FOR REGRESSION INTERCEPT

The simple formula for the SE for the intercept of a simple linear regression $y^E = \alpha + \beta x$ is

$$SE(\hat{\alpha}) = Std[e]\sqrt{\frac{1}{n} + \frac{(\bar{x})^2}{nVar[x]}} \tag{9.16}$$

This is similar to the standard error of the slope coefficient with two differences: the additional $\frac{1}{n}$ term and the $(\bar{x})^2$ in the numerator (note that $\sqrt{Var[x]} = Std[x]$). The first of these two new elements makes the relation of the standard error to the number of observations stronger. The larger n, the smaller $SE(\hat{\alpha})$ because of both n in the denominator and this new $\frac{1}{n}$ term. The second element makes the standard error depend on the mean value of x. One way to understand its role is to acknowledge that the intercept is estimated for y^E when $x = 0$. The larger $(\bar{x})^2$, the further away it is from zero, in any direction. The further away zero is from typical x values in the data, the larger is the uncertainty as to where the intercept could be in the population, or general pattern, represented by the data.

As always, the simple formula is not the right formula in general. In cross-sectional regressions we should use the robust formula instead. We don't show that formula here; standard statistical software packages have it and compute it easily.

9.U2 UNDER THE HOOD: THE LAW OF LARGE NUMBERS FOR $\hat{\beta}$

Recall from Under the Hood section 5.U1 that the Law of Large Numbers tells us that the sample average of a variable is very close to its mean in the population (or general pattern) the data represents if the sample is large; the larger the sample, the closer they are.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \rightarrow E[x] \text{ in probability} \tag{9.17}$$

where $E[x]$ means the expected value of x (its mean value in the population), and $\rightarrow$ in probability ("convergence in probability") means that the left-hand side of the formula approaches the right-hand side, with probability 1 if n goes to infinity. Another notation of the same thing is that the "probability limit," abbreviation *plim*, of the sample average is the expected value:

$$plim(\bar{x}) = E[x] \tag{9.18}$$

Considering the simple linear regression $y^E = \alpha + \beta x$, the corresponding Law of Large Numbers for $\hat{\beta}$ tells us that $\hat{\beta}$ is very close to β in the population (or general pattern) the data represents if the sample is large; the larger the sample, the closer they are.

$$plim(\hat{\beta}) = \beta \tag{9.19}$$

The OLS formula for $\hat{\beta}$ is

$$\hat{\beta} = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{9.20}$$

The important insight is that this is the ratio of two averages. The Law of Large Numbers applies to both the numerator and the denominator:

$$plim\left(\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})\right) = Cov[x, y] \tag{9.21}$$

where $Cov[x, y]$ is the covariance in the population. (Note the specific meaning of *Cov* here: in the text it could refer to the covariance in the data; here it is reserved to refer to the covariance in the population.)

Similarly,

$$plim\left(\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right) = Var[x] \tag{9.22}$$

where $Var[x]$ is the variance in the population (again, note that $Var[x]$ here means the variance in the population).

There is one more result to use: if both the numerator and the denominator converge in probability to some number, their ratio converges in probability to the ratio of those two numbers. This is called the Slutksy theorem (named after the Russian statistician–mathematician–economist Evgeny Slutksy). Thus,

$$plim\hat{\beta} = \frac{Cov[y, x]}{Var[x]} = \beta, \tag{9.23}$$

9.U3 UNDER THE HOOD: DERIVING $SE(\hat{\beta})$ WITH THE CENTRAL LIMIT THEOREM

Recall from Chapter 5, Under the Hood section 5.U1 that the Central Limit Theorem (CLT) states that, in large enough samples, the average is distributed normally around its expected value, and the variance of this normal distribution is related to the variance of the original variable that makes up the average.

Formally, the CLT states that

$$\bar{x} \overset{A}{\sim} N\left(E[x], \frac{1}{n} Var[x]\right) \tag{9.24}$$

Recall that the normal distribution result holds regardless of the distribution of those original variables. The variance of this normal distribution, which is the square of its standard deviation across repeated samples, is equal to the variance of x itself divided by n. Thus we can estimate it from the data. That's why we can have a formula for the SE of $\bar{x}$: all of its elements can be estimated from the data we have.

$$SE(\bar{x}) = \sqrt{\frac{Var[x]}{n}} = \frac{Std[x]}{\sqrt{n}} \tag{9.25}$$

The CLT for regression coefficients follows the same logic. The estimated slope coefficient follows a normal distribution around its value in the population; this distribution is across many repeated samples if those samples are large. Thus the standard error of the regression coefficient is the standard deviation of this distribution, which is the square root of the variance.

It turns out that the general formula for the distribution is

$$\hat{\beta} \overset{A}{\sim} N\left(\beta, \frac{E[e^2(x - E[x])^2]}{n(Var[x])^2}\right) \tag{9.26}$$

where e denotes the regression residual. This is actually the robust formula for the standard error that we need to use in regressions using cross-sectional data. It is complicated and thus we don't interpret it. But we can recognize the three elements in it that we emphasized: the spread of the residuals is embedded in the formula in the numerator, and the number of observations and the variance of x is in the denominator.

The general formula simplifies to the simple formula we show in the text if $E[e^2(x - E[x])^2] = E[e^2]E[(x - E[x])^2]$, which is nothing else than $Var[e]Var[x]$, where $Var[e]$ is the variance of the regression residuals:

$$\hat{\beta} \overset{A}{\sim} N\left(\beta, \frac{Var[e]}{nVar[x]}\right) \tag{9.27}$$

That additional assumption is one of homoskedasticity. The formula $E[e^2(x - E[x])^2]$ is the expected value of the product of e^2 and $(x - E[x])^2$. e^2 is the square of the residual, measuring how far it is from the regression line. If the spread of the residuals around the regression line is the same for all values of x (the assumption of homoskedasticity), the squared value of the residual is expected to be the same for all values of x. This is expected to be the same for all values of $(x - E[x])^2$, too. And then the expected value of the product of the two is equal to the product of the two expected values: $E[e^2(x - E[x])^2] = E[e^2]E[(x - E[x])^2]$.

From this we get the simple formula of the standard error by taking the square root of the variance in the normal distribution:

$$SE(\hat{\beta}) = \frac{Std[e]}{\sqrt{n}Std[x]} \tag{9.28}$$

9.U4 UNDER THE HOOD: DEGREES OF FREEDOM ADJUSTMENT FOR THE SE FORMULA

As we introduced in Chapter 5, the simple standard error formula for the estimated $\bar{x}$ is

$$SE(\bar{x}) = \frac{Std[x]}{\sqrt{n}} \tag{9.29}$$

Here we introduced the simple standard error formula for $\hat{\beta}$ in a simple regression:

$$SE(\hat{\beta}) = \frac{Std[e]}{\sqrt{n}Std[x]} \tag{9.30}$$

Both have $\sqrt{n}$ in the denominator. However, statistical software packages don't divide by $\sqrt{n}$. For $SE(\bar{x})$ the correct formula uses $\sqrt{n-1}$ instead of $\sqrt{n}$. For $SE(\hat{\alpha})$ and $SE(\hat{\beta})$ the correct formula uses $\sqrt{n-2}$. Why, and how much does this matter?

The number less than n that is used in the denominator is called the **degrees of freedom**, abbreviated as **df**. Thus, the degrees of freedom for $SE(\bar{x})$ is $df = n - 1$; the degrees of freedom for both $SE(\hat{\alpha})$ and $SE(\hat{\beta})$ is $df = n - 2$.

The idea is that we need some data to use for estimates that are included in the standard error formula, and thus we are left with less data once we have estimated those. Degrees of freedom means the number of "free" observations we are left with after considering the fact that we need to estimate those other things. For example, we need $\bar{x}$ in the formula for $SE(\bar{x})$. If we didn't have to estimate $\bar{x}$ but knew it somehow, we would have more information in n observations to calculate $SE(\bar{x})$. But we don't know it, so we need to use data to estimate it. Degrees of freedom acknowledges this and takes out one for each and every object within the SE formulae that needs to be estimated.

For $SE(\bar{x})$ we need one thing: $\bar{x}$, because we have $Std[x]$ that in turn needs $\bar{x}$: $Std[x]$ has $\sum(x_i - \bar{x})^2$ in its numerator.

For $SE(\hat{\beta})$ we need $Cov[y, x]$ and $Var[x]$, and they need $\bar{x}$ and $\bar{y}$. That's two things to calculate from the data ($Var[x]$ has $\sum(x_i - \bar{x})^2$ in its numerator and $Cov[y, x]$ has $\sum(y_i - \bar{y})(x_i - \bar{x})$ in its numerator).

How much does this matter? Not much for simple regressions. Dividing by $\sqrt{n}$ or $\sqrt{n-1}$ or $\sqrt{n-2}$ is practically the same for the data we use. In very small datasets this may matter, but in very small datasets the standard error formulae are likely to be wrong anyway. At the same time, the degrees of freedom correction may matter in methods that have many objects estimated at the same time; we will start discussing such models in the next chapter.

10 Multiple Linear Regression

Why and how to carry out multiple linear regression analysis, and how to interpret its results

Motivation

There is a substantial difference in the average earnings of women and men in all countries. You want to understand more about the potential origins of that difference, focusing on employees with a graduate degree in your country. You have data on a large sample of employees with a graduate degree, with their earnings and some of their characteristics, such as age and the kind of graduate degree they have. Women and men differ in those characteristics, which may affect their earnings. How should you use this data to uncover gender difference that are not due to differences in those other characteristics? And can you use regression analysis to uncover patterns of associations between earnings and those other characteristics that may help understand the origins of gender differences in earnings?

You have analyzed your data on hotel prices in a particular city to find hotels that are underpriced relative to how close they are to the city center. But you have also uncovered differences in terms of other features of the hotels that measure quality and are related to price. How would you use this data to find hotels that are underpriced relative to all of their features? And how can you visualize the distribution of hotel prices relative to what price you would expect for their features in a way that helps identify underpriced hotels?

After understanding simple linear regression, we can turn to multiple linear regression, which has more than one explanatory variable. Multiple linear regression is the most used method to uncover patterns of associations between variables. There are multiple reasons to include more explanatory variables in a regression. We may be interested in uncovering patterns of association between y and several explanatory variables, which may help uncover patterns of association that could be investigated in subsequent analysis. Or, we may be interested in the effect of an x variable, but we want to compare observations that are different in x but similar in other variables. Finally, we may want to predict y, and we want to use more x variables to arrive at better predictions.

We discuss why and when we should estimate multiple regression, how to interpret its coefficients, and how to construct and interpret confidence intervals and test the coefficients. We discuss the relationship between multiple regression and simple regression. We explain that piecewise linear splines and polynomial regressions are technically multiple linear regressions without the same interpretation of the coefficients. We discuss how to include categorical explanatory variables as well as interactions that help uncover different slopes for groups. We include an informal discussion on how to decide what explanatory variables to include and in what functional form. Finally, we discuss why a typical multiple regression with observational data can get us closer to causal interpretation without fully uncovering it.

The first case study in this chapter, **Understanding the gender difference in earnings**, uses the `cps-earnings` dataset to illustrate the use of multiple regression to understand potential sources of gender differences in earnings. We also go back to our question on finding underpriced hotels relative to their location and quality in the case study **Finding a good deal among hotels with multiple regression**, using the `hotels-vienna` dataset, to illustrate the use of multiple regression in prediction and residual analysis.

Learning outcomes

After working through this chapter, you should be able to

- identify questions that are best answered with the help of multiple regression from available data;
- estimate multiple linear regression coefficients and present and interpret them;
- estimate appropriate standard errors, create confidence intervals and tests of regression coefficients, and interpret those;
- select the variables to include in a multiple regression guided by the purpose of the analysis;
- understand the relationship between the results of a multiple regression and causal effects when using observational data.

10.1 Multiple Regression: Why and When?

There are three broad reasons to carry out multiple regression analysis instead of simple regression. The first is exploratory data analysis: we may want to uncover more patterns of association, typically to generate questions for subsequent analysis. The other two reasons are the two ultimate aims of data analysis: making a better prediction by explaining more of the variation, and getting closer to establishing cause and effect in observational data by comparing observations that are more comparable.

The first example of the introduction is about understanding the reasons for a difference. It's a causal question of sorts: we are interested in what causes women to earn less than men. The second example is one of prediction: we want to capture average price related to hotel features that customers value in order to identify hotels that are inexpensive compared to what their price "should be."

10.2 Multiple Linear Regression with Two Explanatory Variables

Multiple regression analysis uncovers average y as a function of more than one x variable: $y^E = f(x_1, x_2, ...)$. It can lead to better predictions of $\hat{y}$ by considering more explanatory variables. It may improve the interpretation of slope coefficients by comparing observations that are different in terms of one of the x variables but similar in terms of all other x variables.

Multiple linear regression specifies a linear function of the explanatory variables for the average y. Let's start with the simplest version with two explanatory variables:

$$y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \tag{10.1}$$

This is the standard notation for a multiple regression: all coefficients are denoted by the Greek letter β, but they have subscripts. The intercept has subscript 0, the first explanatory variable and its coefficient have subscript 1, and the second explanatory variable and its coefficient have subscript 2.

Having another right-hand-side variable in the regression means that we further condition on that other variable when we compare observations. The slope coefficient on x_1 shows the difference in average y across observations with different values of x_1 but with the same value of x_2. Symmetrically, the slope coefficient on x_2 shows difference in average y across observations with different values of x_2 but with the same value of x_1. This way, multiple regression with two explanatory variables compares observations that are similar in one explanatory variable to see the differences related to the other explanatory variable.

The interpretation of the slope coefficients takes this into account. β_1 shows how much larger y is on average for observations in the data with one unit larger value of x_1 but the same value of x_2. β_2 shows how much larger y is on average for observations in the data with one unit larger value of x_2 but with the same value of x_1.

Review Box 10.1 Multiple linear regression

Multiple linear regression with two explanatory variables:

$$y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Interpretation of the coefficients:

- β_0 (intercept): average y for observations if both x_1 and x_2 are zero in the data.
- β_1 (slope of x_1): on average, y is β_1 units larger in the data for observations with one unit larger x_1 but with the same x_2.
- β_2 (slope of x_2): on average, y is β_2 units larger in the data for observations with one unit larger x_2 but with the same x_1.

10.3 Multiple Regression and Simple Regression: Omitted Variable Bias

It is instructive to examine the difference in the slope coefficient of an explanatory variable x_1 when it is the only variable on the right-hand side of a regression compared to when another explanatory variable, x_2, is included as well. In notation, the question is the difference between β in the simple linear regression

$$y^E = \alpha + \beta x_1 \tag{10.2}$$

and β_1 in the multiple linear regression

$$y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \tag{10.3}$$

For example, we may use time series data on sales and prices of the main product of our company, and regress month-to-month change in log quantity sold (y) on month-to-month change in log price (x_1). Then β shows the average percentage change in sales when our price increases by 1%.

For example, $\hat{\beta} = -0.5$ would show that sales tend to decrease by 0.5% when our price increases by 1%.

But we would like to know what happens to sales when we change our price but competitors don't. The second regression has change in the log of the average price charged by our competitors (x_2) next to x_1. Here β_1 would answer our question: average percentage change in our sales in months when we increase our price by 1%, but competitors don't change their price. Suppose that in that regression, we estimate $\hat{\beta_1} = -3$. That is, our sales tend to drop by 3% when we increase our price by 1% and our competitors don't change their prices. Also suppose that the coefficient on x_2 is positive, $\hat{\beta_2} = 3$. That means that our sales tend to increase by 3% when our competitors increase their prices by 1% and our price doesn't change.

As we'll see, whether the answer to the two questions is the same will depend on whether x_1 and x_2 are related. To understand that relationship, let us introduce the regression of x_2 on x_1, called the ***x*–*x* regression**, where δ is the slope parameter:

$$x_2^E = \gamma + \delta x_1 \tag{10.4}$$

In our own price–competitor price example, δ would tell us how much the two prices tend to move together. In particular, it tells us about the average percentage change in competitor price in months when our own price increases by 1%. Let's suppose that we estimate it to be $\hat{\delta} = 0.83$: competitors tend to increase their price by 0.83% when our company increases its price by 1%. The two prices tend to move together, in the same direction.

To link the two original regressions, plug this x–x regression back in the multiple regression (this step is fine even though that may not be obvious; see Under the Hood section 10.U1):

$$y^E = \beta_0 + \beta_1 x_1 + \beta_2(\gamma + \delta x_1) = \beta_0 + \beta_2\gamma + (\beta_1 + \beta_2\delta)x_1 \tag{10.5}$$

Importantly, we find that with regards to x_1, the slope coefficients in the simple (β) and multiple regression (β_1) are different:

$$\beta - \beta_1 = \delta\beta_2 \tag{10.6}$$

The slope of x_1 in a simple regression is different from its slope in the multiple regression, the difference being the product of its slope in the regression of x_2 on x_1 and the slope of x_2 in the multiple regression. Or, put simply, the slope in simple regression is different from the slope in multiple regression by the slope in the x–x regression times the slope of the other x in the multiple regression.

This difference is called the **omitted variable bias**. If we are interested in the coefficient on x_1 with x_2 in the regression, too, it's the second regression that we need; the first regression is an incorrect regression as it omits x_2. Thus, the results from that first regression are biased, and the bias is caused by omitting x_2. We will discuss omitted variables bias in detail when discussing causal effects in Chapter 21, Section 21.3.

In our example, we had that $\hat{\beta} = -0.5$ and $\hat{\beta_1} = -3$, so that $\hat{\beta} - \hat{\beta_1} = -0.5 - (-3) = +2.5$. In this case our simple regression gives a biased estimate of the slope coefficient on x_1 compared to the multiple regression, and the bias is positive (the simple regression estimate is less negative). Recall that we had $\hat{\delta} = 0.83$ and $\hat{\beta_2} = 3$. Their product is approximately 2.5. This positive bias is the result of two things: a positive association between the two price changes (δ) and a positive association between competitor price and our own sales (β_2).

In general, the slope coefficient on x_1 in the two regressions is different unless x_1 and x_2 are uncorrelated ($\delta = 0$) or the coefficient on x_2 is zero in the multiple regression ($\beta_2 = 0$). The slope in the

simple regression is more positive or less negative if the correlation between x_2 and x_1 has the same sign as β_2 (both are positive or both are negative).

The intuition is the following. In the simple regression $y^E = \alpha + \beta x_1$, we compare observations that are different in x_1 without considering whether they are different in x_2. If x_1 and x_2 are uncorrelated this does not matter. In this case observations that are different in x_1 are, on average, the same in x_2, and, symmetrically, observations that are different in x_2 are, on average, the same in x_1. Thus the extra step we take with the multiple regression to compare observations that are different in x_1 but similar in x_2 does not matter here.

If, however, x_1 and x_2 are correlated, comparing observations with or without the same x_2 value makes a difference. If they are positively correlated, observations with higher x_2 tend to have higher x_1. In the simple regression we ignore differences in x_2 and compare observations with different values of x_1. But higher x_1 values mean higher x_2 values, too. Corresponding differences in y may be due to differences in x_1 but also due to differences in x_2.

In our sales–own price–competitors' price example, the drop in sales when our own price increases (and competitors do what they tend to do) is smaller than the drop in sales when our own price increases but competitor prices don't change. That's because when our own price increases, competitors tend to increase their prices, too, which in itself would push up our sales. The two work against each other: the increase in our price makes sales decrease, but the increase in competitors' prices that tends to happen at the same time makes sales increase.

Review Box 10.2 Multiple linear regression and simple linear regression

- The difference of slope β in $y^E = \alpha + \beta x_1$ and the slope β_1 in $y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ is

$$\beta - \beta_1 = \delta\beta_2$$

  where δ is the slope in the regression of x_2 on x_1: $x_2^E = \gamma + \delta x_1$
- In words, briefly: slope in the simple regression differs from the slope in the multiple regression by the product of the slope in x–x regression and slope of other x in multiple regression.

10.A1 CASE STUDY – Understanding the Gender Difference in Earnings

Multiple linear regression

We continue investigating patterns in earnings, gender, and age. The data is the same `cps-earnings` dataset that we used earlier in Chapter 9, Section 9.A1: it is a representative sample of all people of age 15 to 85 in the USA in 2014.

Compared to Chapter 9, Section 9.A1, where we focused on a single occupation, we broaden the scope of our investigation here to all employees with a graduate degree – that is, a degree higher than a four-year college degree: these include professional, master's, and doctoral degrees.

We use data on people of age 24 to 65 (to reflect the typical working age of people with these kinds of graduate degrees). We excluded the self-employed (their earnings is difficult to measure) and included those who reported 20 hours or more as their usual weekly time worked. We have 18 241 observations.

The dependent variable is log hourly earnings (ln wage). Table 10.1 shows the results from three regressions: (1) is a simple regression of *ln wage* on a binary *female* variable; (2) is a multiple regression that includes age as well, in a linear fashion; and (3) is a simple regression with *age* as the dependent variable and the *female* binary variable.

Table 10.1 Gender differences in earnings – log earnings and gender

	(1)	(2)	(3)
Variables	**ln wage**	**ln wage**	**age**
female	−0.195**	−0.185**	−1.484**
	(0.008)	(0.008)	(0.159)
age		0.007**	
		(0.000)	
Constant	3.514**	3.198**	44.630**
	(0.006)	(0.018)	(0.116)
Observations	18 241	18 241	18 241
R-squared	0.028	0.046	0.005

Note: Robust standard error estimates in parentheses. ** $p<0.01$, * $p<0.05$.

Source: `cps-earnings` dataset. 2014, USA. Employees of age 24–65 with a graduate degree and 20 or more work hours per week.

According to column (1), women in this sample earn 19.5 log points (around 21%) less than men, on average. Column (2) suggests that when we compare employees of the same age, women in this sample earn 18.5 log points (around 20%) less than men, on average. This is a slightly smaller gender difference than in column (1). While the log approximation is not perfect at these magnitudes, from now on, we will ignore the difference between log units and percent. For example, we will interpret a 0.195 coefficient as a 19.5% difference.

The estimated coefficients differ, and we know where the difference should come from: average difference in age. Let's use the formula for the difference between the coefficient on *female* in the simple regression and in the multiple regression. Their difference is $-0.195 - (-0.185) = -0.01$. This should be equal to the product of the coefficient of *female* in the regression of *age* on *female* (our column (3)) and the coefficient on *age* in column (2): $-1.48 \times 0.007 \approx -0.01$. These two are indeed equal.

Intuitively, we can see that women of the same age have a slightly smaller earnings disadvantage in this data because they are somewhat younger, on average, and employees who are younger tend to earn less. Part of the earnings disadvantage of women is thus due to the fact that they are younger. This is a small part, though: around one percentage point of the 19.5% difference, which is a 5% share of the entire difference.

But why are women employees younger, on average, in the data? It's because there are fewer female employees with graduate degrees over age 45 than below. Figure 10.1 shows two density plots overlaid: the age distributions of male and female employees with graduate degrees. There are relatively few below age 30. From age 30 and up, the age distribution is close to uniform for men. But not for women: the proportion of female employees with graduate degrees drops above age 45, and again above age 55.

In principle, this could be due to two things: either there are fewer women with graduate degrees in the 45+ generation than among the younger ones, or fewer of them are employed (i.e., employed for 20 hours or more for pay, which is the criterion to be in our subsample). Further investigation reveals that it is the former: women are less likely to have a graduate degree if they were born before 1970 (those 45+ in 2014) in the USA. The proportion of women working for pay for more than 20 hours is very similar among those below age 45 and above.

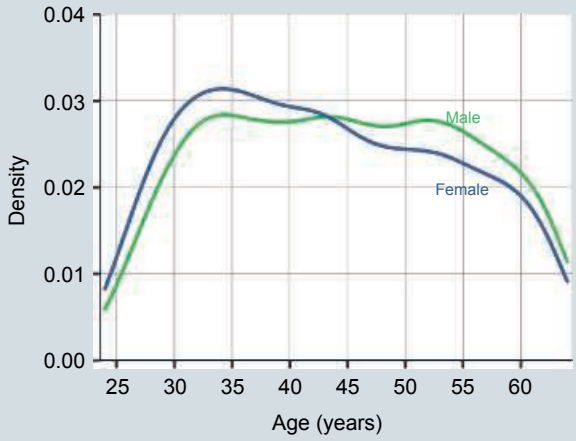


Figure 10.1 Age distribution of employees with graduate degree by gender

Note: Kernel density estimates of the age distribution of employees with a graduate degree; female and male employees separately.

Source: `cps-earnings` dataset. 2014, USA. Employees of age 24–65 with a graduate degree and 20 or more work hours per week. N=18 241.

10.4 Multiple Linear Regression Terminology

Multiple regression with two explanatory variables (x_1 and x_2) allows for assessing the differences in expected y across observations that differ in x_1 but are similar in terms of x_2. This difference is called **conditional** on that other explanatory variable x_2: difference in y by x_1, conditional on x_2. It is also called the controlled difference: difference in y by x_1, controlling for x_2. We often say that we condition on x_2, or control for x_2, when we include it in a multiple regression that focuses on average differences in y by x_1.

When we focus on x_1 in the multiple regression, the other right-hand-side variable, x_2, is called a **covariate**. In some cases, it is also called a **confounder**: if omitting x_2 makes the slope on x_1 different, it is said to confound the association of y and x_1 (we'll discuss confounders in Chapter 19, Section 19.14).

Review Box 10.3 Multiple linear regression terminology

In a regression $y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ that focuses on β_1,

- if we estimate a multiple regression $y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, and we are interested in β_1, x_2 is a covariate, and we say that we condition on x_2 or control for x_2;
- if, instead, we estimate $y^E = \alpha + \beta x_1$, we say x_2 is an omitted variable.

10.5 Standard Errors and Confidence Intervals in Multiple Linear Regression

The concept of statistical inference and the interpretation of confidence intervals in multiple regressions is similar to that in simple regressions. For example, the 95% confidence interval of the slope coefficient of x_1 in a multiple linear regression shows where we can expect the coefficient in the population, or general pattern, represented by the data.

Similarly to the coefficients in the simple regression, the 95% CI of a slope in a multiple regression is the coefficient value estimated from the data plus-or-minus two standard errors. Again similarly to the simple regression case, we can get the standard error either by bootstrap or using an appropriate formula. And, as usual, the simple SE formula is not a good approximation in general: it assumes homoskedasticity (same fit of the regression over the range of the explanatory variables). There is a robust SE formula for multiple regression, too, that works in general, both under homoskedasticity and heteroskedasticity . Thus, just as with simple regressions we should make the software calculate **robust SE** as default.

While not correct in general, the simple formula is good to examine because it shows what makes the SE larger in a simpler more intuitive way than the robust formula. The simple SE formula for the slope $\hat{\beta}_1$ is

$$SE(\hat{\beta}_1) = \frac{Std[e]}{\sqrt{n}Std(x_1)\sqrt{1-R_1^2}} \tag{10.7}$$

Similarly to the simple SE formula for the simple linear regression in Chapter 9, Section 9.2, this formula has $\sqrt{n}$ in its denominator. But, similarly again to the simple linear regression, the correct number to divide with would be slightly different: the degrees of freedom instead of the number of the observations (see Under the Hood section 9.U4). Here that would be $\sqrt{n-k-1}$, where k is the number of right-hand-side variables in the regression. Similarly to the simple regression, this makes little practical difference in most cases. However, in contrast with the simple regression case, it may make a difference not only when we have too few observations, but also when we have many right-hand-side variables relative to the number of observations. We'll ignore that issue for most of this textbook, but it will come back in Chapter 21, Section 21.4.

This formula is very similar to what we have for simple regressions in other details, too, except for that new $\sqrt{1-R_1^2}$ term in the denominator. R_1^2 is the R-squared of the regression of x_1 on x_2. Recall that the R-squared of a simple regression is the square of the correlation between the two variables in the regression. Thus, R_1^2 is the square of the correlation between x_1 and x_2. The stronger this correlation, the larger R_1^2, the smaller $\sqrt{1-R_1^2}$, but then the larger $1/\sqrt{1-R_1^2}$ ($\sqrt{1-R_1^2}$ is in the denominator). So, the stronger the correlation between x_1 and x_2, the larger the SE of $\hat{\beta}_1$. Note the

symmetry: the same would apply to the SE of $\hat{\beta}_2$. As for the familiar terms in the formula: the SE is smaller, the smaller the standard deviation of the residuals (the better the fit of the regression), the larger the sample, and the larger the standard deviation of x_1.

At the polar case of a correlation of one (or negative one) that corresponds to $R_1^2 = 1$, the SE of the two coefficients does not exist. A correlation of one means that x_1 and x_2 are linear functions of each other. It is not only the SE formulae that cannot be computed in this case; the regression coefficients cannot be computed either. In this case the explanatory variables are said to be **perfectly collinear**.

Strong but imperfect correlation between explanatory variables is called **multicollinearity**. It allows for calculating the slope coefficients and their standard errors, but it makes the standard errors large. Intuitively, this is because we would like to compare observations that are different in one of the variables but similar in the other. But strong correlation between the two implies that there are not many observations that are the same in one variable but different in the other variable. Indeed, the problem of multicollinearity is very similar to the problem of having too few observations in general. We can see it in the formula as well: the role of $(1 - R^2)$ and n are the same.

Consider our example of using monthly data to estimate how sales of the main product of our company tend to change when our price changes but the prices of competitors do not. In that example our own price and the competitors' prices tended to move together. One consequence of this is that omitting the change in the competitors' price would lead to omitted variable bias; thus we need to include that in our regression. But here we see that it has another consequence. Including both price variables in the regression makes the SE of the coefficient of our own price larger, and its confidence interval wider, too. Intuitively, that's because there are fewer months when our price changes but the competitors' prices don't change, and it is changes in sales in those months that contain the valuable information for estimating the coefficient on our own price. Months when our own and competitors' prices change the same way don't help. So the reason why we want competitors' price in our regression (strong co-movement) is exactly the reason for having imprecise estimates with wide confidence intervals.

That's true in general, too. Unfortunately, there is not much we can do about multicollinearity in the data we have, just as there is not much we can do about having too few observations. More data helps both, of course, but that is not much help when we have to work with the data that's available. Alternatively, we may decide to change the specification of the regression and drop one of the strongly correlated explanatory variables. However, that results in a different regression. Whether we want a different regression or not needs to be evaluated keeping the substantive question of the analysis in mind.

Review Box 10.4 Inference in multiple regression

- In the linear regression $y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, the 95% confidence interval (CI) of $\hat{\beta}_1$ tells us about the range in which we can expect, with 95% confidence, the difference in y to fall in the general pattern, or population, that our data represents, when comparing observations with the same x_2 but differing in x_1 by one unit.
- In the linear regression $y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, the simple SE formula of $\hat{\beta}_1$ is

$$SE(\hat{\beta}_1) = \frac{Std[e]}{\sqrt{n}Std(x_1)\sqrt{1 - R_1^2}}$$

where e is the residual $e = y - \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ and R_1^2 is the R-squared in the simple linear regression of x_1 on x_2.

- The standard error of $\hat{\beta}_1$ is smaller:
  - the smaller the standard deviation of the residual (the better the fit of the regression);
  - the larger the sample;
  - the larger the variance of x_1;
  - the smaller the correlation between x_1 and x_2 (the smaller R_1^2).

10.6 Hypothesis Testing in Multiple Linear Regression

Testing hypotheses about coefficients in a multiple regression is also very similar to that in a simple regression. The standard errors are estimated in a different way but with the appropriate SE, all works just the same. For example, testing whether $H_0: \beta_1 = 0$ against $H_A: \beta_1 \neq 0$, we need the p-value or the t-statistic. Standard regression output produced by most statistical software shows those statistics. If our level of significance is 0.05, we reject H_0 if the p-value is less than 0.05, or – which is the same information in a different form – the t-statistic is less than -2 or greater than +2.

Besides testing a hypothesis that involves a single coefficient, we sometimes test a hypothesis that involves more coefficients. As we explained in Chapter 9, Section 9.5, these come in two forms: a single null hypothesis about two or more coefficients (e.g., if they are equal), or a list of null hypotheses (e.g., that several slope coefficients are zero). The latter is called testing joint hypotheses .

Testing joint hypotheses are based on a test statistic called the F-statistic, and the related test is called the **F-test**. The underlying logic of hypothesis testing is the same here: reject the null if the test statistic is larger than a critical value, which shows that the estimated coefficients are too far from what's in the null. The technical details are different. But the meaning of the p-value is the same as always. Thus, we advise getting the p-value when testing a joint hypothesis.

In fact, the test that asks whether all slope coefficients are zero in the regression has its own name: the **global F-test**, or simply "the" F-test. Its results are often shown by statistical software by default. More frequently, we use joint testing of joint hypotheses to decide whether a subset of the coefficients (such as the coeffients on all geographical variables) are all zero.

Similarly to testing hypotheses about single coefficients, the F-test needs appropriate standard error estimates. In cross-sectional data, those appropriate estimates are usually the robust SE estimates.

10.A2 CASE STUDY – Understanding the Gender Difference in Earnings

Statistical inference

Let's revisit the results in Table 10.1, taking statistical inference into account. The data represents employees with a graduate degree in the USA in 2014. According to the estimate in column (1), women in this sample earn 19.5 percent less than men, on average. The appropriately estimated (robust) standard error is 0.008, implying a 95% CI of approximately [−0.21, −0.18]. We can be

95% confident that women earned 18 to 21 percent less, on average, than men among employees with graduate degrees in the USA in 2014.

Column (2) suggests that when we compare employees of the same age, women in this sample earn approximately 18.5 percent less than men, on average. The 95% CI is approximately [−0.20,−0.17]. It turns out that the estimated −0.195 in column (1) is within this CI, and the two CIs overlap. Thus it is very possible that there is no difference between these two coefficients in the population. We uncovered a difference in the data between the unconditional gender wage gap and the gender gap conditional on age. However, that difference is small. Moreover, it may not exist in the population. These two facts tend to go together: small differences are harder to pin down in the population, or general pattern, represented by the data. Often, that's all right. Small differences are rarely very important. When they are, we need more precise estimates, which may come with larger sample size.

10.7 Multiple Linear Regression with Three or More Explanatory Variables

We spent a lot of time on multiple regression with two right-hand-side variables. That's because that regression shows all the important differences between simple regression and multiple regression in intuitive ways. In practice, however, we rarely estimate regressions with exactly two right-hand-side variables. The number of right-hand-side variables in a multiple regression varies from case to case, but it's typically more than two. In this section we describe multiple regressions with three or more right-hand-side variables. Their general form is

$$y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots \tag{10.8}$$

All of the results, language, and interpretations discussed so far carry forward to multiple linear regressions with three or more explanatory variables. Interpreting the slope of x_1: on average, y is β_1 units larger in the data for observations with one unit larger x_1 but with the same value for all other x variables. The interpretation of the other slope coefficients is analogous. The language of multiple regression is the same, including the concepts of **conditioning**, **controlling**, **omitted**, or **confounder variables**.

The standard error of coefficients may be estimated by bootstrap or a formula. As always, the appropriate formula is the robust SE formula. But the simple formula contains the things that make even the robust SE larger or smaller. For any slope coefficient $\hat{\beta}_k$ the simple SE formula is

$$SE(\hat{\beta}_k) = \frac{Std[e]}{\sqrt{n}Std[x_k]\sqrt{1-R_k^2}} \tag{10.9}$$

Almost all is the same as with two right-hand-side variables. In particular, the SE is smaller, the smaller the standard deviation of the residuals (the better the fit of the regression), the larger the sample, and the larger the standard deviation of x_k. The new-looking thing is R_k^2. But that's simply the generalization of R_1^2 in the previous formula. It is the R-squared of the regression of x_k on all other x variables. The smaller that R-squared, the smaller the SE.

Review Box 10.5 Multiple linear regression with three or more explanatory variables

- Equation: $y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots$
- Interpretation of β_k (slope of x_k):
  - On average, y is β_k units larger in the data for observations with one unit larger x_k but with the same value for all other x variables.
  - $SE(\hat{\beta}_k) = \frac{Std[e]}{\sqrt{n}Std[x_k]\sqrt{1-R_k^2}}$ where e is the regression residual and R_k^2 is the R-squared of the regression of x_k on all other x variables.

10.8 Nonlinear Patterns and Multiple Linear Regression

In Chapter 8 we introduced piecewise linear splines, quadratics, and other polynomials to approximate a nonlinear $y^E = f(x)$ regression.

From a substantive point of view, piecewise linear splines and polynomials of a single explanatory variable are not multiple regressions. They do not uncover differences with respect to one right-hand-side variable conditional on one or more other right-hand-side variables. Their slope coefficients cannot be interpreted as the coefficients of multiple regressions: it does not make sense to compare observations that have the same x but a different x^2. But such regressions are multiple linear regressions from a technical point of view. This means that the way their coefficients are calculated is the exact same way the coefficients of multiple linear regressions are calculated. Their standard errors are calculated the same way, too and so are their confidence intervals, test statistics, and p-values.

Testing hypotheses can be especially useful here, as it can help choose the functional form. With a piecewise linear spline, we can test whether the slopes are the same in adjacent line segments. If we can't reject the null that they are the same, we may as well join them instead of having separate line segments. Testing hypotheses helps in choosing a polynomial, too. Here an additional complication is that the coefficients don't have an easy interpretation in themselves. At the same time, testing if all nonlinear coefficients are zero may help decide whether to include them at all.

However, testing hypotheses to decide whether to include a higher-order polynomial has its issues. Recall that a multiple linear regression requires that the right-hand-side variables are not perfectly collinear. In other words, they cannot be linear functions of each other. With a polynomial on the right-hand side, those variables are exact functions of each other: x^2 is the square of x. But they are not a linear function of each other, so, technically, they are not perfectly collinear. That's why we can include both x and x^2 and, if needed, its higher-order terms, in a linear regression. While they are not perfectly collinear, explanatory variables in a polynomial are often highly correlated. That multicollinearity results in high standard errors, wide confidence intervals, and high p-values . As with all kinds of multicollinearity, there isn't anything we can do about that once we have settled on a functional form.

Importantly, when thinking about functional form, we should always keep in mind the substantive focus of our analysis. As we emphasized in Chapter 8, Section 8.1, we should go back to that original focus when deciding whether we want to include a piecewise linear spline or a polynomial to approximate a nonlinear pattern. There we said that we want our regression to have a good

approximation to a nonlinear pattern in x if our goal is prediction or analyzing residuals. We may not want that if all we care about is the average association between x and y, except if that nonlinearity messes up the average association. This last point is a bit subtle, but usually means that we may want to transform variables to relative changes or take logs if the distribution of x or y is very skewed.

Here we have multiple x variables. Should we care about whether each is related to average y in a nonlinear fashion? The answer is the same as earlier: yes, if we want to do prediction or analyze residuals; no, if we care about average associations (except we may want to have transformed variables here, too). In addition, when we focus on a single average association (with, say, x_1) and all the other variables (x_2, x_3, ...) are covariates to condition on, the only thing that matters is the coefficient on x_1. Even if nonlinearities matter for x_2 and x_3 themselves, they only matter for us if they make a difference in the estimated coefficient on x_1. Sometimes they do; very often they don't.

10.A3 CASE STUDY – Understanding the Gender Difference in Earnings

Nonlinear patterns and multiple linear regression

This step in our case study illustrates the point we made in the previous section. The regressions in Table 10.2 enter age in linear ways. Using part of the same data, in Chapter 9, Section 9.A2 we found that log earnings and age follow a nonlinear pattern. In particular, there we found that average log earnings are a positive and steep function of age for younger people, but the pattern becomes gradually flatter for the middle-aged and may become completely flat, or even negative, among older employees.

Should we worry about the nonlinear age–earnings pattern when our question is the average earnings difference between men and women? We investigated the gender gap conditional on age. Table 10.2 shows the results for multiple ways of doing it. Column (1) shows the regression with the unconditional difference that we showed in Table 10.1, for reference. Column (2) enters age in linear form. Column (3) enters it as quadratic. Column (4) enters it as a fourth-order polynomial.

The unconditional difference is −19.5%; the conditional difference is −18.5% according to column (2), and −18.3% according to columns (3) and (4). The various estimates of the conditional difference are very close to each other, and all of them are within each others' confidence intervals. Thus, apparently, the functional form for age does not really matter if we are interested in the average gender gap.

At the same time, all coefficient estimates of the high order polynomials are statistically significant, meaning that the nonlinear pattern is very likely true in the population and not just a chance event in the particular dataset. The R-squared of the more complicated regressions are larger. These indicate that the complicated polynomial specifications are better at capturing the patterns. That would certainly matter if our goal was to predict earnings. But it does not matter for uncovering the average gender difference in earnings.

Table 10.2 Gender differences in earnings – log earnings and age, various functional forms

Variables	(1) ln wage	(2) ln wage	(3) ln wage	(4) ln wage
female	−0.195**	−0.185**	−0.183**	−0.183**
	(0.008)	(0.008)	(0.008)	(0.008)
age		0.007**	0.063**	0.572**
		(0.000)	(0.003)	(0.116)
age^2			−0.001**	−0.017**
			(0.000)	(0.004)
age^3				0.000**
				(0.000)
age^4				−0.000**
				(0.000)
Constant	3.514**	3.198**	2.027**	−3.606**
	(0.006)	(0.018)	(0.073)	(1.178)
Observations	18 241	18 241	18 241	18 241
R-squared	0.028	0.046	0.060	0.062

Note: Robust standard error estimates in parentheses. ** $p<0.01$, * $p<0.05$.

Source: `cps-earnings` dataset. 2014 USA. Employees of age 24–65 with a graduate degree and 20 or more work hours per week.

10.9 Qualitative Right-Hand-Side Variables

A great advantage of multiple linear regression is that it can deal with binary and other qualitative explanatory variables (also called categories, factor variables), together with quantitative variables, on the right-hand side.

To include such variables in the regression, we need to have them as binary, zero–one variables – also called **dummy variables** in the regression context. That's straightforward for variables that are binary to begin with: assign values zero and one (as we did with *female* in the case study). We need to transform other kinds of qualitative variables into binary ones, too, each denoting whether the observation belongs to that category (one) or not (zero). Then we need to include all those binary variables in the regression. Well, all except one.

We should select one binary variable denoting one category as a **reference category**, or **reference group** – also known as the "left-out category." Then we have to include the binary variables for all other categories but not the reference category. That way the slope coefficient of a binary variable created from a qualitative variable shows the difference between observations in the category captured by the binary variable and the reference category. If we condition on other explanatory variables, too, the interpretation changes in the usual way: we compare observations that are similar in those other explanatory variables.

As an example, suppose that x is a categorical variable measuring the level of education with three values $x =$ *low*, *medium*, *high*. We need to create binary variables and include two of the three in the

regression. Let the binary variable x_{med} denote if $x = \textit{medium}$, and let the binary x_{high} variable denote if $x = \textit{high}$. Include x_{med} and x_{high} in the regression. The third potential variable for $x = \textit{low}$ is not included. It is the reference category.

$$y^E = \beta_0 + \beta_1 x_{med} + \beta_2 x_{high} \tag{10.10}$$

Let us start with the constant, β_0; this shows average y in the reference category. Here, β_0 is average y when both $x_{med} = 0$ and $x_{high} = 0$: this is when $x = \textit{low}$. β_1 is the difference in average y between observations that are different in x_{med} but the same in x_{high}. Thus β_1 shows the difference of average y between observations with $x = \textit{medium}$ and $x = \textit{low}$, the reference category. Similarly, β_2 shows the difference of average y between observations with $x = \textit{high}$ and $x = \textit{low}$, the reference category.

Which category to choose for the reference? In principle that should not matter: choose a category and all others are compared to that, but we can easily compute other comparisons from those. For example, the difference in y^E between observations with $x = \textit{high}$ and $x = \textit{medium}$ in the example above is simply $\beta_2 - \beta_1$ (both coefficients compare to $x = \textit{low}$, and that drops out of their difference). But the choice may matter for practical purposes. Two guiding principles may help this choice, one substantive, one statistical. The substantive guide is simple: we should choose the category to which we want to compare the rest. Examples include the home country, the capital city, the lowest or highest value group. The statistical guide is to choose a category with a large number of observations. That is relevant when we want to infer differences from the data for the population, or general pattern, it represents. If the reference category has very few observations, the coefficients that compare to it will have large standard errors, wide confidence intervals, and large p-values.

Review Box 10.6 Qualitative right-hand-side variables in multiple linear regression

- We should include qualitative right-hand-side variables with more categories as a series of binary ("dummy") variables.
- For a qualitative right-hand-side variable with k categories, we should enter $k-1$ binary variables; the category not represented by those binary variables is the reference category.
- Coefficients on each of the $k - 1$ binary variables show average differences in y compared to the reference category.

10.A4 CASE STUDY – Understanding the Gender Difference in Earnings

Qualitative variables

Let's use our case study to illustrate qualitative variables as we examine earnings differences by categories of educational degree. Recall that our data contains employees with graduate degrees. The dataset differentiates three such degrees: master's (including graduate teaching degrees, MAs, MScs, MBAs), professional (including MDs), and PhDs.

Table 10.3 shows the results from three regressions. As a starting point, column (1) repeats the results of the simple regression with *female* on the right-hand side; column (2) includes two education categories *ed_Profess* and *ed_PhD*; and column (3) includes another set of education categories, *ed_Profess* and *ed_MA*. The reference category is MA degree in column (2) and PhD in column (3).

Table 10.3 Gender differences in earnings – log earnings, gender and education

	(1)	(2)	(3)
Variables	**ln wage**	**ln wage**	**ln wage**
female	−0.195**	−0.182**	−0.182**
	(0.008)	(0.009)	(0.009)
ed_Profess		0.134**	−0.002
		(0.015)	(0.018)
ed_PhD		0.136**	
		(0.013)	
ed_MA			−0.136**
			(0.013)
Constant	3.514**	3.473**	3.609**
	(0.006)	(0.007)	(0.013)
Observations	18 241	18 241	18 241
R-squared	0.028	0.038	0.038

Note: MA, Professional, and PhD are three categories of graduate degree. Column (2): MA is the reference category. Column (3): the reference category is Professional or PhD. Robust standard error estimates in parentheses. ** $p<0.01$, * $p<0.05$.

Source: `cps-earnings` dataset. USA, 2014. Employees of age 24–65 with a graduate degree and 20 or more work hours per week.

The coefficients in column (2) of Table 10.3 show that comparing employees of the same gender, those with a professional degree earn, on average, 13.4% more than employees with an MA degree, and those with a PhD degree earn, on average, 13.6% more than employees with an MA degree. The coefficients in column (3) show that, among employees of the same gender, those with an MA degree earn, on average, 13.6% less than those with a PhD degree, and those with a professional degree earn about the same on average as those with a PhD degree. These differences are consistent with each other.

This is a large dataset so confidence intervals are rather narrow whichever group we choose as a reference category. Note that the coefficient on *female* is smaller, −0.182, when education is included in the regression. This suggests that part of the gender difference is due to the fact that women are somewhat more likely to be in the lower-earner MA group than in the higher-earner professional or PhD groups. But only a small part. We shall return to this finding later when we try to understand the causes of gender differences in earnings.

10.10 Interactions: Uncovering Different Slopes across Groups

Including binary variables for various categories of a qualitative variable uncovers average differences in y. But sometimes we want to know something more: whether and how much the slope with respect to a third variable differs by those categories. Multiple linear regression can uncover that too, with appropriate definition of the variables.

More generally, we can use the method of linear regression analysis to uncover how association between y and x varies by values of a third variable z. Such variation is called an **interaction**, as it shows how x and z interact in shaping average y. In medicine, when estimating the effect of x on y, if that effect varies by a third variable z, that z is called a **moderator variable**. Examples include whether malnutrition, immune deficiency, or smoking can decrease the effect of a drug to treat an illness. Non-medical examples of interactions include whether and how the effect of monetary policy differs by the openness of a country, or whether and how the way customer ratings are related to hotel prices differs by hotel stars.

Multiple regression offers the possibility to uncover such differences in patterns. For the simplest case, consider a regression with two explanatory variables: x_1 is quantitative; x_2 is binary. We wonder if the relationship between average y and x_1 is different for observations with $x_2 = 1$ than for $x_2 = 0$. How shall we uncover that difference?

A multiple regression that includes x_1 and x_2 estimates two parallel lines for the y–x_1 pattern: one for those with $x_2 = 0$ and one for those with $x_2 = 1$.

$$y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \tag{10.11}$$

The slope of x_1 is β_1 and is the same in this regression for observations in the $x_2 = 0$ group and observations in the $x_2 = 1$ group. β_2 shows the average difference in y between observations that are different in x_2 but have the same x_1. Since the slope of x_1 is the same for the two x_2 groups, this β_2 difference is the same across the range of x_1. This regression does not allow for the slope in x_1 to be different for the two groups. Thus, this regression cannot uncover whether the y–x_1 pattern differs in the two groups.

Denote the expected y conditional on x_1 in the $x_2 = 0$ group as y_0^E, and denote the expected y conditional on x_1 in the $x_2 = 1$ group as y_1^E. Then, the regression above implies that the intercept is different (higher by β_2 in the $x_2 = 1$ group) but the slopes are the same:
First group, $x_2 = 0$

$$y_0^E = \beta_0 + \beta_1 x_1 \tag{10.12}$$

Second group, $x_2 = 1$

$$y_1^E = \beta_0 + \beta_2 \times 1 + \beta_1 x_1 \tag{10.13}$$

If we want to allow for different slopes in the two x_2 groups, we have to do something different. That difference is including the interaction term. An **interaction term** is a new variable that is created from two other variables, by multiplying one by the other. In our case:

$$y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 \tag{10.14}$$

Not only are the intercepts different; the slopes are different, too:

$$y_0^E = \beta_0 + \beta_1 x_1 \tag{10.15}$$

$$y_1^E = \beta_0 + \beta_2 + (\beta_1 + \beta_3) x_1 \tag{10.16}$$

It turns out that the coefficients of this regression can be related to the coefficients of two simple regressions of y on x_1, estimated separately in the two x_2 groups:

$$y_0^E = \gamma_0 + \gamma_1 x_1 \tag{10.17}$$

$$y_1^E = \gamma_2 + \gamma_3 x_1 \tag{10.18}$$

What we have is $\gamma_0 = \beta_0$; $\gamma_1 = \beta_1$; $\gamma_2 = \beta_0 + \beta_2$; and $\gamma_3 = \beta_1 + \beta_3$.

In other words, the separate regressions in the two groups and the regression that pools observations but includes an interaction term yield exactly the same coefficient estimates. The coefficients of the separate regressions are easier to interpret. But the pooled regression with interaction allows for a direct test of whether the slopes are the same. $H_0: \beta_3 = 0$ is the null hypothesis for that test; thus the simple t-test answers this question.

We can mix these tools to build ever more complicated multiple regressions. Binary variables can be interacted with other binary variables . Binary variables created from qualitative explanatory variables with multiple categories can all be interacted with other variables. Piecewise linear splines or polynomials may be interacted with binary variables. More than two variables may be interacted as well. Furthermore, quantitative variables can also be interacted with each other, although the interpretation of such interactions is more complicated.

Review Box 10.7 Interactions of right-hand-side variables in multiple linear regression

- Interactions between right-hand-side variables in a linear regression allow for the slope coefficient of a variable to differ by values of another variable.
- Interactions between two right-hand-side variables are modeled in a linear regression as $y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$.
- β_1 shows average differences in y corresponding to a one-unit difference in x_1 when $x_2 = 0$.
- β_2 shows average differences in y corresponding to a one-unit difference in x_2 when $x_1 = 0$.
- β_3 is the coefficient on the interaction term. It shows the additional average differences in y corresponding to a one-unit difference in x_1 when x_2 is one unit larger, too. (It's symmetrical in x_1 and x_2 so it also shows the additional average differences in y corresponding to a one-unit difference in x_2 when x_1 is one unit larger, too.)
- When one of the two right-hand-side variables is binary, a simpler interpretation is also true. Say, $x_2 = 0$ *or* 1. Then,
  - β_1 shows the average difference in y corresponding to a one-unit difference in x_1 when $x_2 = 0$;
  - $\beta_1 + \beta_3$ shows the average difference in y corresponding to a one-unit difference in x_1 when $x_2 = 1$;
  - the coefficients of the regression are the same as the coefficients of two separate regressions on two parts of the data, one with $x_2 = 0$ and one with $x_2 = 1$:
    if $x_2 = 0$: $y^E = \beta_0 + \beta_1 x_1$
    if $x_2 = 1$: $y^E = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) x_1$

10.A5 CASE STUDY – Understanding the Gender Difference in Earnings

Interactions

We turn to illustrating the use of interactions as we consider the question of whether the patterns with age are similar or different for men versus women. As we discussed, we can investigate this in two ways that should lead to the same result: estimating regressions separately for men and women and estimating a regression that includes age interacted with gender. This regression model with an interaction is

$$(\ln w)^E = \beta_0 + \beta_1 \times age + \beta_2 \times female + \beta_3 \times age * female \tag{10.19}$$

Table 10.4 shows the results with age entered in a linear fashion. Column (1) shows the results for women, column (2) for men, column (3) for women and men pooled, with interactions. To have a better sense of the differences, which are often small, the table shows coefficients up to three digits.

Table 10.4 Gender differences in earnings – log earnings, gender, age, and their interaction

Variables	(1) Women	(2) Men	(3) All
female			−0.036
			(0.035)
age	0.006**	0.009**	0.009**
	(0.001)	(0.001)	(0.001)
female × age			−0.003**
			(0.001)
Constant	3.081**	3.117**	3.117**
	(0.023)	(0.026)	(0.026)
Observations	9 685	8 556	18 241
R-squared	0.011	0.028	0.047

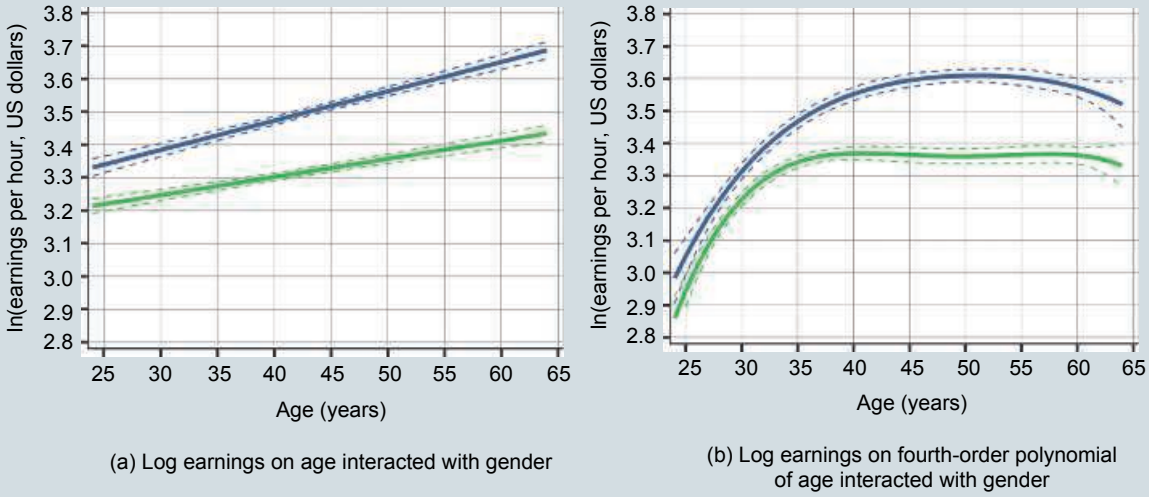
Note: Column (1) is women only; column (2) is men only; column (3) includes all employees. Robust standard error estimates in parentheses. ** $p<0.01$, * $p<0.05$.

Source: `cps-earnings` dataset. USA, 2014. Employees of age 24–65 with a graduate degree and 20 or more work hours per week.

According to column (1) of Table 10.4, women who are one year older earn 0.6% more, on average. According to column (2), men who are one year older earn 0.9% more, on average. Column (3) repeats the age coefficient for men. Then it shows that the slope of the log earnings–age pattern is, on average, 0.003 less positive for women, meaning that the earnings advantage of women who are one year older is 0.3 percentage points smaller than the earnings advantage of men who are one year older.

The advantage of the pooled regression, is its ability to allow for direct inference about gender differences. The 95% CI of the gender difference in the average pattern of ln wages and age is [−0.005, −0.001]. Among employees with a post-graduate degree in the USA in 2014, the wage difference corresponding to a one year difference in age was 0.1 to 0.5 percentage points less for women than for men. This confidence interval does not include zero. Accordingly, the t-test of whether the difference is zero rejects its null at the 5% level, suggesting that we can safely consider the difference as real in the population (as opposed to a chance event in the particular dataset); we are less than 5% likely to make a mistake by doing so.

The coefficient on the *female* variable in the pooled regression is −0.036. This is equal to the difference of the two regression constants: 3.081 for women and 3.117 for men. Those regression constants do not have a clear interpretation here (average log earnings when age is zero are practically meaningless). Their difference, which is actually the coefficient on *female* in the pooled regression, shows the average gender difference between employees with zero age. Similarly to the constants in the separate regressions, the coefficient is meaningless for any substantive purpose. Nevertheless, the regression needs it to have an intercept with the ln *w* axis.



(a) Log earnings on age interacted with gender

(b) Log earnings on fourth-order polynomial of age interacted with gender

Figure 10.2 Earning differences by gender as function of age

Note: Regression lines (curves) and 95% confidence intervals from regression of log earnings on age interacted with gender.
Source: `cps-earnings` dataset. USA, 2014. Employees of age 24–65 with a graduate degree and 20 or more work hours per week. N=18 241.

Taking the coefficients on *female*, *age*, and *female* × *age* together, the regression allows us to calculate the average gender difference by age. This exercise takes the linear functional form seriously, an assumption we know is false. We shall repeat this exercise with a better approximation of the nonlinear patterns. For now, let's stick to the linear specification, for educational purposes. The youngest people in our sample are 25 years old. Starting with the separate regressions, the predicted log wage for women of age 25 is $3.081 + 25 \times 0.006 \approx 3.231$. For men, $3.117 + 25 \times 0.009 \approx 3.342$. The difference is −0.11. We get the same number from the pooled regression: the gender difference at age 25 should be the gender difference at age

zero implied by the coefficient on *female* plus 25 times the difference in the slope by age, the coefficient on the interaction term *female* × *age*: $-0.036 + 25 \times -0.003 \approx -0.11$. Carrying out the same calculations for age 45 yields a difference of -0.17. These results imply that the gender difference in average earnings is wider for older ages.

Figure 10.2a shows the relationship graphically. It includes two lines with a growing gap: earnings difference is higher for older age. Remember, our regression can capture this growing gap because it includes the interaction. Without the interaction, we would not be able to see this, as that specification would force two parallel lines at constant distance.

However, we know that the pattern on age and (log) earnings is not linear. Our earlier results indicate that a fourth-order polynomial is a better approximation to that pattern. To explore whether the shapes of the age–earnings profiles are different between women and men, we re-estimated the regression with age in a fourth-order polynomial interacted with gender:

$$(\ln w)^E = \beta_0 + \beta_1 age + \beta_2 age^2 + \beta_3 age^3 + \beta_4 age^4 + \beta_5 female + \beta_6 female \times age + \beta_7 female \times age^2 + \beta_8 female \times age^3 + \beta_9 female \times age^4 \quad (10.20)$$

This is a complicated regression with coefficients that are practically impossible to interpret. We don't show the coefficient estimates here. Instead we visualize the results. The graph in Figure 10.2b shows the predicted pattern (the regression curves) for women and men, together with the confidence intervals of the regression lines (curves here), as introduced in Chapter 9, Section 9.3.

Figure 10.2a suggests that the average earnings difference is a little less than 10% between ages 25 and 30, increases to around 15% by age 40, and reaches 22% by age 50, from where it decreases slightly to age 60 and more by age 65. These differences are likely similar in the population represented by the data as the confidence intervals around the regression curves are rather narrow, except at the two ends.

These results are very informative. Many factors may cause women with a graduate degree to earn less than men. Some of those factors are present at young age, but either they are more important in middle age, or additional factors start playing a role by then.

What can students infer from these results about the gender differences they may experience through their careers? Statistical inference established that the patterns are very likely present in the population represented by the data: employees with graduate degrees in the USA in 2014. The first question of external validity is whether similar patterns are likely to be true in the future as well and, if we are interested in another country, whether the patterns are similar there. The second question is the extent to which differences by age in the cross-section are informative about what we can expect to happen through time as people age. As we discussed earlier, those questions are impossible to answer with this data. Analyzing more data may help some but will never give a definitive answer to all questions. Nevertheless, the information produced by our analysis is a good starting point.

10.11 Multiple Regression and Causal Analysis

When interpreting regression coefficients, we advise being careful with the language, talking about differences and associations not effects and causation. But, can we say anything regarding the extent to which our results may indicate a causal link?

This question is all the more relevant because one main reason to estimate multiple regressions is to get closer to a causal interpretation. By conditioning on other observable variables, we can get closer to comparing similar objects – "apples to apples" – even in observational data. But getting closer is not the same as getting there.

For example, estimating the effect of a training program at a firm on the performance of employees would require comparing participants to non-participants who would perform similarly to how participants would without the program. A randomized experiment ensures such comparability. By randomly deciding who participates and who does not participate, we get two groups that are very similar in everything that is relevant, including what their future performance would be without the program. If, instead of a random rule, employees decided for themselves whether they participate in the program, a simple comparison of participants to non-participants would not measure the effect of the program because participants may have achieved different performance without the training.

The difference is between data from controlled experiments and observational data . Simple comparisons don't uncover causal relations in observational data. In principle, we may improve this by conditioning on every potential confounder: variables that would affect y and the causal variable x_1 at the same time. (In the training example, these are variables that would make participants and non-participants achieve different performance without the training, such as skills and motivation.) Such a comparison is called **ceteris paribus**.

But, importantly, conditioning on everything is impossible in general. Ceteris paribus prescribes what we want to condition on; a multiple regression can condition on what's in the data the way it is measured.

One more caveat. Not all variables should be included as covariates even if correlated both with the causal variable and the dependent variable. Such variables are called **bad conditioning variables**, or bad control variables. Examples include variables that are actually part of the **causal mechanism**, for example, how much participants in the training program actually learn.

What variables to include in a multiple regression and what variables not to include when aiming to estimate the effect of x on y is a difficult question. Chapter 19 will discuss this question along with the more general question of whether and when conditioning on other variables can lead to a good estimate of the effect of x on y, and what we mean by such an effect in the first place.

10.A6 CASE STUDY – Understanding the Gender Difference in Earnings

Thinking about cause and effect and getting closer to estimating it via multiple regression

Figure 10.2a showed a large and relatively stable average gender difference in earnings between ages 40 and 60 in the data and the population it represents (employees with a graduate degree in the USA in 2014). What might cause that difference?

One potential explanation is labor market discrimination. Labor market discrimination means that members of a group (women, minorities) earn systematically less per hour than members of another group (men, the majority) even if they have the same marginal product. Marginal product simply means their contribution to the sales of their employer by working one additional hour.

If one hour of work by women brings as much for the employer as one hour of work by men, they should earn the same, at least on average. There may be individual deviations for various reasons due to mistakes and special circumstances, but there should not be systematic differences in earnings per hour.

Note that this concept of labor market discrimination is quite narrow. For example, women may earn less on average because they are less frequently promoted to positions in which their work could have a higher effect on company sales. That would not count as labor market discrimination according to this narrow definition. A broader notion of discrimination would want to take that into account. An even broader concept of social inequality may recognize that women may choose occupations with flexible or shorter hours of work due to social norms about division of labor in the family. That may result in the over-representation of women in jobs that offer lower wages in return for more flexible hours.

Let's use our data to shed some light on these issues. Starting with the narrow definition of labor market discrimination, we have a clear steer as to what ceteris paribus analysis would be: condition on marginal product, or everything that matters for marginal product (and may possibly differ by gender). These may include cognitive skills, motivation, the ability to work efficiently in teams, and so on. Real-life data does not include all those variables. Indeed, our data has very little on skills: three broad categories of graduate degree and age. We may add race, ethnicity, and whether a person was born in the USA that may be related to the quality of education as well as other potential sources of discrimination.

To shed light on broader concepts of discrimination, we may want to enrich our regression by including more covariates. One example is occupation. Women may choose occupations that offer shorter and more flexible hours in exchange for lower wages. For the narrow concept of discrimination, we would like to condition on occupation, because we would want to compare women and men with the same work tasks. For the broad concept, we would not want to condition on it, because choice of occupation is affected by social norms about gender. Similar variables are industry, union status, hours worked, or whether the employer is private, nonprofit, or government.

Table 10.5 shows the results of those regressions. Some regressions have many explanatory variables. Instead of showing the coefficients of all, we show the coefficient and standard error of the variable of focus: *female*. The subsequent rows of the table indicate which variables are included as covariates. This is in fact a standard way of presenting results of large multiple regressions that focus on a single coefficient.

The data used for these regressions in Table 10.5 is a subset of the data used previously: it contains employees of age 40 to 60 with a graduate degree who work 20 hours per week or more. We focus on this age group because, as we have seen, this group has the largest average gender difference in earnings. We have 9816 such employees in our data.

Column (1) shows that women earn 22.4% less than men, on average, in the data (employees of age 40 to 60 with a graduate degree who work 20 hours or more). When we condition on age and the two binary variables of education, the difference is only slightly less, 21.2% (column (2)). This small difference appears to suggest that differences in age and education do not contribute to gender differences in earnings. However, our measures of education are only two binary variables of degree level, and more detailed data may imply a larger role of educational differences.

Table 10.5 Gender differences in earnings – regression with many covariates on a narrower sample

Variables	(1) ln wage	(2) ln wage	(3) ln wage	(4) ln wage
female	−0.224**	−0.212**	−0.151**	−0.141**
	(0.012)	(0.012)	(0.012)	(0.012)
Age and education		YES	YES	YES
Family circumstances			YES	YES
Demographic background			YES	YES
Job characteristics			YES	YES
Age in polynomial				YES
Hours in polynomial				YES
Observations	9816	9816	9816	9816
R-squared	0.036	0.043	0.182	0.195

Note: Education: professional, PhD. Family circumstances: marital status and number of children. Demographic background: race, ethnicity, whether US-born. Job characteristics: hours worked, whether employer is federal, state, local government, or nonprofit; union membership, two-digit industry, and two-digit occupation codes. Age and hours polynomials are fourth-order. Robust standard error estimates in parentheses. ** $p<0.01$, * $p<0.05$.

Source: `cps-earnings` dataset. USA, 2014. All employees of age 40–60 with a graduate degree and 20 or more work hours per week.

Column (3) includes all other covariates. The gender difference is 15.1%. When we compare people with the same personal and family characteristics and job features as measured in the data, women earn 15.1% less than men. Some of these variables are meant to measure job flexibility, but they are imperfect. Omitted variables include flexibility of hours and commuting time. Column (4) includes the same variables but pays attention to the potentially nonlinear relations with the two continuous variables, age and hours worked. The gender difference is very similar, 14.1%. The confidence intervals are reasonably narrow around these coefficients (±2%). They suggest that the average gender difference in the data, unconditional or conditional on the covariates, is of similar magnitude in the population represented by our data to what's in the data.

What did we learn from this exercise? We certainly could not safely pin down the role of labor market discrimination versus other reasons in driving the gender inequality in pay. Even their relative role is hard to assess from these results as the productivity measures are few, and the other covariates may be related to discrimination as well as preferences or other aspects of productivity. Thus, we cannot be sure that the 14.1% in column (4) is due to discrimination, and we can't even be sure if the role of discrimination is larger or smaller than that.

Nevertheless, our analysis provided some useful facts. The most important of them is that the gender difference is quite small below age 30, and it's the largest among employees between ages 40 and 60. Thus, gender differences, whether due to discrimination or other reasons, tend to be small among younger employees. In contrast, the disadvantages of women are large among middle-aged employees who also tend to be the highest earning employees. This is consistent with many potential explanations, such as the difficulty of women to advance their careers

relative to men due to "glass ceiling effects" (discrimination at promotion to high job ranks), or differences in preferences for job flexibility versus career advancement, which, in turn, may be due to differences in preferences or differences in the constraints the division of labor in families put on women versus men.

On the methods side, this case study illustrated how to estimate multiple linear regressions, and how to interpret and generalize their results. It showed how we can estimate and visualize different patterns of association, including nonlinear patterns, between different groups. It highlighted the difficulty of drawing causal conclusions from regression estimates using cross-sectional data. Nevertheless, it also illustrated that, even in the absence of clear causal conclusions, multiple regression analysis can advance our understanding of the sources of a difference uncovered by a simple regression.

10.12 Multiple Regression and Prediction

One frequent reason to estimate a multiple regression is to make a **prediction**: find the best guess for the dependent variable, or target variable y_j for a particular target observation j, for which we know the right-hand-side variables x but not y. Multiple regression offers a better prediction than a simple regression because it includes more x variables.

The predicted value of the dependent variable in a multiple regression for an observation j with known values for the explanatory variables $x_{1j}, x_{2j}, \ldots$ is simply

$$\hat{y}_j = \hat{\beta}_0 + \hat{\beta}_1 x_{1j} + \hat{\beta}_2 x_{2j} + \cdots \tag{10.21}$$

When the goal is prediction, we want the regression to produce as good a fit as possible. More precisely, we want as good a fit as possible to the general pattern that is representative of the target observation j. Good fit in a dataset is a good starting point – that is, of course, if our data is representative of that general pattern. But it's not necessarily the same. A regression with a very good fit in our data may not produce a similarly good fit in the general pattern. A common danger is **overfitting** the data: finding patterns in the data that are not true in the general pattern. Thus, when using multiple regression for prediction, we want a regression that provides good fit without overfitting the data. Finding a multiple regression means selecting right-hand-side variables and functional forms for those variables. We'll discuss this issue in more detail when we introduce the framework for prediction in Chapter 13.

But how can we assess the fit of multiple regressions? Just like with simple regressions, the most commonly used measure is the R-squared. The R-squared in a multiple regression is conceptually the same as in a simple regression that we introduced in Chapter 7:

$$R^2 = \frac{Var[\hat{y}]}{Var[y]} = 1 - \frac{Var[e]}{Var[y]} \tag{10.22}$$

where $Var[y] = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2$, $Var[\hat{y}] = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$, and $Var[e] = \sum_{i=1}^{n}(e_i)^2$. Note that $\bar{\hat{y}} = \bar{y}$, and $\bar{e} = 0$.

The R-squared is a useful statistic to describe the fit of regressions. For that reason, it is common practice to report the R-squared in standard tables of regression results.

Unfortunately, the R-squared is an imperfect measure for selecting the best multiple regression for prediction purposes. The reason is that regressions with the highest R-squared tend to overfit the data.

When we compare two regressions, and one of them includes all the right-hand-side variables in the other one plus some more, the regression with more variables always produces a higher R-squared. Thus, regressions with more right-hand-side variables tend to produce higher R-squared. But that's not always good: regressions with more variables have a larger risk of overfitting the data. To see this, consider an extreme example. A regression with a binary indicator variable for each of the observations in the data (minus one for the reference category) produces a perfect fit with an R-squared of one. But such a regression would be completely useless to predict values outside the data. Thus, for variable selection, alternative measures are used, as we shall discuss it in Chapter 14.

Until we learn about more systematic methods to select the right-hand-side variables in the regression for prediction, all we can do is to use our intuition. The goal is to have a regression that captures patterns that are likely to be true for the general pattern for our target observations. Often, that means including variables that capture substantial differences in y, and not including variables whose coefficients imply tiny differences. That implies leaving out variables that capture detailed categories of qualitative variables or complicated interactions. To do really well, we will need the systematic tools we'll cover in Chapters 13 and 14.

The last topic in prediction is how we can visualize the fit of our regression. The purpose of such a graph is to compare values of y to the regression line. We visualized the fit of a simple regression with a scatterplot and the regression line in the x–y coordinate system. We did something similar with the age–gender interaction, too. However, with a multiple regression with more variables, we can't produce such a visualization because we have too many right-hand-side variables.

Instead, we can visualize the fit of a multiple regression by the **$\hat{y}$—y plot**. This plot has $\hat{y}$ on the horizontal axis and y on the vertical axis. The plot features the 45 degree line and the scatterplot around it. The 45 degree line is also the regression line of y regressed on $\hat{y}$. To see this consider that the regression of y on $\hat{y}$ shows the expected value of y for values of $\hat{y}$. But $\hat{y}$ is already the expected value of y conditional on the right-hand-side variables, so the expected value of y conditional on $\hat{y}$ is the same as $\hat{y}$. Therefore this line connects points where $\hat{y} = y$, so it is the 45 degree line.

The scatterplot around this line shows how actual values of y differ from their predicted value $\hat{y}$. The better the fit of the regression, the closer this scatterplot is to the 45 degree line (and the closer R-squared is to one). This visualization is more informative than the R-squared. For example, we can use the $\hat{y}$—y plot to identify observations with especially large positive or negative residuals. In this sense, it generalizes the scatterplot with a regression line when we only had a single x.

Review Box 10.8 Prediction with multiple linear regression

- The predicted value of the y variable from a multiple regression is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x1 + \hat{\beta}_2 x_2 + \cdots$$

- The $\hat{y}$—y plot is a good way to visualize the fit of a prediction. It's a scatterplot with $\hat{y}$ on the horizontal axis and y on the vertical axis, together with the 45 degree line, which is the regression line of y on $\hat{y}$.
  - observations to the right of the 45 degree line show overpredictions ($\hat{y} > y$).
  - observations to the left of the 45 degree line show underpredictions ($\hat{y} < y$).

10.B1 CASE STUDY – Finding a Good Deal among Hotels with Multiple Regression

Prediction with multiple regression

Let's return once more to our example of hotel prices and distance to the city center. Recall that the goal of the analysis is to find a good deal from among the hotels for the date contained in the data. A good deal is a hotel that is inexpensive relative to its characteristics. Of those characteristics two are especially important: the distance of the hotel to the city center and the quality of the hotel. In the earlier chapters we considered simple regressions with the distance to the city center as the only explanatory variable. Here we add measures of quality and consider a multiple regression. Those measures of quality are stars (3, 3.5, or 4) and rating (average customer rating, ranging from 2 to 5).

With prediction, capturing the functional form is often important. Based on earlier explorations of the price–distance relationship and similar explorations of the price–stars and price–ratings relationships, we arrived at the following specification. The regression has log price as the dependent variable, a piecewise linear spline in distance (knots at 1 and 4 miles), a piecewise linear spline in rating (one knot at 3.5), and binary indicators for stars (one for 3.5 stars, one for 4 stars; 3 stars is the reference category).

From a statistical point of view, this is prediction analysis. The goal is to find the best predicted (log) price that corresponds to distance, stars, and ratings of hotels. Then we focus on the difference of actual (log) price from its predicted value.

Good deals are hotels with large negative residuals from this regression. They have a (log) price that is below what's expected given their distance, stars, and rating. The more negative the residual, the lower their log price, and thus their price, compared to what's expected for them. Of course, our measures of quality are imperfect. The regression does not consider information on room size, view, details of location, or features that only photos can show. Therefore the result of this analysis should be a shortlist of hotels that the decision maker should look into in more detail.

Table 10.6 shows the five best deals: these are the hotels with the five most negative residuals. We may compare this list with the list in Chapter 7, Section 4.U1, that was based on the residuals of a simple linear regression of hotel price on distance. Only two hotels "21912" and "22080" featured on both lists; hotel "21912" is the best deal now, there it was the second best deal. The rest of the hotels from Chapter 7 did not make it to the list here. When considering stars and rating, they do not appear to be such good deals anymore because their ratings and stars are low. Instead, we have three other hotels that have good measures of quality and are not very far yet they have relatively low price. This list is a good shortlist to find the best deal after looking into specific details and photos on the price comparison website.

How good is the fit of this regression? Its R-squared is 0.55: 55 percent in the variation in log price is explained by the regression. In comparison, a regression with log price and piecewise linear spline in distance would produce an R-squared of 0.37. Including stars and ratings improved the fit by 18 percentage points.

The $\hat{y}-y$ plot in Figure 10.3 visualizes the fit of this regression. The plot features the 45 degree line. Dots above the line correspond to observations with a positive residual: hotels that have

Table 10.6 Good deals for hotels: the five hotels with the most negative residuals

Hotel name	Price	Residual in ln(price)	Distance	Stars	Rating
21912	60	−0.565	1.1	4	4.1
21975	115	−0.405	0.1	4	4.3
22344	50	−0.385	3.9	3	3.9
22080	54	−0.338	1.1	3	3.2
22184	75	−0.335	0.7	3	4.1

Note: List of the five observations with the smallest (most negative) residuals from the multiple regression with log *price* on the left-hand side; right-hand-side variables are distance to the city center (piecewise linear spline with knots at 1 and 4 miles), average customer rating (piecewise linear spline with knot at 3.5), binary variables for 3.5 stars and 4 stars (reference category is 3 stars).

Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. Hotels with 3 to 4 stars within 8 miles of the city center, N=217.

higher price than expected based on the right-hand-side variables. Dots below the line correspond to observations with a negative residual: hotels that have lower price than expected. The dots that are furthest down from the line are the candidates for a good deal.

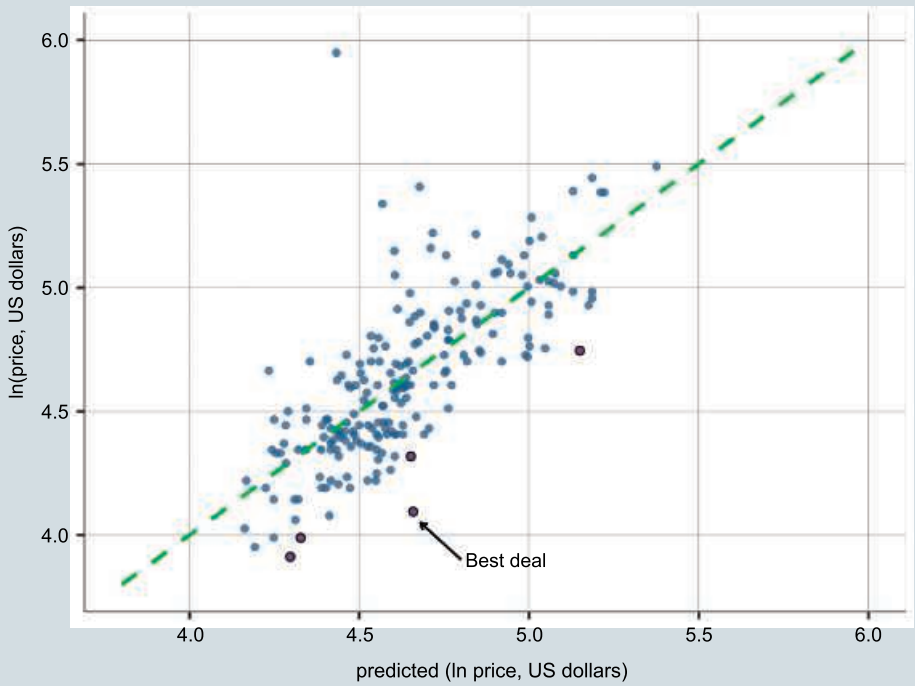


Figure 10.3 $\hat{y}$–y plot for log hotel price

Note: Results from a regression of ln price on distance to the city center (piecewise linear spline with knots at 1 and 4 miles), average customer rating (piecewise linear spline with knot at 3.5), binary variables for 3.5 stars and 4 stars (reference category is 3 stars). y is ln ***price***; $\hat{y}$ is predicted ln ***price*** from the regression. Five best deals denoted with purple.

Source: `hotels-vienna` dataset. Vienna, November 2017, weekday. Hotels with 3 to 4 stars within 8 miles of the city center; N=207.

This concludes the series of case studies using the `hotels-vienna` dataset to identify the hotels that are the best deals. We produced a shortlist of hotels that are the least expensive relative to their distance to the city center and their quality, measured by average customer ratings and stars.

10.13 Main Takeaways

Multiple regression allows for comparing mean y for different values of x for observations with the same values for the other variables.

- Doing so leads to better predictions and estimates of the slope on x that are usually closer to its true effect.
- Qualitative variables should be entered as binary variables on the right-hand side of multiple regressions.
- Interactions can uncover different slopes of one variable by values of another variable (e.g., denoting different groups).

PRACTICE QUESTIONS

1. The 95% CI of a slope coefficient in a multiple regression is narrower, the larger the R-squared of the regression. Why?
2. The 95% CI of a slope coefficient in a multiple regression is wider, the more the variable is correlated with the other right-hand-side variables. Why?
3. What's the difference between β in $y^E = \alpha + \beta x_1$ and β_1 in $y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2$? Why is this difference called omitted variable bias?
4. You want to estimate differences in spending on cigarettes by family income, and you are also interested how this difference in spending varies by the level of education of the adults in the family. Write down a regression that can uncover those differences, and interpret the coefficients of that regression. (Hint: you may define education as a binary variable, high versus low.)
5. Give an example of a multiple regression with two binary right-hand-side variables and their interaction. Write down the regression and interpret its coefficients.
6. What's a $\hat{y}$–y plot, and what is it good for? Give an example.
7. You want to predict y with the help of ten x variables using multiple linear regression. A regression that includes the ten variables produces an R-squared of 0.4. A regression that includes those ten variables together with many of their interactions has 100 variables altogether, and it produces an R-squared of 0.43. List pros and cons for choosing each of the two regressions for your prediction.
8. You want to estimate the effect of x on y with the help of a multiple regression. You have 10 z variables that you want to condition on to get closer to causality. A regression that includes the ten variables produces a coefficient on x that is 0.40, $SE = 0.05$. A regression that includes those

ten variables together with many of their interactions gives a coefficient estimate of 0.35 and $SE = 0.10$. List pros and cons for choosing each of the two regressions for your causal analysis.

9. True or false? Why? Multiple regression analysis of observational data shows expected differences in the dependent variable corresponding to differences of an explanatory variable, ceteris paribus.
10. In a dataset of a cross-section of cars advertised on an online platform, log price is regressed on a binary variable that is one if the ad comes from a dealer and zero if it comes from the owner of the car. The slope coefficient is 0.1. When we add the age of the car to the regression, the slope coefficient on the same binary variable is 0.05. Interpret both numbers.
11. In the previous example the coefficient on the age of the car is −0.1. Are dealer-advertised cars older or younger on average? By how much?
12. Log hotel price is regressed on distance to the city center, average customer rating, and number of stars in a linear regression, using data on a cross-section of hotels in a city. The slope coefficient on stars is 0.2. Interpret this number.
13. The standard error on the slope coefficient in the previous question is 0.05. What's its 95% CI and what does that mean?
14. We are interested in how airline prices are related to the distance traveled and the size of the market (number of passengers). The data consists of average prices on the most popular markets (routes) of the USA (e.g., Boston–Chicago) for the year 2018. OLS estimates of our regression are the following:

$$(\ln price)^E = 4.4 + 0.3 distance - 0.06 passengers \tag{10.23}$$

where ln *price* is log price, *distance* is measured in thousand miles, and *passengers* is the number of passengers per day (thousands). Interpret the two slope coefficient estimates.
15. We include the interaction of distance and passengers in the same regression and get a coefficient estimate of −0.002. Interpret this number. What can we conclude from its inclusion about the interaction of distance and passengers if the SE is 0.002?

DATA EXERCISES

Easier and/or shorter exercises are denoted by [*]; harder and/or longer exercises are denoted by [**].

1. Re-do the case study on gender difference in earnings by age – using a different group of employees or a different year in the USA, a different educational group, or a different country. Compare your results to those in the text and try to explain what you see. [*]
2. Use the `hotels-europe` dataset and pick a different city or a different date than used in the case study. Estimate a regression to predict hotel prices (or their log) using stars of the hotel, average customer ratings, and distance to the city center. Pay attention to functional forms. Argue for the best regression specification, and use its results to create a shortlist of five hotels that are underpriced. [**]
3. Use the same data as in the previous exercise and consider adding other variables in the data in an appropriate functional form. Argue for the best regression specification. Use the results to create a shortlist of five hotels that are underpriced, and compare this list to the list you produced in the previous exercise. [**]

4. Use the `worldbank-lifeexpectancy` dataset on a cross-section of countries. Pick a year. Regress life expectancy on log GDP per capita separately for different groups of countries (e.g., by continent). Then estimate a regression with the group dummies and their interactions with log GDP per capita. Interpret the coefficients and their 95% CI, and visualize the regression lines. What do you conclude from this exercise? [**]
5. Football is a global sport; FIFA, its governing body has 211 countries. There is plenty of evidence that countries' success in football is correlated with many socio-economic variables. Collect data on the results of all international games in a recent year, and pick a few socio-economic variables such as GDP/capita (pick three or more variables). Build a linear regression model to see which variables are correlated with the goal difference between teams. Create a $\hat{y}-y$ graph to see which countries perform better or worse than expected. [**]

REFERENCES AND FURTHER READING

A great reading on the role of regressions in arguing social change is Golbeck (2017).

One example of multivariate regression used in real life is the widespread use of the hedonic price index. For more on price indexes and hedonic regression, a great resource is Eurostat's *Handbook on Residential Property Prices Indices (RPPIs)* (Eurostat, 2013).

10.U1 UNDER THE HOOD: A TWO-STEP PROCEDURE TO GET THE MULTIPLE REGRESSION COEFFICIENT

We can get β_1 in the multiple regression from a two-step procedure that involves two simple regressions. There is no practical use for this two-step procedure: we get the exact coefficient we need by running multiple regression in software. In fact, this procedure is inferior because it produces standard errors that are wrong. Nevertheless, this procedure may highlight the intuition of how multiple regression works and how we should interpret its results. Moreover, the procedure, or its underlying logic, may become useful in substantially more complicated models.

We can get coefficient $\hat{\beta}_1$ in

$$y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \tag{10.24}$$

by (1) regressing x_1 on x_2:

$$x_1^E = \kappa + \lambda x_2 \tag{10.25}$$

and saving the residual $e = x_1 - \hat{\kappa} - \hat{\lambda} x_2$ and then (2) regressing y on this residual:

$$y^E = \pi + \rho e \tag{10.26}$$

The estimated slope coefficients are exactly the same:

$$\hat{\beta}_1 = \hat{\rho} \tag{10.27}$$

The procedure is analogous with more right-hand-side variables, only we have to regress x_1 on all other right-hand-side variables in step (1).